

GATE-BT

2011-2019

Old PAPERS

The Biomics

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BT : BIOTECHNOLOGY

Duration: Three Hours *Maximum Marks*: 100

Read the following instructions carefully.

1. This question paper contains **16** pages including blank pages for rough work. Please check all pages and report discrepancy, if any.

2. Write your registration number, your name and name of the examination centre at the specified locations on the right half of the Optical Response Sheet (**ORS**).

3. Using HB pencil, darken the appropriate bubble under each digit of your registration number and the letters corresponding to your paper code.

4. All questions in this paper are of objective type.

5. Questions must be answered on the **ORS** by darkening the appropriate bubble (marked A, B, C, D) using HB pencil against the question number on the left hand side of the ORS. **For each question darken the bubble of the correct answer**. In case you wish to change an answer, erase the old answer completely. More than one answer bubbled against a question will be treated as an incorrect response.

6. There are a total of 65 questions carrying 100 marks.

7. Questions Q.1 – Q.25 will carry 1-mark each, and questions Q.26 – Q.55 will carry 2-marks each.

8. Questions Q.48 – Q.51 (2 pairs) are common data questions and question pairs (Q.52, Q.53) and (Q.54, Q.55) are linked answer questions. The answer to the second question of the linked answer questions depends on the answer to the first question of the pair. If the first question in the linked pair is wrongly answered or is un-attempted, then the answer to the second question in the pair will not be evaluated.

9. Questions Q.56 – Q.65 belong to General Aptitude (GA). Questions Q.56 – Q.60 will carry 1-mark each, and questions Q.61 – Q.65 will carry 2-marks each. The GA questions will begin on a fresh page starting from page **11**.

10. Un-attempted questions will carry zero marks.

11. Wrong answers will carry **NEGATIVE** marks. For Q.1 – Q.25 and Q.56 – Q.60, ⅓ mark will be deducted for each wrong answer. For Q.26 – Q.51 and Q.61 – Q.65, ⅔ mark will be deducted for each wrong answer. The question pairs (Q.52, Q.53), and (Q.54, Q.55) are questions with linked answers. There will be negative marks only for wrong answer to the first question of the linked answer question pair i.e. for Q.52 and Q.54. ⅔ mark will be deducted for each wrong answer. There is no negative marking for Q.53 and Q.55.

12. Calculator (without data connectivity) is allowed in the examination hall.

13. Charts, graph sheets or tables are NOT allowed in the examination hall.

14. Rough work can be done on the question paper itself. Additionally, blank pages are provided at the end of the question paper for rough work.

Q.1 – Q.25 carry one mark each.

Q.1 Hybridoma technology is used to produce

(A) monoclonal antibodies
(B) polyclonal antibodies
(C) both monoclonal and polyclonal antibodies
(D) B cells

Q.2 Ames test is used to determine

(A) the mutagenicity of a chemical
(B) carcinogenicity of a chemical
(C) both mutagenicity and carcinogenicity of a chemical
(D) toxicity of a chemical

Q.3 The bacteria known to be naturally competent for transformation of DNA is

(A) *Escherichia coli*
(B) *Bacillus subtilis*
(C) *Mycobacterium tuberculosis*
(D) *Yersinia pestis*

Q.4 Antibiotic resistance marker that CANNOT be used in a cloning vector in Gram negative bacteria is

(A) Streptomycin
(B) Ampicillin
(C) Vancomycin
(D) Kanamycin

Q.5 Program used for essentially local similarity search is

(A) BLAST
(B) RasMol
(C) ExPASY
(D) SWISS-PROT

Q.6 Peptidyl transferase activity resides in

(A) 16S rRNA
(B) 23S rRNA
(C) 5S rRNA
(D) 28S rRNA

Q.7 In transgenics, alterations in the sequence of nucleotide in genes are due to

P. Substitution
Q. Deletion
R. Insertion
S. Rearrangement

(A) P and Q
(B) P, Q and R
(C) Q and R
(D) R and S

Q.8 During transcription

(A) DNA Gyrase introduces negative supercoils and DNA Topoisomerase I removes negative supercoils
(B) DNA Topoisomerase I introduces negative supercoils and DNA Gyrase removes negative supercoils
(C) both DNA Gyrase and DNA Topoisomearse I introduce negative supercoils
(D) both DNA Gyrase and DNA Topoisomearse I remove negative supercoils

Q.9 Under stress conditions bacteria accumulate

(A) ppGpp (Guanosine tetraphosphate)
(B) pppGpp (Guanosine pentaphosphate)
(C) both ppGpp and pppGpp
(D) either ppGpp or pppGpp

Q.10 An example for template independent DNA polymerase is

(A) DNA Polymerase I
(B) RNA polymerase
(C) Terminal deoxynucleotidyl transferase
(D) DNA polymerase III

Q.11 Which one of the following DOES NOT belong to the domain of Bacteria?

(A) Cyanobacteria
(B) Proteobacteria
(C) Bacteroids
(D) Methanobacterium

Q.12 Interferon-β is produced by

(A) bacteria infected cells
(B) virus infected cells
(C) both virus and bacteria infected cells
(D) fungi infected cells

Q.13 A culture of bacteria is infected with bacteriophage at a multiplicity of 0.3. The probability of a single cell infected with 3 phages is

(A) 0.9
(B) 0.27
(C) 0.009
(D) 0.027

Q.14 A neonatally thymectomized mouse, immunized with protein antigen shows

(A) both primary and secondary responses to the antigen
(B) only primary response to the antigen
(C) delayed type hypersensitive reactions
(D) no response to the antigen

Q.15 Lymphocytes interact with foreign antigens in

(A) Bone marrow
(B) Peripheral blood
(C) Thymus
(D) Lymph nodes

Q.16 Somatic cell gene transfer is used for

P. transgenic animal production
Q. transgenic diploid cell production
R. *in-vitro* fertilization
S. classical breeding of farm animals

(A) P, R and S
(B) P, Q and R
(C) P and R
(D) P only

Q.17 Accession number is a unique identification assigned to a

(A) single database entry for DNA/Protein
(B) single database entry for DNA only
(C) single database entry for Protein only
(D) multiple database entry for DNA/Protein

Q.18 Expressed Sequence Tag is defined as

(A) a partial sequence of a codon randomly selected from cDNA library
(B) the characteristic gene expressed in the cell
(C) the protein coding DNA sequence of a gene
(D) uncharacterized fragment of DNA presence in the cell

Q.19 In a chemostat operating under steady state, a bacterial culture can be grown at dilution rate higher than maximum growth rate by

(A) partial cell recycling
(B) using sub-optimal temperature
(C) pH cycling
(D) substrate feed rate cycling

Q.20 During lactic acid fermentation, net yield of ATP and NADH per mole of glucose is

(A) 2 ATP and 2 NADH
(B) 2 ATP and 0 NADH
(C) 4 ATP and 2 NADH
(D) 4 ATP and 0 NADH

Q.21 Identify the enzyme that catalyzes the following reaction

$$\alpha\text{-Ketoglutarate} + NADH + NH_4^+ + H^+ \xrightarrow{?} \text{Glutamate} + NAD^+ + H_2O$$

(A) Glutamate synthetase
(B) Glutamate oxoglutarate aminotransferase
(C) Glutamate dehydrogenase
(D) α-ketoglutarate deaminase

Q.22 The degree of inhibition for an enzyme catalyzed reaction at a particular inhibitor concentration is independent of initial substrate concentration. The inhibition follows

(A) competitive inhibition
(B) mixed inhibition
(C) un-competitive inhibition
(D) non-competitive inhibition

Q.23 Oxidation reduction reactions with positive standard redox potential (ΔE^0) have

(A) positive ΔG^0
(B) negative ΔG^0
(C) positive ΔE^*
(D) negative ΔE^*

Q.24 Nuclease-hypersensitive sites in the chromosomes are sites that appear to be

(A) H2 and H4 histone free
(B) H1 and H2 histone free
(C) H3 and H4 histone free
(D) Nucleosome free

Q.25 The formation of peptide cross-links between adjacent glycan chains in cell wall synthesis is called

(A) Transglycosylation
(B) Autoglycosylation
(C) Autopeptidation
(D) Transpeptidation

Q.26 – Q.55 carry two marks each.

Q.26 Determine the correctness or otherwise of the following **Assertion** (a) and the **Reason** (r)

Assertion: Somatic embryogenesis in plants is a two step process comprising of embryo initiation followed by embryo production.

Reason: Embryo initiation is independent of the presence of 2, 4-dichlorophenoxyacetic acid whereas embryo production requires a high concentration of 2, 4-dichlorophenoxyacetic acid.

(A) both (a) and (r) are true and (r) is the correct reason for (a)
(B) both (a) and (r) are true and (r) is not the correct reason for (a)
(C) (a) is true but (r) is false
(D) (a) is false but (r) is true

Q.27 An immobilized enzyme being used in a continuous plug flow reactor exhibits an effectiveness factor (η) of 1.2. The value of η being greater than 1.0 could be apparently due to

(A) substrate inhibited kinetics with internal pore diffusion limitation
(B) external pore diffusion limitation
(C) sigmoidal kinetics
(D) unstability of the enzyme

Q.28 A roller bottle culture vessel perfectly cylindrical in shape having inner radius (r) = 10 cm and length (l) = 20 cm was fitted with a spiral film of length (L) = 30 cm and width (W) = 20 cm. If the film can support 10^5 anchorage dependent cells per cm^2, the increase in the surface area after fitting the spiral film and the additional number of cells that can be grown respectively are

(A) 1200 cm^2 and 12×10^7 cells
(B) 600 cm^2 and 6×10^7 cells
(C) 600 cm^2 and 8300 cells
(D) 1200 cm^2 and 8300 cells

Q.29 Determine the correctness or otherwise of the following **Assertion** (a) and the **Reason** (r)

Assertion: MTT assay is used to determine cell viability based on the principle of colour formation by DNA fragmentation.

Reason: MTT assay is used to determine cell viability based on the colour development by converting tetrazolium soluble salt to insoluble salt.

(A) both (a) and (r) are true and (r) is the correct reason for (a)
(B) both (a) and (r) are true and (r) is not the correct reason for (a)
(C) (a) is true but (r) is false
(D) (a) is false but (r) is true

Q.30 Match the following antibiotics in **Group I** with their mode of action in **Group II**

Group I	**Group II**
P. Chloramphenicol	1. Binds to DNA gyrase
Q. Norfloxacin	2. Binds to RNA Polymerase
R. Puromycin	3. Inhibits peptidyl transferase
S. Rifampicin	4. Mimics aminoacyl-tRNA

(A) P-1, Q-3, R-2, S-4
(B) P-3, Q-1, R-2, S-4
(C) P-3, Q-1, R-4, S-2
(D) P-4, Q-2, R-3, S-1

Q.31 Match the chemicals in **Group I** with the possible type/class in **Group II**

Group I	**Group II**
P. Picloram	1. Vitamin
Q. Zeatin	2. Auxin
R. Thiamine	3. Amino Acid
S. Glutamine	4. Cytokinin

(A) P-2, Q-4, R-1, S-3
(B) P-4, Q-1, R-2, S-3
(C) P-3, Q-1, R-2, S-4
(D) P-4, Q-2, R-1, S-3

Q.32 Match **Group I** with **Group II**

Group I	**Group II**
P. Fibronectin	1. Uptake of amino acids and glucose
Q. Insulin	2. Trypsin inhibitor
R. α-Macroglobulin	3. Binds iron
S. Transferrin	4. Cell attachment to substratum

(A) P-2, Q-1, R-4, S-3 (B) P-3, Q-2, R-1, S-4
(C) P-4, Q-2, R-1, S-3 (D) P-4, Q-1, R-2, S-3

Q.33 Match the promoters listed in **Group I** with the tissues listed in **Group II**

Group I	**Group II**
P. α–Amylase	1. Endosperm
Q. Glutenin	2. Tuber
R. Phaseollin	3. Aleurone
S. Patatin	4. Cotyledon

(A) P-3, Q-1, R-4, S-2 (B) P-3, Q-4, R-1, S-2
(C) P-4, Q-2, R-1, S-3 (D) P-1, Q-3, R-2, S-4

Q.34 Consider the following statements.
I. T4 DNA ligase can catalyze blunt end ligation more efficiently than *E.coli* DNA ligase
II. The ligation efficiency of T4 DNA ligase can be increased with PEG and ficoll.

(A) only I is true (B) both I and II are true
(C) only II is true (D) I is true and II is false

Q.35 The turnover numbers for the enzymes, E1 and E2 are 150 s^{-1} and 15 s^{-1} respectively. This means

(A) E1 binds to its substrate with higher affinity than E2
(B) The velocity of reactions catalyzed by E1 and E2 at their respective saturating substrate concentrations could be equal, if concentration of E2 used is 10 times that of E1
(C) The velocity of E1 catalyzed reaction is always greater than that of E2
(D) The velocity of E1 catalyzed reaction at a particular enzyme concentration and saturating substrate concentration is lower than that of E2 catalyzed reaction under the same conditions

Q.36 Match the items in **Group I** with **Group II**

Group I *(Vectors)*	**Group II** (Maximum DNA packaging)
P. λ phage	1. 35-45 kb
Q. Bacterial Artificial Chromosomes (BACs)	2. 100-300 kb
R. P1 derived Artificial Chromosomes (PACs)	3. ≤ 300 kb
S. λ cosmid	4. 5 – 25 kb

(A) P-3, Q-4, R-1, S-2 (B) P-1, Q-3, R-2, S-4
(C) P-4, Q-3, R-2, S-1 (D) P-1, Q-2, R-3, S-4

Q.37 Match **Group I** with **Group II**

Group I	**Group II**
P. *Staphylococcus aureus*	1. Biofilms
Q. *Candida albicans*	2. Bacteriocins
R. *Mycobacterium tuberculosis*	3. Methicillin resistance
S. *Lactobacillus lactis*	4. Isoniazid

(A) P-1, Q-4, R-2, S-3 (B) P-2, Q-3, R-1, S-4
(C) P-3, Q-1, R-4, S-2 (D) P-1, Q-2, R-4, S-3

Q.38 A mutant G_α protein with increased GTPase activity would

(A) not bind to GTP (B) not bind to GDP
(C) show increased signaling (D) show decreased signaling

Q.39 Dizygotic twins are connected to a single placenta during their embryonic development. These twins

(A) have identical MHC haplotypes (B) have identical T_H cells
(C) have identical T cells (D) can accept grafts from each other (both (A) and (B))

Q.40 The dissociation constant K_d for ligand binding to the receptor is 10^{-7} M. The concentration of ligand required for occupying 10% of receptors is

(A) 10^{-6} M (B) 10^{-7} M
(C) 10^{-8} M (D) 10^{-9} M

Q.41 Receptor R is over expressed in CHO cells and analysed for expression. 6×10^7 cells were incubated with its radioactive ligand (specific activity 100 counts per picomole). If the total counts present in cell pellet was 1000 cpm, the average number of receptors R per cell is (assume complete saturation of receptors with ligand and one ligand binds to one receptor)

(A) 10^4 (B) 10^5
(C) 10^6 (D) 10^7

Q.42 A cell has five molecules of a rare mRNA. Each cell contains 4×10^5 mRNA molecules. How many clones one will need to screen to have 99% probability of finding at least one recombinant cDNA of the rare mRNA, after making cDNA library from such cell?

(A) 4.50×10^5 (B) 3.50×10^5
(C) 4.20×10^5 (D) 4.05×10^5

Q.43 Match the products in **Group I** with the microbial cultures in **Group II** used for their industrial production

Group I	**Group II**
P. Gluconic acid	1. *Leuconostoc mesenteroids*
Q. L – Lysine	2. *Aspergillus niger*
R. Dextran	3. *Brevibacterium flavum*
S. Cellulase	4. *Trichoderma reesei*

(A) P-2, Q-1, R-3, S-4 (B) P-1, Q-3, R-4, S-2
(C) P-2, Q-3, R-1, S-4 (D) P-3, Q-2, R-4, S-1

Q.44 Determine the correctness or otherwise of the following **Assertion** (a) and the **Reason** (r)

Assertion: Cytoplasmic male sterility (cms) is invariably due to defect(s) in mitochondrial function.

Reason: cms can be overcome by pollinating a fertility restoring (Rf) plant with pollen from a non cms plant.

(A) both (a) and (r) are true and (r) is the correct reason for (a)
(B) both (a) and (r) are true and (r) is not the correct reason for (a)
(C) (a) is false but (r) is true
(D) (a) is true but (r) is false

Q.45 Thermal death of microorganisms in the liquid medium follows first order kinetics. If the initial cell concentration in the fermentation medium is 10^8 cells / ml and the final acceptable contamination level is 10^{-3} cells, for how long should $1m^3$ medium be treated at temperature of 120° (thermal deactivation rate constant = 0.23 / min) to achieve acceptable load?

(A) 48 min (B) 11 min
(C) 110 min (D) 20 min

Q.46 True breeding *Drosophila* flies with curved wings and dark bodies were mated with true breeding short wings and tan body *Drosophila*. The F 1 progeny was observed to be with curved wings and tan body. The F 1 progeny was again allowed to breed and produced flies of the following phenotype, 45 curved wings tan body, 15 short wings tan body, 16 curved wings dark body and, 6 short wings dark body.

The mode of inheritance is

(A) Typical Mendelian with curved wings and tan body being dominant
(B) Typical non-Mendelian with curved wings and tan body not following any pattern
(C) Mendelian with suppression of phenotypes
(D) Mendelian with single crossover

Q.47 Match **Group I** with **Group II**

Group I	**Group II**
P. Real Time-PCR	1. Biochips
Q. 2-D Electrophoresis	2. Syber Green
R. Affinity chromatography	3. Antibody linked sephrose beads
S. Microarray	4. Ampholytes

(A) P-1, Q-2, R-4, S-3 (B) P-2, Q-3, R-4, S-1
(C) P-2, Q-4, R-3, S-1 (D) P-3, Q-2, R-1, S-4

Common Data Questions

Common Data for Questions 48 and 49:

A culture of *Rhizobium* is grown in a chemostat (100 m^3 bioreactor). The feed contains 12 g / L sucrose, K_s for the organism is 0.2 g / L and $\mu_m = 0.3\ h^{-1}$.

Q.48 The flow rate required to result in steady state concentration of sucrose as 1.5 g / L in the bioreactor will be

(A) 15 $m^3\ h^{-1}$ (B) 26 $m^3\ h^{-1}$
(C) 2.6 $m^3\ h^{-1}$ (D) 150 $m^3\ h^{-1}$

Q.49 If $Y_{x/s} = 0.4$ g / g for the above culture and steady state cell concentration in the bioreactor is 4 g / L the resulting substrate concentration will be

(A) 2 g / L (B) 8 g / L
(C) 4 g / L (D) 6 g / L

Common Data for Questions 50 and 51:
The width of the lipid bilayer membrane is 30 Å. It is permeated by a protein which is a right handed α-helix.

Q.50 The number of α-helical turns permeating the membrane is

(A) 5.6 turns (B) 3.5 turns
(C) 6.5 turns (D) 5.0 turns

Q.51 The number of amino acid residues present in the protein is

(A) 15 (B) 18
(C) 17 (D) 20

Linked Answer Questions

Statement for Linked Answer Questions 52 and 53:

The standard redox potential values for two half-reactions are given below. The value for Faraday's constant is 96.48 kJ $V^{-1}\ mol^{-1}$ and Gas constant R is 8.31 J $K^{-1}\ mol^{-1}$.

$NAD^+ + H^+ + 2e^- \leftrightarrow NADH$ -0.315 V
$FAD + 2H^+ + 2e^- \leftrightarrow FADH_2$ -0.219 V

Q.52 The ΔG^0 for the oxidation of NADH by FAD is

(A) − 9.25 kJ mol^{-1} (B) − 103.04 kJ mol^{-1}
(C) + 51.52 kJ mol^{-1} (D) − 18.5 kJ mol^{-1}

Q.53 The value of $\Delta G'$, given K_{eq} as 1.7, at 23°C will be

(A) − 17.19 kJ mol^{-1} (B) −19.8 kJ mol^{-1}
(C) + 52.82 kJ mol^{-1} (D) −117.07 kJ mol^{-1}

Statement for Linked Answer Questions 54 and 55:

During bioconversion of sucrose to citric acid by *Aspergillus niger* final samples of 6 batches of fermentation broth were analyzed for citric acid content. The results (in g/L) were found to be 47.3, 52.2, 49.2, 52.4, 49.1 and 46.3.

Q.54 The mean value of acid concentration will be

(A) 49.4 (B) 51.0
(C) 48.2 (D) 50.8

Q.55 The standard deviation for the above results is

(A) 2.49 (B) 3.0
(C) 1.84 (D) 5.91

GATE BT

General Aptitude (GA) Questions

Q.56 – Q.60 carry one mark each.

Q.56 *Which of the following options is the closest in meaning to the word below:*
Circuitous

(A) cyclic
(B) indirect
(C) confusing
(D) crooked

Q.57 *The question below consists of a pair of related words followed by four pairs of words. Select the pair that best expresses the relation in the original pair.*
Unemployed : Worker

(A) fallow : land
(B) unaware : sleeper
(C) wit : jester
(D) renovated : house

Q.58 *Choose the most appropriate word from the options given below to complete the following sentence:*
If we manage to _____ our natural resources, we would leave a better planet for our children.

(A) uphold
(B) restrain
(C) cherish
(D) conserve

Q.59 *Choose the most appropriate word from the options given below to complete the following sentence:*
His rather casual remarks on politics _____ his lack of seriousness about the subject.

(A) masked
(B) belied
(C) betrayed
(D) suppressed

Q.60 25 persons are in a room. 15 of them play hockey, 17 of them play football and 10 of them play both hockey and football. Then the number of persons playing neither hockey nor football is:

(A) 2 (B) 17 (C) 13 (D) 3

Q.61 – Q.65 carry two marks each.

Q.61 **Modern warfare has changed from large scale clashes of armies to suppression of civilian populations. Chemical agents that do their work silently appear to be suited to such warfare; and regretfully, there exist people in military establishments who think that chemical agents are useful tools for their cause.**

Which of the following statements best sums up the meaning of the above passage:

(A) Modern warfare has resulted in civil strife.
(B) Chemical agents are useful in modern warfare.
(C) Use of chemical agents in warfare would be undesirable.
(D) People in military establishments like to use chemical agents in war.

Q.62 If 137 + 276 = 435 how much is 731 + 672?

(A) 534 (B) 1403 (C) 1623 (D) 1513

Q.63 5 skilled workers can build a wall in 20 days; 8 semi-skilled workers can build a wall in 25 days; 10 unskilled workers can build a wall in 30 days. If a team has 2 skilled, 6 semi-skilled and 5 unskilled workers, how long will it take to build the wall?

(A) 20 days (B) 18 days (C) 16 days (D) 15 days

Q.64 Given digits 2, 2, 3, 3, 3, 4, 4, 4, 4 how many distinct 4 digit numbers greater than 3000 can be formed?

(A) 50 (B) 51 (C) 52 (D) 54

Q.65 Hari (H), Gita (G), Irfan (I) and Saira (S) are siblings (i.e. brothers and sisters). All were born on 1st January. The age difference between any two successive siblings (that is born one after another) is less than 3 years. Given the following facts:

i. Hari's age + Gita's age > Irfan's age + Saira's age.
ii. The age difference between Gita and Saira is 1 year. However, Gita is not the oldest and Saira is not the youngest.
iii. There are no twins.

In what order were they born (oldest first)?

(A) HSIG (B) SGHI (C) IGSH (D) IHSG

END OF THE QUESTION PAPER

GATE BT 2010 Answer Key

1	A	14	B	27	A	40	C	53	A
2	C	15	D	28	A	41	B	54	A
3	B	16	A	29	A	42	D	55	A
4	C	17	A	30	C	43	C	56	
5	A	18	A	31	A	44	A	57	
6	B	19	D	32	D	45	C	58	
7	B	20	B	33	D	46	B	59	
8	D	21	C	34	B	47	C	60	
9	B	22	D	35	C	48	C	61	
10	C	23	B	36	C	49	A	62	
11	B	24	C	37	C	50	A	63	
12	B	25	D	38	D	51	D	64	
13	D	26	C	39	A	52	D	65	

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Read the following instructions carefully.

1. Write your name and registration number in the space provided at the bottom of this page.
2. Take out the **O**ptical **R**esponse **S**heet (**ORS**) from this Question Booklet **without breaking the seal**.
3. Do not open the seal of the Question Booklet until you are asked to do so by the invigilator.
4. Write your registration number, your name and name of the examination centre at the specified locations on the right half of the **ORS**. Also, using HB pencil, darken the appropriate bubble under each digit of your registration number and the letters corresponding to your test paper code (BT).
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Name								
Registration Number	**BT**							

Q. 1 – Q. 25 carry one mark each.

Q.1 Embryonic stem cells are derived from

(A) fertilized embryo
(B) unfertilized embryo
(C) sperm
(D) kidney

Q.2 Members of the antibody protein family that have common structural features are collectively known as

(A) haptens (B) allergens (C) antigens (D) immunoglobulins

Q.3 Apoptosis is characterized by

(A) necrosis
(B) programmed cell death
(C) membrane leaky syndrome
(D) cell cycle arrest process

Q.4 Yeast artificial chromosomes (YAC's) are used for cloning

(A) large segments of DNA
(B) mRNA
(C) bacterial DNA
(D) yeast DNA

Q.5 The product commercially produced by animal cell culture is

(A) insulin
(B) tissue plasminogen activator
(C) interferon
(D) hepatitis B vaccine

Q.6 An alternative to glycolysis pathway is

(A) glyoxylate pathway
(B) pentose phosphate pathway
(C) citric acid cycle
(D) gluconeogenesis

Q.7 A cell in G_1 of interphase has 12 chromosomes. How many chromatids will be found per cell during metaphase II of meiosis?

(A) 6 (B) 12 (C) 18 (D) 24

Q.8 Diploid *Drosophila* has eight chromosomes. Which one of the following terms should **NOT** be used to describe *Drosophila* with sixteen numbers of chromosomes?

(A) Polyploid (B) Aneuploid (C) Euploid (D) Tetraploid

Q.9 Hydrated synthetic seeds which are produced by ion exchange reaction involve mixing the somatic embryos in a solution of

(A) sodium alginate and dropping it in a solution of calcium nitrate
(B) calcium alginate and dropping it in a solution of sodium nitrate
(C) calcium alginate and dropping it in a solution of ammonium nitrate
(D) mannitol and dropping it in a solution of sodium nitrate

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Q.10 Shoot organogenesis by tissue culture results into

(A) a bipolar structure that has no vascular connection with the explant
(B) a monopolar structure that has a strong connection with the pre-existing vascular tissue of the explant
(C) a monopolar structure that has no vascular connection with the explant
(D) a bipolar structure that has a strong connection with the pre-existing vascular tissue of the explant

Q.11 'Hairy roots' induced *in vitro* by the infection of *Agrobacterium rhizogenes,* are characterized by

P. a high degree of lateral branching
Q. genetic instability of culture
R. an absence of geotropism
S. poor biomass production

(A) P and R only
(B) P and Q only
(C) Q and R only
(D) R and S only

Q.12 In balanced growth phase of a cell

P. all components of a cell grow at the same rate
Q. specific growth determined by cell number or cell mass would be the same
R. the growth rate is independent of substrate concentration
S. the growth rate decreases with decreasing substrate concentration

(A) P, Q and S only
(B) Q, R and S only
(C) P, Q and R only
(D) P only

Q.13 In N-linked glycosylation, the oligosaccharide chain is attached to protein by

(A) asparagine
(B) arginine
(C) serine
(D) threonine

Q.14 Restriction endonucleases which recognize and cut same recognition sequences are known as

(A) isoschizomers
(B) isozymes
(C) isoaccepting endonucleases
(D) abzymes

Q.15 Substrate consumption in lag phase of microbial growth is primarily used for

P. turn over of the cell material
Q. maintenance of intracellular pH
R. motility
S. increase in cell number

(A) P, Q and S only
(B) Q, R and S only
(C) P, Q and R only
(D) S only

Q.16 Wash out (as defined by $D=\mu_{max}$) of a continuous stirred tank fermenter is characterized by (X=biomass, S=substrate concentration in bioreactor, S_0=substrate concentration in feed, P=product concentration in bioreactor)

(A) $X = 0, S = 0, P = 0$
(B) $X = 0, S = S_0, P = 0$
(C) $X = 0, S < 0, P = 0$
(D) $X < 0, S < 0, P < 0$

Q.17 The study of evolutionary relationships is known as

(A) genomics
(B) proteomics
(C) phylogenetics
(D) genetics

Q.18 The lipopolysaccharides present in bacterial cell wall has lipid A which is connected to

(A) O-polysaccharide
(B) core polysaccharide
(C) both with O-polysaccharide and core polysaccharide
(D) rhamnose-mannose disaccharide

Q.19 Molecular chaperones are class of proteins that facilitate

(A) the proper folding of newly synthesized proteins
(B) unfolding of newly synthesized proteins
(C) degradation of newly synthesized proteins
(D) targeting of newly synthesized proteins

Q.20 Gas vacuoles are present in

(A) *Anabaena flos-aquae*
(B) *Bacillus subtilis*
(C) *Acanthurus nigrofuscus*
(D) *Mycobacterium tuberculosis*

Q.21 In ABO blood group system, antigenic determinants are

(A) nucleic acid
(B) carbohydrate
(C) lipid
(D) protein

Q.22 The most widely used program for multiple sequence alignment is

(A) BLAST (B) FASTA (C) CLUSTAL (D) Chime

Q.23 Diphtheria toxin, tetracycline and streptomycin inhibit

(A) DNA repair
(B) DNA replication
(C) transcription
(D) translation

Q.24 The polymorphic domains for Class II MHC proteins are

(A) α_1 and β_2 domains only
(B) β_1 and α_2 domains only
(C) α_1 and β_1 domains only
(D) α_2 and β_2 domains only

Q.25 The protein in eukaryotes which is subjected to degradation undergoes

(A) phosphorylation
(B) carboxylation
(C) ubiquitation
(D) methylation

Q. 26 to Q. 55 carry two marks each.

Q.26 Match the viruses in **Group I** with their host cell receptors in **Group II.**

Group I	**Group II**
P. Hepatitis A virus	1. Heparan sulphate
Q. Human immunodeficiency virus	2. Acetylcholine receptor
R. Rabies virus	3. CD4 protein
S. Herpes simplex virus type I	4. Alpha-2 macroglobulin

(A) P-1, Q-3, R-2, S-4
(B) P-3, Q-4, R-1, S-2
(C) P-4, Q-3, R-2, S-1
(D) P-2, Q-3, R-1, S-4

Q.27 Match the microbial growth characteristics in **Group I** with the corresponding features in **Group II.**

Group I	**Group II**
P. Growth associated product formation	1. Specific growth rate decreases with increasing product concentration
Q. Non growth associated product formation	2. Specific product formation rate is constant
R. Product inhibition	3. Specific product formation rate is proportional to specific growth rate
S. Substrate inhibition	4. Specific growth rate decreases with increasing substrate concentration

(A) P-1, Q-2, R-4, S-3
(B) P-3, Q-2, R-1, S-4
(C) P-2, Q-1, R-3, S-4
(D) P-2, Q-3, R-4, S-1

Q.28 Match the items in **Group I** with **Group II.**

Group I	**Group II**
P. Circular dichroism	1. Concentration
Q. X-ray crystallography	2. Sedimentation coefficient
R. Freeze-drying	3. Secondary structure determination
S. Ultracentrifugation	4. Tertiary structure determination

(A) P-4, Q-1, R-2, S-3
(B) P-1, Q-4, R-3, S-2
(C) P-2, Q-3, R-4, S-1
(D) P-3, Q-4, R-1, S-2

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Q.29 Match the products in **Group I** with their respective organisms in **Group II.**

Group I	**Group II**
P. Glycerol	1. *Corynebacterium glutamicum*
Q. Glutamic acid	2. *Alcaligenes faecalis*
R. Curdlan	3. *Dunaliella salina*
S. Amphotericin B	4. *Streptomyces nodosus*

(A) P-2, Q-1, R-3, S-4
(B) P-4, Q-2, R-1, S-3
(C) P-3, Q-1, R-2, S-4
(D) P-2, Q-1, R-4, S-3

Q.30 Determine the correctness or otherwise of the following **Assertion** (a) and the **Reason** (r).

Assertion : I_gM is found in serum as a pentameric protein consisting of five I_gM monomers.

Reason : The pentameric form of I_gM is due to cross-linking of I_gM monomers via peptide bond.

(A) both (a) and (r) are true and (r) is the correct reason for (a)
(B) both (a) and (r) are true but (r) is not the correct reason for (a)
(C) (a) is true but (r) is false
(D) (a) is false but (r) is true

Q.31 Determine the correctness or otherwise of the following **Assertion** (a) and the **Reason** (r).

Assertion : N-methyl-N'-nitro-N-nitrosoguanidine (NTG) is an effective chemical mutagen.

Reason : Mutations induced by NTG mainly are the GC → AT transitions.

(A) both (a) and (r) are true and (r) is the correct reason for (a)
(B) both (a) and (r) are true but (r) is not the correct reason for (a)
(C) (a) is true but (r) is false
(D) (a) is false but (r) is true

Q.32 Determine the correctness of the following statements

I. Enhancer sequences are those DNA sequences that are involved in increasing the rate of DNA replication.

II. Enhancer sequences work by binding with eukaryotic gene activator factors.

(A) only **I** is true
(B) only **II** is true
(C) both **I** and **II** are true
(D) both **I** and **II** are false

Q.33 In a well aerated and agitated microbial culture, the 'supply' of oxygen is equal to 'demand' (uptake) of the growing culture. The K_La for such a system will be (K_La = volumetric mass transfer coefficient, C^* = dissolved oxygen concentration in liquid in equilibrium with gaseous oxygen, C = instantaneous value of dissolved oxygen concentration, 'r' = specific oxygen uptake rate per unit weight of cells, X = dry weight of the cells per unit volume)

(A) $(r\,X) / (C^* - C)$
(B) $(r) / X\,(C^* - C)$
(C) $(C^* - C) / (r\,X)$
(D) $(X) / r\,(C^* - C)$

Q.34 Structured William's model

P. can describe the changes in intracellular components of the cell during growth
Q. can not describe the death phase of the cells
R. can describe the variation of size of cells in the different phases of growth
S. can not describe the lag period of growth

Which one of the following is **CORRECT**?

(A) P, Q and S only
(B) P, Q and R only
(C) Q, R and S only
(D) P, R and S only

Q.35 Match items in **Group I** with **Group II.**

Group I	**Group II**
P. Glycolytic pathway	1. Chloroplast
Q. Eukaryotic oxidative metabolism	2. Glyoxysomes
R. Glyoxylate cycle	3. Mitochondria
S. Calvin cycle	4. Cytosol

(A) P-1, Q-2, R-3, S-4
(B) P-2, Q-3, R-4, S-1
(C) P-4, Q-3, R-2, S-1
(D) P-3, Q-4, R-1, S-2

Q.36 Match items in **Group I** with **Group II.**

Group I	**Group II**
P. Alzheimer's disease	1. H1N1
Q. Mad cow disease	2. Hemoglobin
R. Sickle cell anaemia	3. Prions
S. Swine flu	4. Amyloid

(A) P-4, Q-3, R-2, S-1
(B) P-3, Q-4, R-1, S-2
(C) P-2, Q-1, R-4, S-3
(D) P-1, Q-2, R-3, S-4

Q.37 Determine the correctness or otherwise of the following **Assertion** (a) and the **Reason** (r)

Assertion: The elucidation of ribosome structure helps in the development of new generation drugs.

Reason: The high resolution of macromolecular structure has enabled in structure-based drug design.

(A) both (a) and (r) are true and (r) is the correct reason for (a)
(B) both (a) and (r) are true but (r) is not the correct reason for (a)
(C) (a) is true but (r) is false
(D) (a) is false but (r) is true

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Q.38 Determine the correctness or otherwise of the following **Assertion** (a) and the **Reason** (r)
Assertion : A very low amount of inhibitor can act as an activator for allosteric enzymes.
Reason : Allosteric enzymes follow Michaelis-Menten kinetics.

(A) both (a) and (r) are true and (r) is the correct reason for (a)
(B) both (a) and (r) are true but (r) is not the correct reason for (a)
(C) (a) is true but (r) is false
(D) (a) is false but (r) is true

Q.39 Match the terms in **Group I** with their associated functions in **Group II.**

Group I	**Group II**
P. Shine-Dalgarno sequences	1. Aminoacylation of tRNA
Q. Leucine zipper	2. Gene silencing
R. Aminoacyl tRNA synthetase	3. Ribosome binding and facilitation of translation initiation
S. RNA interference (RNAi)	4. Transcription factors

(A) P-3, Q-4, R-1, S-2
(B) P-4, Q-3, R-2, S-1
(C) P-2, Q-3, R-1, S-4
(D) P-3, Q-2, R-4, S-1

Q.40 Protein-protein interactions are studied by

P. DNA foot printing
Q. Yeast two hybrid system
R. Ligase chain reaction
S. Mass spectrometry

(A) P and S only
(B) Q and S only
(C) P and R only
(D) Q and R only

Q.41 Determine the correctness or otherwise of the following **Assertion** (a) and the **Reason** (r)
Assertion : Isopropylthiogalactoside (IPTG) is a gratuitous inducer of lactose operon.
Reason : Gratuitous inducers are chemical analogs which behave like natural inducer but they do not serve as substrate for the enzymes that are subsequently synthesized.

(A) both (a) and (r) are true and (r) is the correct reason for (a)
(B) both (a) and (r) are true but (r) is not the correct reason for (a)
(C) (a) is true but (r) is false
(D) (a) is false but (r) is true

Q.42 Determine the correctness or otherwise of the following **Assertion** (a) and the **Reason** (r)
Assertion : In synchronous culture, majority of the cells move to next phase of the cell cycle simultaneously.
Reason : Synchronous culture could be obtained by starving cells for essential nutrient components.

(A) both (a) and (r) are true and (r) is the correct reason for (a)
(B) both (a) and (r) are true but (r) is not the correct reason for (a)
(C) (a) is true but (r) is false
(D) (a) is false but (r) is true

Q.43 Which of the following characteristics with respect to bacterial DNA polymerase III are **TRUE**?

P. Initiation of chain synthesis
Q. 5′– 3′ polymerization
R. 3′ – 5′ exonuclease activity
S. 5′ – 3′ exonuclease activity

(A) P and Q only
(B) Q and R only
(C) Q and S only
(D) P and S only

Q.44 Maximum specific growth rate (μ_{max}) of a microorganism is calculated by taking the (ln=$\log_e$, X=biomass, t = time)

(A) slope of ln X *vs* t of the growth cycle
(B) slope of ln X *vs* t during the exponential growth phase
(C) slope of X *vs* t
(D) slope of X *vs* t during the exponential phase of growth

Q.45 Identify the **CORRECT** statements

P. 5′ and 3′ ends of the transcripts can be mapped by utilizing polymerase chain reaction
Q. S_1 nuclease can cleave the DNA strand of a DNA-RNA hybrid
R. T_4 polynucleotide kinase is used for labeling 3′ end of DNA
S. Baculovirus (*Autographa californica*) can be used as an insect expression vector

(A) P and Q only
(B) R and S only
(C) P and S only
(D) Q and R only

Q.46 Value of the determinant mentioned below is

$$\begin{vmatrix} 1 & 0 & -1 & 0 \\ 4 & 7 & 0 & 2 \\ 1 & 1 & -1 & 1 \\ 2 & 0 & 2 & 1 \end{vmatrix}$$

(A) 24 (B) –30 (C) –24 (D) –10

Q.47 HAT (hypoxanthine, aminopterin and thymidine) is used for selecting the hybridomas based on the following

I. Only hybridoma will grow since it inherited the HGPRT genes from B-cells and can synthesize DNA from hypoxanthine.
II. Myeloma cells will not grow in cultures since *de novo* synthesis is blocked by aminopterin and due to the lack of HGPRT enzyme.

(A) only **I** is true
(B) only **II** is true
(C) both **I** and **II** are true
(D) **I** is true and **II** is false

Common Data Questions

Common Data for Questions 48 and 49:

Red-green colour blindness is inherited as a recessive X-linked trait.

Q.48 What will be the probability of having the colour-blind son to a woman with phenotypically normal parents and a colour-blind brother, and married to a normal man? (Assume that she has no previous children)

(A) 100 % (B) 50 % (C) 25 % (D) 12.5 %

Q.49 What will be the probability of having the colour-blind daughter to a phenotypically normal woman, who already had one colour-blind son, and is married to a colour-blind man?

(A) 75 % (B) 50 % (C) 25 % (D) 15%

Common Data for Questions 50 and 51:

A microorganism grows in a continuous 'chemostat' culture of 60 m^3 working volume with sucrose as the growth limiting nutrient at dilution rate, D = 0.55 h^{-1}. The steady state biomass concentration is 4.5 Kg dry biomass m^{-3} and the residual sucrose concentration is 2.0 Kg m^{-3}. The sucrose concentration in the incoming feed medium is 10.0 Kg m^{-3}.

Q.50 What would be the yield $Y_{X/S}$ (Kg biomass/Kg substrate)?

(A) 0.562 (B) 0.462 (C) 0.362 (D) 0.162

Q.51 What would be the sucrose concentration in the input feed for the output to be 45 Kg biomass h^{-1}?

(A) 3.225 Kg m^{-3} (B) 4.425 Kg m^{-3} (C) 5.115 Kg m^{-3} (D) 6.525 Kg m^{-3}

Linked Answer Questions

Statement for Linked Answer Questions 52 and 53:

The abdomen length (in millimeters) was measured in 15 male fruit flies, and the following data were obtained: 1.9, 2.4, 2.1, 2.0, 2.2, 2.4, 1.7, 1.8, 2.0, 2.0, 2.3, 2.1, 1.6, 2.3 and 2.2 .

Q.52 Variance (V_X) for this population of fruit flies as calculated from the above data shall be

(A) 0.85 (B) 0.25 (C) 0.061 (D) 0.08

Q.53 The value of standard deviation (SD) will be

(A) 0.061 (B) 0.25 (C) 0.61 (D) 0.85

Statement for Linked Answer Questions 54 and 55:

A 200 μl of polymerase chain reaction has 100 template DNA molecules and the reaction was performed for 10 cycles.

Q.54 How many molecules of amplicons will be generated?

(A) 1.024×10^4 (B) 1.024×10^5 (C) 2.048×10^4 (D) 2.048×10^5

Q.55 How many molecules of amplicons will be present in 0.1 μl of reaction?

(A) 102.4 (B) 1024 (C) 51.2 (D) 512

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General Aptitude (GA) Questions

Q. 56 – Q. 60 carry one mark each.

Q.56 Which of the following options is the closest in the meaning to the word below:
Inexplicable

(A) Incomprehensible
(B) Indelible
(C) Inextricable
(D) Infallible

Q.57 Choose the word from the options given below that is most nearly opposite in meaning to the given word:
Amalgamate

(A) merge
(B) split
(C) collect
(D) separate

Q.58 Choose the most appropriate word from the options given below to complete the following sentence.
If you are trying to make a strong impression on your audience, you cannot do so by being understated, tentative or _____.

(A) hyperbolic
(B) restrained
(C) argumentative
(D) indifferent

Q.59 Choose the most appropriate word(s) from the options given below to complete the following sentence.
I contemplated _____ Singapore for my vacation but decided against it.

(A) to visit
(B) having to visit
(C) visiting
(D) for a visit

Q.60 If Log (P) = (1/2)Log (Q) = (1/3) Log (R), then which of the following options is **TRUE**?

(A) $P^2 = Q^3R^2$ (B) $Q^2 = PR$ (C) $Q^2 = R^3P$ (D) $R = P^2Q^2$

Q. 61 to Q. 65 carry two marks each.

Q.61 **Few school curricula include a unit on how to deal with bereavement and grief, and yet all students at some point in their lives suffer from losses through death and parting.**

Based on the above passage which topic would not be included in a unit on bereavement?

(A) how to write a letter of condolence
(B) what emotional stages are passed through in the healing process
(C) what the leading causes of death are
(D) how to give support to a grieving friend

Q.62 A container originally contains 10 litres of pure spirit. From this container 1 litre of spirit is replaced with 1 litre of water. Subsequently, 1 litre of the mixture is again replaced with 1 litre of water and this process is repeated one more time. How much spirit is now left in the container?

(A) 7.58 litres (B) 7.84 litres (C) 7 litres (D) 7.29 litres

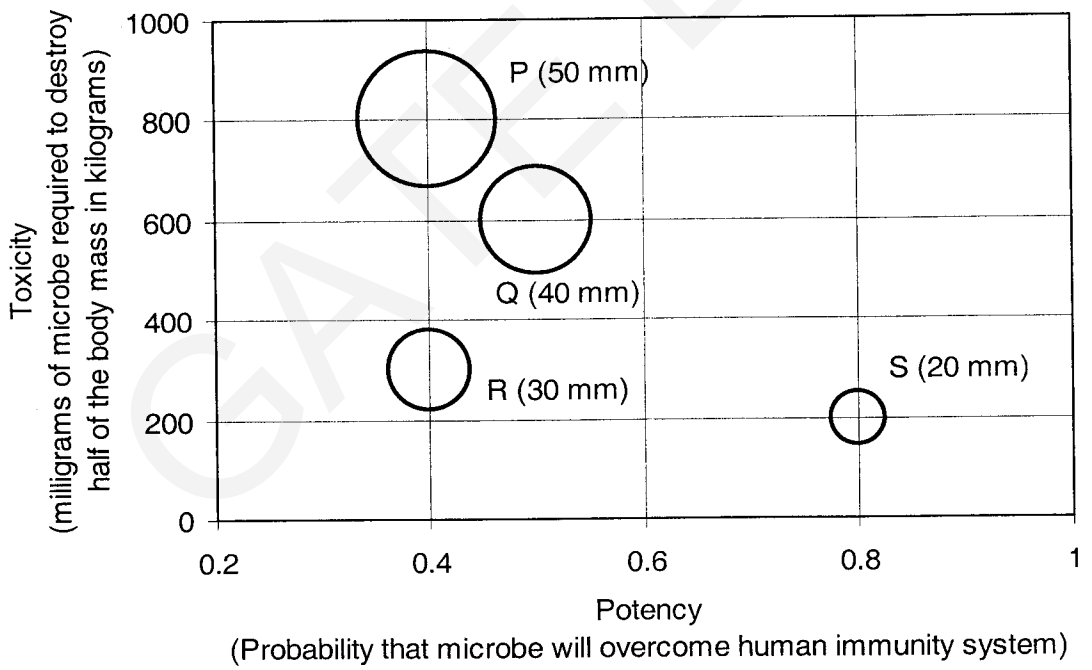
Q.63 A transporter receives the same number of orders each day. Currently, he has some pending orders (backlog) to be shipped. If he uses 7 trucks, then at the end of the 4th day he can clear all the orders. Alternatively, if he uses only 3 trucks, then all the orders are cleared at the end of the 10th day. What is the minimum number of trucks required so that there will be no pending order at the end of the 5th day?

(A) 4 (B) 5 (C) 6 (D) 7

Q.64 The variable cost (V) of manufacturing a product varies according to the equation V= 4q, where q is the quantity produced. The fixed cost (F) of production of same product reduces with q according to the equation F = 100/q. How many units should be produced to minimize the total cost (V+F)?

(A) 5 (B) 4 (C) 7 (D) 6

Q.65 P, Q, R and S are four types of dangerous microbes recently found in a human habitat. The area of each circle with its diameter printed in brackets represents the growth of a single microbe surviving human immunity system within 24 hours of entering the body. The danger to human beings varies proportionately with the toxicity, potency and growth attributed to a microbe shown in the figure below:



A pharmaceutical company is contemplating the development of a vaccine against the most dangerous microbe. Which microbe should the company target in its first attempt?

(A) P (B) Q (C) R (D) S

END OF THE QUESTION PAPER

BT- GATE-2011-Keys

1	A	**2**	D	**3**	B	**4**	A	**5**	B	**6**	D	**7**	B
8	B	**9**	A	**10**	D	**11**	B	**12**	B	**13**	A	**14**	A
15	C	**16**		**17**	C	**18**	B	**19**	A	**20**	A	**21**	B
22	C	**23**	D	**24**	C	**25**	C	**26**	C	**27**	B	**28**	D
29	C	**30**	C	**31**	C	**32**	B	**33**		**34**		**35**	C
36	A	**37**	B	**38**	C	**39**	A	**40**	B	**41**	A	**42**	B
43	B	**44**	B	**45**	C	**46**	D	**47**	C	**48**	C	**49**	B
50		**51**		**52**		**53**		**54**	B	**55**	B	**56**	A
57	B	**58**	B	**59**	C	**60**	B	**61**	C	**62**	D	**63**	C
64	A	**65**	D										

GATE 2012

BIOTECHNOLOGY – BT

Q. 1 - Q. 25 carry one mark each.

Q.1 In mismatch correction repair, the parental DNA strand is distinguished from the daughter strand by

(A) acetylation
(B) phosphorylation
(C) methylation
(D) glycosylation

Q.2 The basis for blue-white screening with pUC vectors is

(A) intraallelic complementation
(B) intergenic complementation
(C) intragenic suppression
(D) extnlgenic suppression

Q.3 Idiotypic determinants of an antibody are associated with the

(A) constant region of the heavy chains
(B) constant region of the light chains
(C) variable region
(D) constant regions of light and heavy chains

Q4. Identification of blood groups involves

(A) precipitation
(B) neutralization
(C) opsonization
(D) agglutination

Q.5 B-lymphocytes originate from the bone marrow whereas T -lymphocytes originate from

(A) thymus
(B) bone marrow
(C) spleen
(D) liver

Q.6 A humanized antibody is one in which the

(A) heavy and light chains are from human
(B) heavy chain is from human and light chain is from mouse
(C) light chain is from human and heavy chain is from mouse
(D) CDRs are from mouse, and the rest is from human

Q.7 Dimethyl sulfoxide (DMSO) is used as a cryopreservant for mammalian cell cultures because

(A) it is an organic solvent
(B) it easily penetrates cells
(C) it protects cells by preventing crystallization of water'
(D) it is also utilized as a nutrient

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Q.8 Nude mice refers to

(A) mice without skin
(B) mice without thymus
(C) knockout mice
(D) transgenic mice

Q.9 Heat inactivation of serum is done to inactivate

(A) prions
(B) mycoplasma
(C) complement
(D) pathogenic bacteria

Q. 10 Choose the correct signal transduction pathway.

(A) Hormone → 7 TM receptor → G protein → cAMP → PKA
(B) Hormone → G protein → 7 TM receptor → cAMP → PKA
(C) Hormone → 7 TM receptor → G protein → PKA → cAMP
(D) Hormone → 7 TM receptor → cAMP → G protein → PKA

Q.11 A protein is phosphorylated at a serine residue. A phosphomimic mutant of the protein can be generated by substituting that serine with

(A) glycine
(B) alanine
(C) Aspartate
(D) threonine

Q.12 A truncated polypeptide is synthesized due to a nonsense mutation. Where would you introduce another mutation to obtain a full-length polypeptide?

(A) Ribosomal protein gene
(B) Transfer RNA gene
(C) DNA repair gene
(D) Ribosomal RNA gene

Q.13 Protein-DNA interactions *in vivo* can be studied by

(A) gel shift assay
(B) Southern hybridization
(C) chromatin immunoprecipitation assay
(D) fluorescence *in situ* hybridization assay

Q.14 The direction of shell coiling in the snail *Limnaea peregra* is a classic example of

(A) chromosomal inheritance
(B) extra-chromosomal inheritance
(C) chromosomal translocation
(D) homologous recombination

Q.15 During photorespiration under low CO2 and high O2 levels, O2 reacts with ribulose 1,5- bisphosphate to yield

(A) one molecule each of 3-phosphoglycerate and 2-phosphoglycolate
(B) two molecules of 3-phosphoglycerate
(C) two molecules of 2-phosphoglycolate
(D) one molecule each of 3-phosphoglycerate and glyoxylate

Q.16 Which one of the following is NOT a protoplast fusion inducing agent?

(A) Inactivated Sendai virus
(B) Ca^{2+} at alkaline pH
(C) Polyethylene glycol
(D) Colchicine

Q.17 The activity of an enzyme is expressed in International Units (IU). However, the SI unit for enzyme activity is Katal. One Katal is

(A) 1.66×10^4 IU

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(B) 60 IU
(C) 6 x 10^7 IU
(D) 10^6 IU

Q.18 Identify the statement that is **NOT** applicable to an enzyme catalyzed reaction.

(A) Enzyme catalysis involves propinquity effects
(B) The binding of substrate to the active site causes a strain in the substrate
(C) Enzymes do not accelerate the rate of reverse reaction
(D) Enzyme catalysis involves acid-base chemistry

Q.19 An example of a derived protein structure database is

(A) Pfam
(B) SCOP
(C) GEO
(D) Prosite

Q.20 An example of a program for constructing a phylogenetic tree is

(A) phylip
(B) phrap
(C) prodom
(D) PHDsec

Q.21 Synteny refers to

(A) gene duplication from a common an~estor
(B) a tree representation of related sequences
(C) the extent of similarity between two sequences
(D) local conservation of gene order

Q.22 While searching a database for similar sequences, E value does **NOT** depend on the

(A) sequence length
(C) scoring system
(B) number of sequences in the database
(D) probability from a normal distribution

Q.23 In transmission electron microscopy, electron opacity is greatly enhanced by treating the specimen with

(A) ferrous ammonium sulfate
(C) sodium chloride
(B) uranium acetate·
(D) basic fuchsin

Q.24 The molarity of water in a water: ethanol mixture (IS : 85, v/v) is approximately

(A) 0.85
(C) 8.5
(B) 5.55
(D) 55.5

Q.25 The helix content of a protein can be determined using

(A) an infrared spectrometer
(B) a fluorescence spectrometer
(C) a circular dichroism spectrometer
(D) a UV-Visible spectrophotometer

Q. 26 to Q. 55 carry two marks each.

Q. 26 Which one of the following DNA sequences carries an invert repeat?

(A) ATGAGCCCCGAGTA
TACTCGGGGCTCAT
(B) ATGAGCCGGCTCTA
TACTCGGCCGAGAT

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(C) ATGAGCCGAGCCTA
ACTCGGCTCGGAT
(D) ATGAGCCTATGGTA
TACTCGGATACCAT

Q.27 In zinc finger proteins, the amino acid residues that coordinate zinc are

(A) Cys and His
(e) Arg and Lys
(B) Asp and Glu
(D) Asp and Arg

Q.28 Match the entries in **Group I** with those in **Group II.**

Group I	**Group II**
P. MTT	1. Dihydrofolate reductase
Q. Annexin V	2. Succinate dehydrogenase
R. Methotrexate	3. Microtubules
S. Taxol	4. Phosphatidylserine

(A) P-3, Q-1, R-4, S-2
(B) P-2, Q-4. R-1, S-3
(C) P-2, Q-3, R-4, S-1
(D) P-4, Q-2, R-1, S-3

Q.29 In an exponentially growing batch culture of *Saccharomyces cerevisiae,* the cell density is 20 gl^{-1} (DCW), the specific growth rate (μ) is 0.4 h^{-1} and substrate uptake rate (ν) is 16 $gl^{-1}h^{-1}$. The cell yield coefficient $Y_{x/s}$ will be

(A) 0.32
(B) 0.64
(C) 0.80
(D) 0.50

Q.30 A single base pair of DNA weighs 1.1 x 10^{-21} grams. How many picomoles of a plasmid vector of length 2750 bp are contained in L μg of purified DNA?

(A) 0.30
(B) 0.55
(C) 0.25
(D) 0.91

Q.31 Match the terms in **Group I** with the ploidy in **Group II.**

Group I	**Group II**
P. Disome	1. 2n + 1
Q. Monosome	2. 2n - 1
R. Nullisome	3. n- 1
S. Trisome	4. n + 1

(A) P-4, Q-2, R-3, S-1
(B) P-4, Q-3, R-1, S-2
(C) P-2, Q-3, R-4, S-1
(D) P-1, Q-4, R-3, S-2

Q.32 What is the rank of the following matrix?

$$\begin{pmatrix} 5 & 3 & -1 \\ 6 & 2 & -4 \\ 14 & 10 & 0 \end{pmatrix}$$

(A) 0
(B) 1
(C) 2
(D) 3

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Q.33 Match the products in **Group I** with the applications in **Group II.**

Group I	**Group II**
P. Digoxin	1. Muscle relaxant
Q. Stevioside	2. Anti-cancer agent
R. Atropine	3. Cardiovascular disorder
S. Vinblastine	4. Sweetener

(A) P-1, Q-4, R-3, S-2
(B) P-3, Q-2, R-1, S-4
(C) P-3, Q-4, R-1, S-2
(D) P-2, Q-3, R-1, S-4

Q.34 Determine the correctness or otherwise of the following **Assertion** (a) and **Reason** (r).

Assertion: The production of secondary metabolites in plant cell cultures is enhanced by the addition of elicitors.
Reason: Elicitors induce the expression of enzymes responsible for the biosynthesis of secondary metabolites.

(A) Both (a) and (r) are true but (r) is not the correct reason for (a)
(B) Both (a) and (r) are true and (r) is the correct reason for (a)
(C) (a) is true but (r) is false
(D) (a) is false but (r) is true

Q.35 Determine the correctness or otherwise of the following **Assertion** (a) and **Reason** (r).

Assertion: Plants convert fatty acids into glucose.
Reason: Plants have peroxisomes.

(A) Both (a) and (r) are true but (r) is not the correct reason for (a)
(B) Both (a) and (r) are true and (r) is the correct reason for (a)
(C) (a) is true but (r) is false
(D) (a) is false but (r) is true

Q.36 Determine the correctness or otherwise of the following Assertion (a) and Reason (r).

Assertion: In direct somatic embryogenesis, embryos are developed without going through callus formation.
Reason: This is possible due to the presence of pre-embryonically determined cells.

(A) Both (a) and (r) are true but (r) is not the correct reason for (a)
(B) (a) is false but (r) is true
(C) (a) is true but (r) is false
(D) Both (a) and (r) are true and (r) is the correct reason for (a)

Q.37 Match the entries in **Group I** with the process parameters in **Group II.**

Group I	**Group II**
P. Clark electrode	1. Liquid level
Q. Redox probe	2. Dissolved oxygen concentration
R. Load cell	3. Vessel pressure
S. Diaphragm gauge	4. pH (anaerobic process)

(A) P-2, Q-1, R-3, S-4
(B) P-4, Q-2, R-3, S-1
(C) P-2, Q-4, R-1, S-3
(D) P-2, Q-1, R-4, S-3

Q.38 Match the downstream processes in **Group I** with the products in **Group II.**

Group I	**Group II**
P. Solvent extraction	1. Lactic acid
Q. Protein-A linked affinity chromatography	2. Penicillin
R. Extractive distillation	3. Monoclonal antibody
S. Salting out	4. Lipase

(A) P-2, Q-3, R-1, S-4
(B) P-4, Q-1, R-2, S-3

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(C) P-4, Q-1, R-3, S-2
(D) P-2, Q-4, R-1, S-3

Q.39 Determine the correctness or otherwise of the following Assertion (a) and Reason (r).

Assertion: Cell mass yield of a methylotrophic yeast is more on methanol compared to glucose.
Reason: Methanol has a greater degree of reductance compared to glucose.

(A) Both (a) and (r) are correct and (r) is the correct reason for (a)
(B) (a) is correct (r) is false
(C) (a) is false, (r) is correct
(D) Both (a) and (r) are correct but (r) is not the correct reason for (a)

Q.40 A disease is inherited by a child with a probability of 114. In a family with two children, the probability that exactly one sibling is affected by this disease is

(A) 1/4
(B) 3/8
(C) 7/16
(D) 9/16

Q.41 Match the organisms in **Group I** with the entries in **Group II.**

Group I	**Group II**
P. *Clostridium*	1. Rods with teichoic acid in the cell wall
Q. *Escherichia*	2. Rods with endospores
R. *Vibrio*	3. Helical rods with flagella
S. *Bacillus*	4. Rods with LPS in the outer membrane
	5. Curved rods with polar flagella

(A) P-2, Q-4, R-5, S-1
(B) P-2, Q-1, R-5, S-4
(C) P-5, Q-4, R-2, S-3
(D) P-3, Q-2, R-1, S-4

Q.42 Match the entries in **Group I** with the methods of sterilization in **Group II.**

Group I	**Group II**
P. Serum	1. Autoclave
Q. Luria broth	2. Membrane filtration
R. Polypropylene tubes	3. UV irradiation
S. Biological safety cabinets	4. Gamma irradiation
	5. Dry heat

(A) P-5, Q-3, R-1, S-4
(B) P-1, Q-4, R-5, S-3
(C) P-2, Q-1, R-4, S-3
(D) P-4, Q-1, R-3, S-5

Q.43 Match the high energy compounds in **Group I** with the biosynthetic pathways for the molecules in

Group I	**Group II.**
P. GTP	
Q. UTP	
R. CTP	
S. Acyl coenzyme A	

(A) P-3, Q-2, R-4, S-1
(B) P-2, Q-4, R-3, S-1
(C) P-4, Q-3, R-l, S-2
(D) P-3, Q-4, R-2, S-1

1. Fatty acid
2. Phospholipid
3. Protein
4. Peptidoglycan

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Q44. Match the vitamins in **Group I** with the processes/reactions in **Group II.**

Group I **Group II**

P. Pantothenic acid
Q. Vitamin B2
R. Vitamin B6
S. Folic acid

(A) P-5, Q-2, R-4, S-1
(B) P-4, Q-1, R-3, S-2
(C) P-4, Q-2, R-1, S-3
(D) P-2, Q-1, R-3, S-5

1. Electron transport
2. Transfer of 1-C units
3. Decarboxylation
4. Fatty acid metabolism
5. Hydrolysis

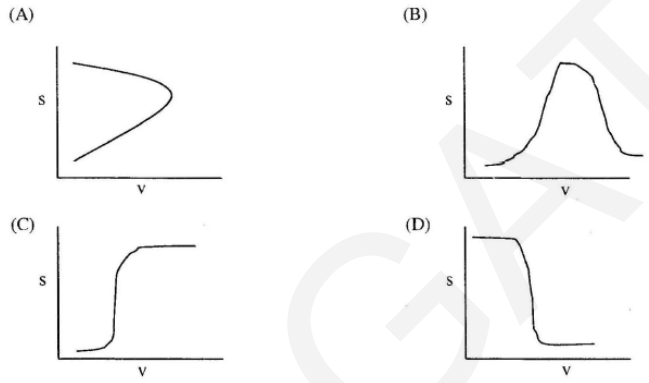
Q.45 Consider the data set 14, 18, 14, 14, 10, 29, 33, 31, 25. If you add 20 to each of the values, then

(A) both mean and variance change
(B) both mean and variance are unchanged
(C) the mean is unchanged, variance changes
(D) the mean changes, the variance is unchanged

Q.46 An enzymatic reaction is described by the following rate expression.

$$v = \frac{v_m s}{k_m + s + s^2 / k_s}$$

Which one of the following curves represents this expression?



Q.47 A bacterial culture (200 μl containing 1.8 x 10^9 cells) was treated with an antibiotic Z (50 μg per ml) for 4 h at 37 °C. After this treatment, the culture was divided into two equal aliquots.

Set A: 100 μl was plated on Luria agar.
Set B: 100 μl was centrifuged, the cell pellet washed and plated on Luria agar.

After incubating these two plates for 24 h at 37 °C. Set A plate showed no colonies, whereas the Set B plate showed 0.9 x 10^9 cells. This experiment showed that the antibiotic Z is

(A) bacteriostatic
(B) bacteriocidal
(C) bacteriolytic
(D) apoptotic

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Common Data Questions

Common Data for Questions 48 and 49 :

In a muscle, the extracellular and intracellular concentrations of Na^+ are 150 mM and 12 mM, and those of K+ are 2.7 mM and 140 mM, respectively. Assume that the temperature is 25 °C and that the membrane potential is -60 mY, with the interior more negatively charged than the exterior. (R = 8.314 J mol^{-1} K^{-1}; F = 96.45 kJ mol^{-1} V^{-1})

Q.48 The free energy change for the transport of three Na+ out of the cell is

(A) + 1.5 kJ/mol
(B) +17.4 kJ/mol
(C) +18.9 kJ/mol
(D) +36.3 kJ/mol

Q.49 The free energy change for the transport of two K+ into the cell is

(A) +8.0 kJ/mol
(B) + 11.6 kJ/mol
(C) + 19.6 kJ/mol
(D) +31.2 kJ/mol

Common Data for Questions 50 and 51:

The purification data for an enzyme is given below:

	Step	Volume (ml)	Total protein (mg)	Total activity (Units)	Specific activity (Units/mg)
P	Cell-free extract	17	177	102	0.58
Q	Q- Sepharose	14	18.8	72	3.83
R	Phenyl Sepharose	26	9.2	45	4.89
S	Sephacryl S-200	7	4.1	30	7.32

Q.50 The fold purification for each step is

(A) P-0.1, Q-0.66, R-0.84, S-1.26
(B) P-1.0, Q-0.52, R-0.67, S-0.8
(C) P-1, Q-6.6, R-8.4, S-12.6
(D) P-100, Q-66, R-84, S-12

Q.51 The yield (%) for each step is

(A) P-10, Q-7.2, R-4.5, S-2.0
(B) P-34, Q-24, R-15, S-1
(C) P-3.4, Q-2.4, R-1.5, S-0.1
(D) P-100, Q-71, R-44, S-29

Linked Answer Questions

Statement for Linked Answer Questions 52 and 53:

An *E. coli* cell of volume 10^{-12} cm^3 contains 60 molecules of lac-repressor. The repressor has a binding affinity (Kd) of 10^{-8} M and 10^{-9} M with and without lactose respectively, in the medium.

Q.52 The molar concentration of the repressor in the cell is

(A) 0.1 nM
(B) 1 nM
(C) 10 nM
(D) 100 nM

Q.53 Therefore the lac-operon is

(A) repressed and can only be induced with lactose.
(B) repressed and cannot be induced with lactose.
(C) not repressed.
(D) expressed only when glucose and lactose are present.

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Statement for Linked Answer Questions 54 and 55:

β-Galactosidase bound to DEAE-cellulose is used to hydrolyze lactose to glucose and galactose in a plug flow bioreactor with a packed bed of volume 100 liters and a voidage (ε) of 0.55. The K'_m and V'_{max}, for the immobilized enzyme are 0.72 gl^{-1} and 18 $gl^{-1}h^{-1}$, respectively. The lactose concentration in the field stream is 20 gl^{-1}, and a fractional conversion of 0.90 is desired. Diffusional limitations may be ignored.

Q.54 The residence time required for the steady state reactor operation will be

(A) 0.1 h
(B) 0.4 h
(C) 1.0 h
(D) 1.1 h

Q.55 The feed flow rate required for the above bioconversion will be

(A) 50 lh^{-1}
(B) 55 lh^{-1}
(C) 137 lh^{-1}
(D) 550 lh^{-1}

General Aptitude (GA) Questions

Q. 56 - Q. 60 carry one mark each.

Q.56 The cost function for a product in a firm is given by $5q^2$, where q is the amount of production. The firm can sell the product at a market price of Rs 50 per unit. The number of units to be produced by the firm such that the profit is maximized is

(A) 5
(B) 1o
(C) 15
(D) 25

Q.57 Choose the most appropriate alternative from the options given below to complete the following sentence:

Suresh's dog is the one ____ was hurt in the stampede.

(A) that
(B) which
(C) who
(D) whom

Q.58 Choose the grammatically INCORRECT sentence:

(A) They gave us the money back less the service charges of Three Hundred rupees.
(B) This country's expenditure is not less than that of Bangladesh.
(C) The committee initially asked for a funding of Fifty Lakh rupees, but later settled for a lesser sum.
(D) This country's expenditure on educational reforms is very less.

Q.59 Which one of the following options is the closest in meaning to the word given below?

Mitigate

(A) Diminish
(B) Divulge
(C) Dedicate
(D) Denote

Q.60 Choose the most appropriate alternative from the options given below to complete the following sentence:

Despite several ____ the mission succeeded in its attempt to resolve the conflict.

(A) attempts
(B) setbacks
(C) meetings
(D) delegations

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Q. 61 - Q. 65 carry two marks each.

Q.61 **Wanted Temporary, Part-time persons for the post of Field Interviewer to conduct personal interviews to collect and collate economic data. Requirements: High School-pass, must be available for Day, Evening and Saturday work. Transportation paid, expenses reimbursed.**

Which one of the following is the best inference from the above advertisement?

(A) Gender-discriminatory
(B) Xenophobic
(C) Not designed to make the post attractive
(D) Not gender-discriminatory

Q.62 Given the sequence of terms, AD CG FK JP, the next term is

(A) OV
(B) OW
(C) PV
(D) PW

Q.63 Which of the following assertions are **CORRECT**?

P: Adding 7 to each entry in a list adds 7 to the mean of the list
Q: Adding 7 to each entry in a list adds 7 to the standard deviation of the list
R: Doubling each entry in a list doubles the mean of the list
S: Doubling each entry in a list leaves the standard deviation of the list unchanged

(A) P, Q
(B) Q, R
(C) P, R
(D) R, S

Q.64 An automobile plant contracted to buy shock absorbers from two suppliers X and Y. X supplies 60% and Y supplies 40% of the shock absorbers. All shock absorbers are subjected to a quality test. The ones that pass the quality test are considered reliable. Of X's shock absorbers, 96% are reliable. Of Y's shock absorbers, 72% are reliable.

The probability that a randomly chosen shock absorber, which is found to be reliable, is made by Y is

(A) 0.288
(B) 0.334
(C) 0.667
(D) 0.720

Q.65 A political party orders an arch for the entrance to the ground in which the annual convention is being held. The profile of the arch follows the equation $y = 2x - 0.1x^2$ where y is the height of the arch in meters. The maximum possible height of the arch is

(A) 8 meters
(B) 10 meters
(C) 12 meters
(D) 14 meters

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GATE BT 2012 Answer Key

Answers

1C 2A 3C 4D 5B 6D 7C 8B 9C 10A 11C 12B 13C 14B 15A 16D 17C 18C 19B 20A 21D 22D 23B 24C 25C 26B 27A 28B 29D 30B 31A 32C 33C 34B 35A 36D 37C 38A 39A 40B 41A 42C 43D 44B 45D 46A 47A 48D 49A 50C 51D 52D 53B 54 Marks to All 55 Marks to All 56A 57 Marks to All 58D 59A 60B 61D 62A 63C 64B 65B

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BT:BIOTECHNOLOGY

Duration: Three Hours *Maximum Marks*:100

Please read the following instructions carefully:

General Instructions:

1. Total duration of examination is 180 minutes (3 hours).

2. The clock will be set at the server. The countdown timer in the top right corner of screen will display the remaining time available for you to complete the examination. When the timer reaches zero, the examination will end by itself. You will not be required to end or submit your examination.

3. The Question Palette displayed on the right side of screen will show the status of each question using one of the following symbols:



You have not visited the question yet.

You have not answered the question.

You have answered the question.

You have NOT answered the question, but have marked the question for review.

You have answered the question, but marked it for review.

The Marked for Review status for a question simply indicates that you would like to look at that question again. ***If a question is answered and Marked for Review, your answer for that question will be considered in the evaluation.***

Navigating to a Question

4. To answer a question, do the following:

   a. Click on the question number in the Question Palette to go to that question directly.
   b. Select an answer for a multiple choice type question. Use the virtual numeric keypad to enter a number as answer for a numerical type question.
   c. Click on **Save and Next** to save your answer for the current question and then go to the next question.
   d. Click on **Mark for Review and Next** to save your answer for the current question, mark it for review, and then go to the next question.
   e. **Caution: Note that your answer for the current question will not be saved, if you navigate to another question directly by clicking on its question number.**

5. You can view all the questions by clicking on the **Question Paper** button. Note that the options for multiple choice type questions will not be shown.

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Answering a Question

6. Procedure for answering a multiple choice type question:
    a. To select your answer, click on the button of one of the options
    b. To deselect your chosen answer, click on the button of the chosen option again or click on the **Clear Response** button
    c. To change your chosen answer, click on the button of another option
    d. To save your answer, you MUST click on the **Save and Next** button
    e. To mark the question for review, click on the **Mark for Review and Next** button**.** ***If an answer is selected for a question that is Marked for Review, that answer will be considered in the evaluation.***

7. Procedure for answering a numerical answer type question:
    a. To enter a number as your answer, use the virtual numerical keypad
    b. A fraction (eg.,-0.3 or -.3) can be entered as an answer with or without '0' before the decimal point
    c. To clear your answer, click on the **Clear Response** button
    d. To save your answer, you MUST click on the **Save and Next** button
    e. To mark the question for review, click on the **Mark for Review and Next** button**.** ***If an answer is entered for a question that is Marked for Review, that answer will be considered in the evaluation.***

8. To change your answer to a question that has already been answered, first select that question for answering and then follow the procedure for answering that type of question.

9. Note that ONLY Questions for which answers are saved or marked for review after answering will be considered for evaluation.

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Paper specific instructions:

1. There are a total of 65 questions carrying 100 marks. Questions are of multiple choice type or numerical answer type. A multiple choice type question will have four choices for the answer with only **one** correct choice. For numerical answer type questions, the answer is a number and no choices will be given**. A number as the answer should be entered** using the virtual keyboard on the monitor.
2. Questions Q.1 – Q.25 carry 1mark each. Questions Q.26 – Q.55 carry 2marks each. The 2marks questions include two pairs of common data questions and two pairs of linked answer questions. The answer to the second question of the linked answer questions depends on the answer to the first question of the pair. If the first question in the linked pair is wrongly answered or is not attempted, then the answer to the second question in the pair will not be evaluated.
3. Questions Q.56 – Q.65 belong to General Aptitude (GA) section and carry a total of 15 marks. Questions Q.56 – Q.60 carry 1mark each, and questions Q.61 – Q.65 carry 2marks each.
4. Questions not attempted will result in zero mark. Wrong answers for multiple choice type questions will result in **NEGATIVE** marks. For all 1 mark questions, ⅓ mark will be deducted for each wrong answer. For all 2 marks questions, ⅔ mark will be deducted for each wrong answer. However, in the case of the linked answer question pair, there will be negative marks only for wrong answer to the first question and no negative marks for wrong answer to the second question. There is no negative marking for questions of numerical answer type.
5. Calculator is allowed. Charts, graph sheets or tables are **NOT** allowed in the examination hall.
6. Do the rough work in the Scribble Pad provided.

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Q. 1 – Q. 25 carry one mark each.

Q.1 Under alkaline conditions, DNA is more stable than RNA because

(A) RNA forms secondary structures
(B) RNA is a single stranded molecule
(C)RNA has uracil in place of thymidine
(D)RNA is susceptible to hydrolysis

Q.2 Which one of the following modifications is common toboth protein and DNA?

(A) SUMOylation (B) Nitrosylation (C)Methylation (D) Ubiquitination

Q.3 Protein A, which has strong affinity to F_c region of immunoglobulin, is extracted from

(A)*Saccharomyces cerevisiae*
(B)*Staphylococcus aureus*
(C)*Streptococcus pyogenes*
(D)*Streptococcus sanguis*

Q.4 The first humanized monoclonal antibody approved for the treatment of breast canceris

(A) Rituximab (B) Cetuximab (C) Bevacizumab (D) Herceptin

Q.5 Which one of the following aminoacids in proteins does **NOT** undergo phosphorylation?

(A) Ser (B) Thr (C) Pro (D) Tyr

Q.6 The role of an adjuvant isto

(A) prolong the persistence of antigen
(B) cross link the antigen
(C) increase the size of antigen
(D) avoid inflammation

Q.7 Endogenous antigens are presented on to the cell surface along with

(A) MHC-II (B) MHC-I (C) $F_c\gamma$receptor (D) complement receptor

Q.8 Human genome sequencing project involved the construction of genomic library in

(A)bacterial artificial chromosome (B)pBR322
(C) bacteriophage (D) pcDNA3.1

Q.9 The nucleotide analogue used in DNA sequencing by chain termination method is

(A) 1',3'-dideoxy nucleoside triphosphate
(B) 2',3'-dideoxy nucleoside triphosphate
(C) 2',4'-dideoxy nucleoside triphosphate
(D) 2',5'-dideoxy nucleoside triphosphate

Q.10 In nature, the horizontal gene transfer across bacteria is mediated by

(A) gene cloning followed by transformation (B)conjugation and transformation
(C) conjugation only (D) transformation only

Q.11 Phylum proteobacteria is subdivided into α-, β-, γ-, δ- and ε-proteobacteria based on

(A) G+C content (B) 23S rRNA sequences
(C)tRNA sequences (D)16S rRNA sequences

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Q.12 Which one of the following is an ABC transporter?

(A)multidrug resistance protein (B) acetylcholine receptor
(C)bacteriorhodopsin (D) ATP synthase

Q.13 The catalytic efficiency for an enzyme is definedas

(A) k_{cat} (B) $\frac{V_{max}}{k_{cat}}$ (C) $\frac{k_{cat}}{K_m}$ (D) $\frac{k_{cat}}{V_{max}}$

Q.14 Of the two diploid species, species I has 36 chromosomes and species II has 28 chromosomes. How many chromosomes would be found in an allotriploid individual?

(A) 42 or 54 (B)46 or 50 (C) 74 or 86 (D) 84 or 108

Q.15 The RNA primer synthesized during the replication process in bacteria is removed by

(A) DNA gyrase (B)primase
(C) DNA polymeraseI (D) DNA polymerase II

Q.16 The suitable substitution matrix to align closely related sequences is

(A) PAM 250 or BLOSSUM 80 (B) PAM 40 or BLOSSUM 80
(C) PAM 120 or BLOSSUM 40 (D) PAM 250 or BLOSSUM 40

Q.17 If $P=\begin{bmatrix}1 & 1\\ 2 & 2\end{bmatrix}, Q=\begin{bmatrix}2 & 1\\ 2 & 2\end{bmatrix}$ and $R=\begin{bmatrix}3 & 0\\ 1 & 3\end{bmatrix}$,which one of the following statements is **TRUE**?

(A) PQ = PR (B) QR= RP
(C) QP = RP (D) PQ = QR

Q.18 If u = log $(e^x + e^y)$, then $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y}$=

(A) $e^x + e^y$ (B) $e^x - e^y$
(C) $\frac{1}{e^x + e^y}$ (D) 1

Q.19 Hypophosphatemia is manifested by an X-linked dominant allele.What proportion of the offsprings from a normal male and an affected heterozygous female will manifest the disease?

(A) ½ sons and ½ daughters (B) all daughters and no sons
(C) all sons and no daughters (D) ¼ daughters and ¼ sons

Q.20 One of theeigen values of $P=\begin{bmatrix}10 & -4\\ 18 & -12\end{bmatrix}$ is

(A) 2 (B) 4 (C) 6 (D) 8

Q. 21 – Q. 25 are of numerical answer type.

Q.21 A callus of 5 g dry weight was inoculated on semi-solid medium for growth. The dry weight of the callus was found to increase by 1.5 fold after 10 days of inoculation. The growth index of the culture is _____

Q.22 A chemostat is operated at a dilution rate of 0.6 h^{-1}. At steady state, the biomass concentration in the exit stream was found to be 30 g l^{-1}. The biomass productivity (g $l^{-1}h^{-1}$) after 3h of steady state operation will be _____

Q.23 A batch bioreactor is to be scaled up from 10 to 10,000 liters. The diameter of the large bioreactor is 10 times that of the small bioreactor. The agitator speed in the small bioreactor is 450 rpm. Determine the agitator speed (rpm) of the large bioreactor with same impeller tip speed as that of the small bioreactor. _____

Q.24 Calculate the percentage sequence identity for the pairwise alignment given below. _____

H E L L O –
Y E L L O W

Q.25 In a batch culture, the specific rate of substrate utilization is 0.25 g (g cell mass)$^{-1}$ h^{-1} and specific rate of product formation is 0.215 g (g cell mass)$^{-1}$ h^{-1}. Calculate the yield of product from the substrate($Y_{p/s}$). _____

Q. 26 to Q. 55 carry two marks each.

Q.26 Match the commercial microbial sources in **Group I** with the products in **Group II.**

Group I	**Group II**
P. *Corynebacteriumlilium*	1. 2,3-Butane di-ol
Q. *Klebsiellaoxytoca*	2. Poly-β-hydroxybutyric acid
R. *Aspergillusniger*	3. Glutamic acid
S. *Alcaligeneseutrophus*	4. Citric acid

(A) P-3,Q-1,R-2,S-4 (B) P-3,Q-1,R-4,S-2
(C) P-1,Q-3,R-2,S-4 (D) P-1,Q-3,R-4,S-2

Q.27 Match the entries in the **Group I** with the elution conditions in **Group II.**

Group I	**Group II**
P. Ion-exchange chromatography	1. Isocratic solvent
Q. Hydrophobic column chromatography	2. Ampholytes
R. Gel filtration chromatography	3. Increasing gradient of salt
S. Chromatofocusing	4. Decreasing gradient of polarity

(A) P-4,Q-1,R-2,S-3 (B) P-4,Q-3,R-1,S-2
(C) P-3,Q-4,R-1,S-2 (D) P-3,Q-4,R-2,S-1

Q.28 Determine the correctness or otherwise of the following **Assertion** (a) and **Reason** (r).

Assertion: Immobilization of plant cells can enhance secondary metabolite production during bioreactor cultivation.

Reason: Immobilization protects the plant cells from shear forces in the bioreactor.

(A) Both (a) and (r) are true and (r) is the correct reason for (a).
(B) Both (a) and (r) are true but (r) is not the correct reason for (a).
(C) (a) is true but (r) is false.
(D) (a) is false but (r) is true.

Q.29 Match the cell structures in **Group I** with the organismsin **Group II.**

Group I	**Group II**
P. Endospores	1. *Methanobacterium*
Q. Bipolar flagella	2. *Treponema*
R. Pseudomurine in cell wall	3. *Spirillum*
S. Periplasmic flagella	4. *Clostridium*

(A) P-4, Q-3, R-1, S-2 (B) P-4, Q-3, R-2, S-1
(C) P-3, Q-4, R-1, S-2 (D) P-4, Q-1, R-3, S-2

Q.30 Match the antibioticsin**Group I**with the targets in**Group II.**

Group I	**Group II**
P. Sulfonamide	1. Peptidoglycan synthesis
Q. Quinolones	2. Peptide chain elongation
R. Erythromycin	3. Folic acid biosynthesis
S. Cephalosporin	4. Topoisomerase

(A) P-3, Q-4, R-1, S-2 (B) P-2, Q-4, R-3, S-1
(C) P-4, Q-1, R-2, S-3 (D) P-3, Q-4, R-2, S-1

Q.31 In nature, *Agrobacterium tumefaciens* mediated infection of plant cells leads to

P. crown gall disease in plants
Q. hairy root disease in plants
R. transfer of T-DNA into the plant chromosome
S. transfer of Ri-plasmid into the plant cell

(A) S only (B) P and R only (C) Q and S only (D) Q only

Q.32 Match the entries in**Group I** with the enzymes in**Group II.**

Group I	**Group II**
P. NAD^+	1. Glutathione peroxidase
Q. Selenium	2. Nitrogenase
R. Pyridoxal phosphate	3. Lactate dehydrogenase
S. Molybdenum	4. Glycogen phosphorylase

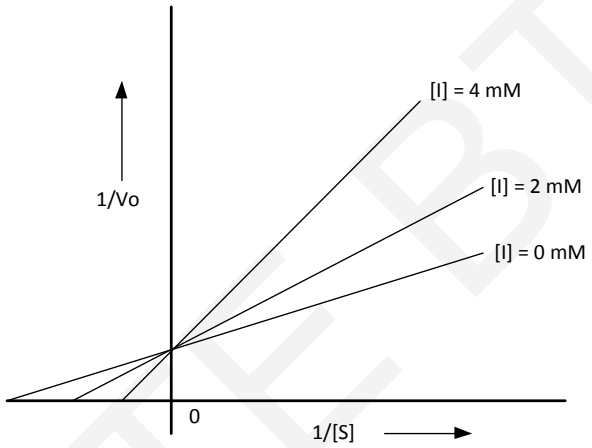
(A) P-3, Q-2, R-4, S-1 (B) P-4, Q-1, R-3, S-2
(C) P-3, Q-1, R-4, S-2 (D) P-3, Q-4, R-2, S-1

Q.33 Match the herbicides in **Group I**with the target enzymesin **Group II.**

Group I	**Group II**
P. Glyphosate	1. Nitrilase
Q. Bromoxynil	2. Acetolactatesynthetase
R. Sulphonylureas	3. Dehalogenase
S. Dalapon	4. 5-Enolpyruvyl shikimate3-phosphate synthase

(A) P-4, Q-1, R-2, S-3
(B) P-2, Q-1, R-4, S-3
(C) P-4, Q-3, R-2, S-1
(D) P-3, Q-2, R-4, S-1

Q.34 The activity of an enzyme was measured by varying the concentration of the substrate (S) in the presence of three different concentrations of inhibitor (I) 0, 2 and 4 mM.The double reciprocal plot given belowsuggests that the inhibitor (I) exhibits



(A) substrate inhibition
(B) uncompetitive inhibition
(C) mixed inhibition
(D) competitive inhibition

Q.35 Match the entries in **Group I**with the entries in **Group II.**

Group I	**Group II**
P. RNAse P	1. Polyadenylation
Q. RNase H	2. Splicing
R. snRNAs	3. Ribozymes
S. CstF	4. DNA-RNAhybrids

(A) P-3, Q-4, R-2, S-1
(B) P-4, Q-3, R-2, S-1
(C) P-3, Q-2, R-1, S-4
(D) P-2, Q-4, R-1, S-3

Q.36 Determine the correctness or otherwise of the following **Assertion** (a) and **Reason** (r).

Assertion:UPGMA method produces ultrametric tree.

Reason: Sequence alignment is converted into evolutionary distances in UPGMA method.

(A) Both (a) and (r) are true and (r) is the correct reason for (a)
(B) Both (a) and (r) are true and (r) is not the correct reason for (a)
(C) (a) is true but (r) is false
(D) (a) is false but (r) is true

Q.37 Match the entries in the **Group I**with the entries in**Group II.**

Group I	**Group II**
P. Threading	1. Gene duplication
Q. FASTA	2. Fold prediction
R. Profile	3. HMM
S. Paralogs	4. k-tuple

(A) P-2, Q-1, R-3, S-4 (B) P-2, Q-4, R-3, S-1
(C) P-3, Q-4, R-2, S-1 (D) P-1, Q-4, R-3, S-2

Q.38 Evaluate $\lim_{x\to\infty} x \tan\frac{1}{x}$

(A) ∞ (B) 1 (C) 0 (D)-1

Q.39 The Laplace transform of $f(t) = 2t + 6$ is

(A) $\frac{1}{s} + \frac{2}{s^2}$ (B) $\frac{3}{s} - \frac{6}{s^2}$ (C) $\frac{6}{s} + \frac{2}{s^2}$ (D) $-\frac{6}{s} + \frac{2}{s^2}$

Q.40 The solution of the following set of equations is

$$x + 2y + 3z = 20$$
$$7x + 3y + z = 13$$
$$x + 6y + 2z = 0$$

(A) $x = -2,\ y = 2,\ z = 8$ (B) $x = 2,\ y = -3,\ z = 8$
(C) $x = 2,\ y = 3,\ z = -8$ (D) $x = 8,\ y = 2,\ z = -3$

Q.41 The solution to $\frac{dy}{dx} + y\ cot\ x = cosec\ x$ is

(A) $y = (c+x)\ cot\ x$ (B) $y = (c+x)\ cosec\ x$
(C) $y = (c + x)\ cosec\ x\ cot\ x$ (D) $y = (c+x)\frac{cosec\ x}{\cot x}$

Q.42 A complete restriction digestion of a circular plasmid (5000bp) was carried out with *Hin*dIII,*Bam*HIand*Eco*RIindividually. Restriction digestion yielded following fragments.

Plasmid + *Hin*dIII→ 1200bp and3800bp
Plasmid + *Bam*HI→5000bp
Plasmid + *Eco*RI→ 2500bp

The number of sites for *Eco*RI,*Bam*HI*andHin*dIIIpresent on this plasmid are

(A)*Eco*RI-2, *Bam*HI-1, *Hin*dIII-2 (B)*Eco*RI-1, *Bam*HI-1, *Hin*dIII-2
(C)*Eco*RI-3, *Bam*HI-2, *Hin*dIII-1 (D)*Eco*RI-2, *Bam*HI-2, *Hin*dIII-1

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Q. 43 – Q. 47 are of numerical answer type.

Q.43 The total number of fragments generated by the complete and sequential cleavage of the polypeptide given below by Trypsin followed by CNBr is _____

Phe-Trp-Met-Gly-Ala-Lys-Leu-Pro-Met-Asp-Gly-Arg-Cys-Ala-Gln

Q.44 In a genetic study, 80 people were found to have alleles for polydactyly. Only 36 of them were polydactylous. What is the extent of penetrance percentage? _____

Q.45 One percent of the cars manufactured by a company are defective. What is the probability (upto four decimals) that more than two cars are defective, if 100 cars are produced? _____

Q.46 The maximum cell concentration (g l^{-1}) expected in a bioreactor with initial cell concentration of 1.75 g l^{-1} and an initial glucose concentration of 125 g l^{-1} is ($Y_{x/s}$ = 0.6 g cell/g substrate) _____

Q.47 A fed batch culture was operated with intermittent addition of glucose solution at a flow rate of 200 ml h^{-1}. The values of K_s, μ_m and D are 0.3 g l^{-1}, 0.4 h^{-1} and 0.1 h^{-1}, respectively. Determine the concentration of growth limiting substrate (gl^{-1}) in the reactor at quasi-steady state. _____

Common Data Questions

Common Data for Questions 48 and 49:

A solution was prepared by dissolving 100 mg of protein X in 100 ml of water.
Molecular weight of protein X is 15,000 Da; Avogadro's number = 6.022×10^{23}.

Q.48 Calculate the molarity(μM) of the resulting solution.

(A)66.6 (B)6.6
(C)0.67 (D)0.067

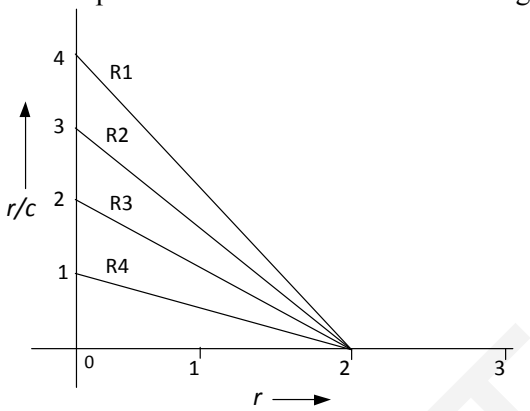
Q.49 The number of moleculespresent in this solution is

(A) 40.15×10^{19} (B) 6.023×10^{19} (C)4.015×10^{19} (D) 0.08×10^{19}

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Common Data for Questions 50 and 51:

The binding efficiency of three different receptorsR1, R2 and R3 were tested against a ligand using equilibrium dialysis, with a constant concentration of receptor and varying concentrations of ligand. The Scatchard plot of receptor titration with different concentration of ligand is given below
(*r*is moles of bound ligand per moles of receptor and *c* is concentration of free ligand)



Q.50 The number of ligand binding sites present on receptors R1 and R3, respectively are

(A)1 and 4 (B)1 and 1 (C)4 and 1 (D)2 and 2

Q.51 Which one of the receptors has thehighestaffinity for the ligand?

(A) R1 (B) R2 (C) R3 (D) R4

Linked Answer Questions

Statement for Linked Answer Questions 52 and 53:

A DNA fragment of 5000bp needs to be isolated from *E.coli* (genome size $4x10^3$kb) genomic library.

Q.52 The minimum number of independent recombinant clones required to represent this fragment in genomic library are

(A) $16x10^2$ (B)$12x10^2$ (C) $8x10^2$ (D) $1.25x10^2$

Q.53 The number of clones to represent this fragment in genomic library with a probability of 95% are

(A) $5.9\ x10^3$ (B) $4.5x10^3$ (C) $3.6\ x10^3$ (D) $2.4x10^3$

Statement for Linked Answer Questions 54 and 55:

During sterilization of a fermentation medium in a given bioreactor$\nabla_{heating}$= 12.56, $\nabla_{cooling}$ = 7.48 and the total value of ∇ required for whole sterilization process is 52, where ∇ is the design criteria.

Q.54 What is the value of $\nabla_{holding}$?

(A)31.96 (B) 42.32 (C) 52.43 (D) 61.18

Q.55 What is the holding period (min) at a k value of $3.36min^{-1}$?

(A) 10.6 (B) 9.5 (C) 8.4 (D) 7.2

General Aptitude (GA) Questions

Q. 56 – Q. 60 carry one mark each.

Q.56 If $3 \leq X \leq 5$ and $8 \leq Y \leq 11$ then which of the following options is TRUE?

(A) $\frac{3}{5} \leq \frac{X}{Y} \leq \frac{8}{5}$

(B) $\frac{3}{11} \leq \frac{X}{Y} \leq \frac{5}{8}$

(C) $\frac{3}{11} \leq \frac{X}{Y} \leq \frac{8}{5}$

(D) $\frac{3}{5} \leq \frac{X}{Y} \leq \frac{8}{11}$

Q.57 The Headmaster _____ to speak to you.

Which of the following options is incorrect to complete the above sentence?

(A) is wanting

(B) wants

(C) want

(D) was wanting

Q.58 Mahatama Gandhi was known for his humility as

(A) he played an important role in humiliating exit of British from India.

(B) he worked for humanitarian causes.

(C) he displayed modesty in his interactions.

(D) he was a fine human being.

Q.59 <u>All engineering students</u> <u>should learn mechanics,</u> <u>mathematics and</u> <u>how to do computation</u>.
I II III IV

Which of the above underlined parts of the sentence is not appropriate?

(A) I (B) II (C) III (D) IV

Q.60 Select the pair that best expresses a relationship similar to that expressed in the pair:
water: pipe::

(A) cart: road
(B) electricity: wire
(C) sea: beach
(D) music: instrument

GATE BT

Q. 61 to Q. 65 carry two marks each.

Q.61 Velocity of an object fired directly in upward direction is given by $V = 80 - 32\,t$, where t (time) is in seconds. When will the velocity be between 32 m/sec and 64 m/sec?

(A) (1, 3/2) (B) (1/2, 1)

(C) (1/2, 3/2) (D) (1, 3)

Q.62 In a factory, two machines M1 and M2 manufacture 60% and 40% of the autocomponents respectively. Out of the total production, 2% of M1 and 3% of M2 are found to be defective. If a randomly drawn autocomponent from the combined lot is found defective, what is the probability that it was manufactured by M2?

(A) 0.35 (B) 0.45 (C) 0.5 (D) 0.4

Q.63 Following table gives data on tourists from different countries visiting India in the year 2011.

Country	Number of Tourists
USA	2000
England	3500
Germany	1200
Italy	1100
Japan	2400
Australia	2300
France	1000

Which two countries contributed to the one third of the total number of tourists who visited India in 2011?

(A) USA and Japan
(B) USA and Australia
(C) England and France
(D) Japan and Australia

Q.64 If $|-2X + 9| = 3$ then the possible value of $|-X| - X^2$ would be:

(A) 30 (B) -30 (C) -42 (D) 42

Q.65 All professors are researchers
Some scientists are professors

Which of the given conclusions is logically valid and is inferred from the above arguments:

(A) All scientists are researchers
(B) All professors are scientists
(C) Some researchers are scientists
(D) No conclusion follows

END OF THE QUESTION PAPER

GATE 2013 : Answer keys for BT - Biotechnology

Paper	Q.No	Key(s)/Value(s)
BT	1	D
BT	2	C
BT	3	B
BT	4	D
BT	5	C
BT	6	A
BT	7	B
BT	8	A
BT	9	B
BT	10	B
BT	11	D
BT	12	A
BT	13	C
BT	14	B
BT	15	C
BT	16	B
BT	17	A
BT	18	D
BT	19	A
BT	20	C
BT	21	0.5
BT	22	18
BT	23	45
BT	24	66.5 to 66.7
BT	25	0.81 to 0.91
BT	26	B
BT	27	C
BT	28	A
BT	29	A
BT	30	D
BT	31	B
BT	32	C
BT	33	A
BT	34	D
BT	35	A

Paper	Q.No	Key(s)/Value(s)
BT	36	B
BT	37	B
BT	38	B
BT	39	C
BT	40	B
BT	41	B
BT	42	A
BT	43	5
BT	44	45
BT	45	0.075 to 0.085
BT	46	76.7 to 76.9
BT	47	0.1
BT	48	A
BT	49	Marks to All
BT	50	D
BT	51	A
BT	52	C
BT	53	D
BT	54	A
BT	55	B
BT	56	B
BT	57	C
BT	58	C
BT	59	D
BT	60	B
BT	61	C
BT	62	C
BT	63	C
BT	64	B
BT	65	C

GATE

GATE 2014 Examination

BT: Biotechnology

Duration: **180 minutes** *Maximum Marks*: **100**

Read the following instructions carefully.

1. To login, enter your Registration Number and password provided to you. Kindly go through the various symbols used in the test and understand their meaning before you start the examination.
2. Once you login and after the start of the examination, you can view all the questions in the question paper, by clicking on the **View All Questions** button in the screen.
3. This question paper consists of **2 sections**, General Aptitude (GA) for **15 marks** and the subject specific GATE paper for **85 marks**. Both these sections are compulsory.
   The GA section consists of **10** questions. Question numbers 1 to 5 are of 1-mark each, while question numbers 6 to 10 are of 2-mark each.
   The subject specific GATE paper section consists of **55** questions, out of which question numbers 1 to 25 are of 1-mark each, while question numbers 26 to 55 are of 2-mark each.
4. Depending upon the GATE paper, there may be useful common data that may be required for answering the questions. If the paper has such useful data, the same can be viewed by clicking on the **Useful Common Data** button that appears at the top, right hand side of the screen.
5. The computer allotted to you at the examination center runs specialized software that permits only one answer to be selected for multiple-choice questions using a mouse and to enter a suitable number for the numerical answer type questions using the virtual keyboard and mouse.
6. Your answers shall be updated and saved on a server periodically and also at the end of the examination. The examination will **stop automatically** at the end of **180 minutes**.
7. In each paper a candidate can answer a total of 65 questions carrying 100 marks.
8. The question paper may consist of questions of **multiple choice type** (MCQ) and **numerical answer type**.
9. Multiple choice type questions will have four choices against A, B, C, D, out of which only **ONE** is the correct answer. The candidate has to choose the correct answer by clicking on the bubble (○) placed before the choice.
10. For numerical answer type questions, each question will have a numerical answer and there will not be any choices**. For these questions, the answer should be entered**by using the virtual keyboard that appears on the monitor and the mouse.
11. All questions that are not attempted will result in zero marks. However, wrong answers for multiple choice type questions (MCQ) will result in **NEGATIVE** marks. For all MCQ questions a wrong answer will result in deduction of⅓ marks for a 1-mark question and ⅔ marks for a 2-mark question.
12. There is **NO NEGATIVE MARKING** for questions of **NUMERICAL ANSWER TYPE**.
13. Non-programmable type Calculator is allowed. Charts, graph sheets, and mathematical tables are **NOT** allowed in the Examination Hall. You must use the Scribble pad provided to you at the examination centre for all your rough work. The Scribble Pad has to be returned at the end of the examination.

Declaration by the candidate:

"I have read and understood all the above instructions. I have also read and understood clearly the instructions given on the admit card and shall follow the same. I also understand that in case I am found to violate any of these instructions, my candidature is liable to be cancelled. I also confirm that at the start of the examination all the computer hardware allotted to me are in proper working condition".

Q. 1 – Q. 5 carry one mark each.

Q.1 Choose the most appropriate word from the options given below to complete the following sentence.

A person suffering from Alzheimer's disease _____ short-term memory loss.

(A) experienced (B) has experienced
(C) is experiencing (D) experiences

Q.2 Choose the most appropriate word from the options given below to complete the following sentence.

_____ is the key to their happiness; they are satisfied with what they have.

(A) Contentment (B) Ambition (C) Perseverance (D) Hunger

Q.3 Which of the following options is the closest in meaning to the sentence below?

"As a woman, I have no country."

(A) Women have no country.
(B) Women are not citizens of any country.
(C) Women's solidarity knows no national boundaries.
(D) Women of all countries have equal legal rights.

Q.4 In any given year, the probability of an earthquake greater than Magnitude 6 occurring in the Garhwal Himalayas is 0.04. The average time between successive occurrences of such earthquakes is ____ years.

Q.5 The population of a new city is 5 million and is growing at 20% annually. How many years would it take to double at this growth rate?

(A) 3-4 years (B) 4-5 years (C) 5-6 years (D) 6-7 years

Q. 6 – Q. 10 carry two marks each.

Q.6 In a group of four children, Som is younger to Riaz. Shiv is elder to Ansu. Ansu is youngest in the group. Which of the following statements is/are required to find the eldest child in the group?

Statements

1. Shiv is younger to Riaz.
2. Shiv is elder to Som.

(A) Statement 1by itself determines the eldest child.
(B) Statement 2 by itself determines the eldest child.
(C) Statements 1 and 2 are both required to determine the eldest child.
(D) Statements 1 and 2 are not sufficient to determine the eldest child.

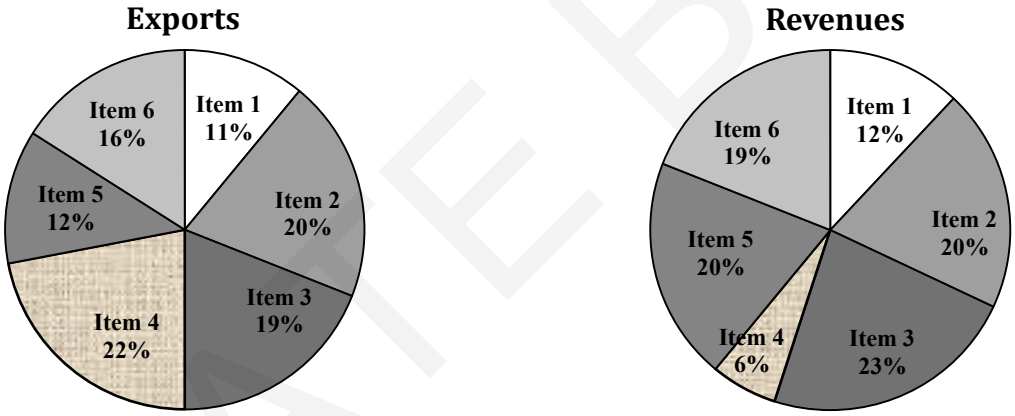
GATEBT

Q.7 Moving into a world of big data will require us to change our thinking about the merits of exactitude. To apply the conventional mindset of measurement to the digital, connected world of the twenty-first century is to miss a crucial point. As mentioned earlier, the obsession with exactness is an artefact of the information-deprived analog era. When data was sparse, every data point was critical, and thus great care was taken to avoid letting any point bias the analysis.
From "BIG DATA" Viktor Mayer-Schonberger and Kenneth Cukier

The main point of the paragraph is:

(A) The twenty-first century is a digital world
(B) Big data is obsessed with exactness
(C) Exactitude is not critical in dealing with big data
(D) Sparse data leads to a bias in the analysis

Q.8 The total exports and revenues from the exports of a country are given in the two pie charts below. The pie chart for exports shows the quantity of each item as a percentage of the total quantity of exports. The pie chart for the revenues shows the percentage of the total revenue generated through export of each item. The total quantity of exports of all the items is 5 lakh tonnes and the total revenues are 250 crore rupees. What is the ratio of the revenue generated through export of Item 1 per kilogram to the revenue generated through export of Item 4 per kilogram?



(A) 1:2 (B) 2:1 (C) 1:4 (D) 4:1

Q.9 X is 1 km northeast of Y. Y is 1 km southeast of Z. W is 1 km west of Z. P is 1 km south of W. Q is 1 km east of P. What is the distance between X and Q in km?

(A) 1 (B) $\sqrt{2}$ (C) $\sqrt{3}$ (D) 2

Q.10 10% of the population in a town is HIV^+. A new diagnostic kit for HIV detection is available; this kit correctly identifies HIV^+ individuals 95% of the time, and HIV^- individuals 89% of the time. A particular patient is tested using this kit and is found to be positive. The probability that the individual is actually positive is _____

END OF THE QUESTION PAPER

Q. 1 – Q. 25 carry one mark each.

Q.1 The eigenvalues of $\mathbf{A} = \begin{bmatrix} 1 & -4 \\ 2 & -3 \end{bmatrix}$ are

(A) $2 \pm i$ (B) -1, -2

(C) $-1 \pm 2i$ (D) non-existent

Q.2 If an unbiased coin is tossed 10 times, the probability that all outcomes are same will be ___ $x10^{-5}$

Q.3 The solution for the following set of equations is,

$$5x + 4y + 10z = 13$$
$$x + 3y + z = 7$$
$$4x - 2y + z = 0$$

(A) x=2, y=1, z=1 (B) x=1, y=2, z=0

(C) x=1, y=0, z=2 (D) x=0, y=1, z=2

Q.4 The limit of the function $e^{-2t} sin\ (t)$ as t $\rightarrow \infty$, is _____

Q.5 The solution to the following set of equations is

$$2x + 3y = 4$$
$$4x + 6y = 0$$

(A) x=0, y=0 (B) x=2, y=0 (C) 4x= -6y (D) No solution

Q.6 The unit for specific substrate consumption rate in a growing culture is

(A) $\frac{g}{L \cdot h}$ (B) $\frac{g}{h}$

(C) $\frac{g}{g \cdot h}$ (D) $\frac{gmoles}{L \cdot h}$

Q.7 If the dissociation constant for solute-adsorbent binding is K_D, the retention time of the solute in a chromatography column

(A) increases with increasing K_D
(B) decreases with increasing K_D
(C) passes through minimum with increasing K_D
(D) is independent of K_D

Q.8 In a batch culture of *Penicillium chrysogenum*, the maximum penicillin synthesis occurs during the

(A) lag phase
(B) exponential phase
(C) stationary phase
(D) death phase

GATE BT

Q.9 The most plausible explanation for a sudden increase of the respiratory quotient (RQ) of a microbial culture is that

(A) cells are dying
(B) yield of biomass is increasing
(C) the fermentation rate is increasing relative to respiration rate
(D) the maintenance rate is decreasing

Q.10 Which of the following is employed for the repeated use of enzymes in bioprocesses?

(A) polymerization (B) immobilization (C) ligation (D) isomerization

Q.11 Since mammalian cells are sensitive to shear, scale-up of a mammalian cell process must consider, among other parameters, the following (given N = rotations/time, D=diameter of impeller)

(A) πND (B) $\pi N^2 D$

(C) πND^2 (D) none of these

Q.12 The degree of reduction of ethanol is _____

Q.13 Gram-positive bacteria are generally resistant to complement-mediated lysis because

(A) thick peptidoglycan layer prevents insertion of membrane attack complex into the inner membrane
(B) Gram-positive bacteria import the membrane attack complex and inactivate it
(C) membrane attack complex is degraded by the proteases produced by the Gram-positive bacteria
(D) Gram-positive bacteria cannot activate the complement pathway

Q.14 A bacterium belonging to cocci group has a diameter of 2 μm. The numerical value of the ratio of its surface area to volume (in μm^{-1}) is _____

Q.15 Which of the following essential element(s) is/are required as major supplement to enhance the bioremediation of oil spills by the resident bacteria?

(A) Sulfur (B) Nitrogen and phosphorus

(C) Iron (D) Carbon

Q.16 The 4-amino or 4-keto group of pyrimidine bases is located in the

(A) major groove of the double stranded DNA
(B) minor groove of the double stranded DNA
(C) minor groove of the B form DNA but not the A form DNA
(D) major groove of the B form DNA but not the A form DNA

Q.17 The product(s) resulting from the hydrolysis of maltose is/are

(A) a mixture of α-D-Glucose and β-D-Glucose
(B) a mixture of D-Glucose and L-Glucose
(C) α-D-Glucose only
(D) β-D-Glucose only

Q.18 Amino acid residue which is most likely to be found in the interior of water-soluble globular proteins is

(A) Threonine (B) Aspartic acid (C) Valine (D) Histidine

Q.19 The 5' ends of the mature forms of the prokaryotic mRNAs and tRNAs are

(A) a triphosphate group in mRNAs and a monophosphate group in tRNAs
(B) triphosphate groups in both mRNAs and tRNAs
(C) monophosphate groups in both mRNAs and tRNAs
(D) a monophosphate group in mRNAs and a triphosphate group in tRNAs

Q.20 Prior exposure of plants to pathogens is known to increase resistance to future pathogen attacks. This phenomenon is known as

(A) systemic acquired resistance (B) hypersensitive response

(C) innate immunity (D) antibody mediated response

Q.21 Reactions between antibodies and antigens that are detected by precipitate formation in an agar gel are referred as

(A) immunoprecipitation assay (B) immunodiffusion assay

(C) immunoaggregation assay (D) immunofixation assay

Q.22 The algorithm for BLAST is based on

(A) Dynamic Programming (B) Hidden Markov Model

(C) k-tuple analysis (D) Neural Network

Q.23 The statistical frequency of the occurrence of a particular restriction enzyme cleavage site that is 6 bases long can be estimated to be

(A) once every 24 bases (B) once every 256 bases

(C) once every 1024 bases (D) once every 4096 bases

Q.24 The reactions leading to the formation of amino acids from the TCA cycle intermediates are

(A) carboxylation (B) isomerization (C) transamination (D) decarboxylation

Q.25 The growth medium for mammalian cells contains serum. One of the major functions of serum is to stimulate cell growth and attachment. However, it must be filter sterilized to

(A) remove large proteins
(B) remove collagen only
(C) remove mycoplasma and microorganisms
(D) remove foaming agents

GATE BT

Q. 26 – Q. 55 carry two marks each.

Q.26 The concentration profile of a chemical at a location x and time t, denoted by $c(x,t)$, changes as per the following equation,

$$c(x,t) = \frac{c_0}{\sqrt{2\,\pi\,D\,t}}\; exp[-\frac{x^2}{2\,D\,t}]$$

where $D\ and\ c_0$ are assumed to be constant. Which of the following is correct?

(A) $\frac{\partial c}{\partial t} = D\ \frac{\partial^2 c}{\partial x^2}$ (B) $\frac{\partial c}{\partial t} = \frac{D}{2}\frac{\partial^2 c}{\partial x^2}$

(C) $\frac{\partial^2 c}{\partial t^2} = D\ \frac{\partial^2 c}{\partial x^2}$ (E) $\frac{\partial^2 c}{\partial t^2} = \frac{D}{2}\frac{\partial^2 c}{\partial x^2}$

Q.27 If $y = x^x$, then $\frac{dy}{dx}$ is

(A) $x^x(x-1)$ (B) x^{x-1}

(C) $x^x(1+logx)$ (D) $e^x(1+logx)$

Q.28 Which of the following statements is true for the series given below?

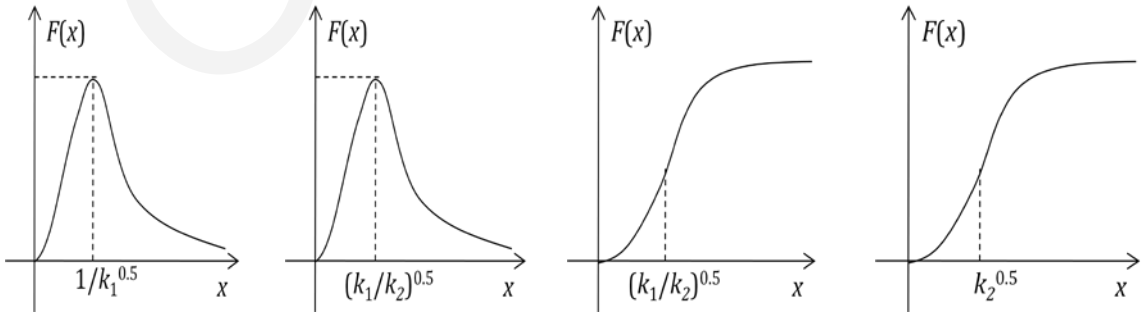
$$s_n = 1 + \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{3}} + \frac{1}{\sqrt{4}} + \cdots + \frac{1}{\sqrt{n}}$$

(A) s_n converges to log(√n) (B) s_n converges to √n

(C) s_n converges to exp(√n) (D) s_n diverges

Q.29 The graph of the function $F(x) = \frac{x}{k_1x^2+k_2x+1}$ for $0 < x < \infty$ is

(A) (B) (C) (D)



Q.30 A T-flask is seeded with 10^5 anchorage-dependent cells. The available area of the T-flask is 25 cm^2 and the volume of the medium is 25 ml. Assume that the cells are rectangles of size 5 µm × 2 µm. If the cells grow to monolayer confluence after 50 h, the growth rate in number of cells/(cm^2.h) is ____ x10^5.

Q.31 Consider a continuous culture provided with a sterile feed containing 10 mM glucose. The steady state cell density and substrate concentration at three different dilution rates are given in the table below.

Dilution rate (h^{-1})	Cell density ($g\ L^{-1}$)	Substrate concentration (mM)
0.05	0.248	0.067
0.5	0.208	1.667
5	0	10

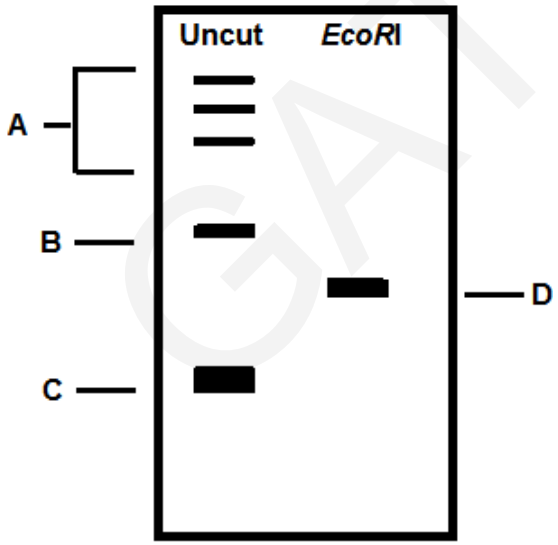
The maximum specific growth rate μ_m (in h^{-1}), will be _____.

Q.32 Cholera toxin increases *c*AMP levels by

(A) modifying G_i protein
(B) modifying G_s protein
(C) binding to adenylate cyclase
(D) activating *c*AMP phosphodiesterase

Q.33 Triose phosphate isomerase converts dihydroxy acetone phosphate (DHAP) to glyceraldehyde-3-phosphate (G-3-P) in a reversible reaction. At 298 K and pH 7.0, the equilibrium mixture contains 40 mM DHAP and 4 mM G-3-P. Assume that the reaction started with 44 mM DHAP and no G-3-P. The standard free-energy change in kJ/mol for the formation of G-3-P [R = 8.315 J/mol.K] is _____.

Q.34 The plasmid DNA was subjected to restriction digestion using the enzyme *EcoR*I and analysed on an agarose gel. Assuming digestion has worked (the enzyme was active), match the identity of the DNA bands shown in the image in **Group I** with their identity in **Group II**.



Group I	**Group II**
P. Bands labeled as A	1. Nicked
Q. Band labeled as B	2. Supercoiled
R. Band labeled as C	3. Concatemers
S. Band labeled as D	4. Linear

(A) P-3, Q-1, R-2, S-4
(B) P-1, Q-4, R-3, S-2
(C) P-4, Q-3, R-1, S-2
(D) P-4, Q-1, R-2, S-3

Q.35 In a relatively large but finite and closed population of sexually reproducing diploid organisms, the frequency of homozygous genotype PP changes from 0.40 to 0.50 and that of pp changes from 0.40 to 0.41 in a span of 10 generations. Which of the following is the most likely cause for the above change in frequency of the PP genotype?

(A) Non-random mating (B) Random genetic drift

(C) Selection (D) Combination of non-random mating and random genetic drift

Q.36 Topological winding number of a 2.0 kb covalently closed circular DNA was found to be 191 with a writhing number of -4. Hence, its LINKING NUMBER and the NUMBER OF BASE PAIR PER TURN when the molecule is laid flat on the surface is _____ and_____, respectively.

(A) 187, 10.69 (B) 195, 10.25 (C) 200, 10.00 (D) 187, 10.50

Q.37 Consider a population of 10,000 individuals, of which 2500 are homozygotes (PP) and 3000 are heterozygotes (Pp) genotype. The frequency of allele p in the population is _____ .

Q.38 Match the following photoreceptors with their prosthetic groups and spectral specificity

Photoreceptor		**Moiety that absorbs light**		**Absorption (nm)**	
P.	Phototropin	1.	Chromobilin	a.	400-500
Q.	Cryptochrome	2.	FAD	b.	600-800
R.	Phytochrome	3.	FMN	c.	500-600

(A) P-3-a, Q-2-a, R-1-b (B) P-1-b, Q-1-a, R-3-b

(C) P-3-a, Q-1-a, R-1-c (D) P-2-c, Q-1-c, R-1-a

Q.39 Match the following plant sources with their secondary metabolites and medical uses

Source plant	**Secondary metabolites**	**Medical use**
P. Belladona	1. Menthol	a. Cancer treatment
Q. Foxglove	2. Atropine	b. Heart disease
R. Pacific yew	3. Digitalin	c. Eye examination
S. Eucalyptus	4. Taxol	d. Cough

(A) P-2-c, Q-3-b, R-4-a, S-1-d (B) P-3-c, Q-2-a, R-1-d, S-4-b

(C) P-2-c, Q-4-b, R-1-a, S-3-d (D) P-1-b, Q-4-c, R-2-d, S-3-a

Q.40 The pungency of mustard seeds is primarily due to secondary metabolites such as isothiocyanate and nitrile. The pungency is usually felt only when the seeds are crushed. This is because of

(A) the coat of the intact seeds blocks the pungent volatiles from being released
(B) the pungent chemicals are stored as inactive conjugates and compartmentalized from the enzymes that convert them into active chemicals
(C) the pungent chemicals are formed only after the reaction with atmospheric oxygen
(D) the pungent chemicals are formed only after the reaction with atmospheric carbondioxide

Q.41 In a mouse genome, the numbers of functional Vα, Jα, Vβ, Dβ, Jβ gene segments are 79, 38, 21, 2 and 11, respectively. The total number of possible combinations for αβ T cell receptors are _____x10^6.

Q.42 The percentage SIMILARITIES and IDENTITIES, respectively, between the two peptide sequences given below will be _____% and _____%.

```
Peptide I : Ala-Ala-Arg-Arg-Gln-Trp-Leu-Thr-Phe-Thr-Lys-Ile-Met-Ser-Glu
Peptide II: Ala-Ala-Arg-Glu-Gln-Tyr-Ile-Ser-Phe-Thr-Lys-Ile-Met-Arg-Asp
```

(A) 80, 80 (B) 80, 60 (C) 60, 60 (D) 90, 60

Q.43 In an affine gap penalty model, if the gap opening penalty is -20, gap extension penalty is -4 and gap length is 8, the gap score is _____.

Q.44 For their efficient translation, eubacterial mRNAs possess a Shine-Dalgarno sequence for its recognition by an anti-Shine-Dalgarno sequence (ASD) in the ribosomes. The correct statement is

(A) ASD is present in 5S rRNA
(B) ASD is present in 23S rRNA
(C) ASD is present in 16S rRNA
(D) ASD is formed by the interaction of the 16S rRNA with the 23S rRNA upon docking of the 50S subunit on the 30S subunit of the ribosomes

Q.45 Match the items in **Group I** with **Group II**

	Group I		**Group II**
P	Receptor tyrosine kinase	1	Inactivation of G-proteins
Q	Cyclic GMP (cGMP)	2	Reception of insulin signal
R	GTPase activating protein (GAP)	3	Thyroid hormone
S	Nuclear receptor	4	Receptor guanylyl cyclase

(A) P-1, Q-3, R-4, S-2 (B) P-2, Q-4, R-3, S-1

(C) P-3, Q-1, R-4, S-2 (D) P-2, Q-4, R-1, S-3

Q.46 Match the immunoglobulin class in **Group I** with its properties in **Group II**

Group I	**Group II**
P. IgG	1. Major antibody in external secretions such as bronchial mucus
Q. IgA	2. Protects against parasites
R. IgE	3. Antibody that appears first in serum after exposure to an antigen
S. IgM	4. Antibody present in highest concentration in serum

(A) P-4, Q-1, R-2, S-3 (B) P-3, Q-2, R-1, S-4

(C) P-4, Q-3, R-1, S-2 (D) P-1, Q-4, R-3, S-2

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Q.47 In a genetic cross between the genotypes WWXX and wwxx, the following phenotypic distributions were observed among the F_2 progeny: WX, 562; wx, 158; Wx, 38; and wX, 42. Likewise, a cross between XXYY and xxyy yielded the following results: XY, 675; xy, 175; Xy, 72; and xY, 78. Similarly, a cross between WWYY and wwyy yielded: WY, 292; wy, 88; Wy, 12; and wY, 8. In all the genotypes, capital letters denote the dominant allele. Assume that the F_1 progeny were self-fertilized in all three crosses. Also, double cross-over does not occur in this species. Which of the following is correct?

(A) Relative position: W-X-Y Distances: W-X = 5 map units, X-Y = 17 map units
(B) Relative position: X-Y-W Distances: X-Y = 15 map units, Y-W = 11 map units
(C) Relative position: Y-W-X Distances: Y-W = 5 map units, W-X = 11 map units
(D) Relative position: X-W-Y Distances: X-W = 5 map units, W-Y = 10 map units

Q.48 The length of the minimum unique stretch of DNA sequence that can be found only once in a 3 billion base pairs long genome is

(A) 14 (B) 15 (C) 16 (D) 18

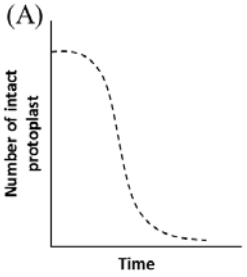
Q.49 Lysine is being produced in a lab-scale reactor by a threonine auxotroph. After 2 weeks of operation it was observed that the concentration of lysine in the reactor was gradually decreasing. Microbiological assays of reactor samples showed absence of contamination and recorded data showed no change in the operating conditions. The most probable reason for decrease in lysine concentration may be attributed to

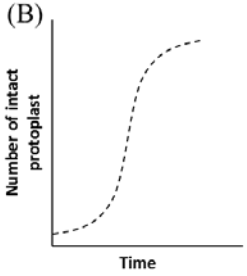
(A) accumulation of ethanol
(B) growth of revertants
(C) production of citric acid
(D) unutilized phosphoenol pyruvate

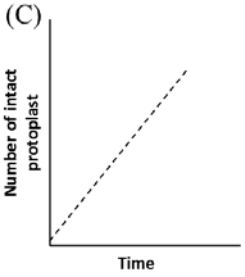
Q.50 If a plant is shifted to cold temperature, which of the following changes would take place in its membrane?

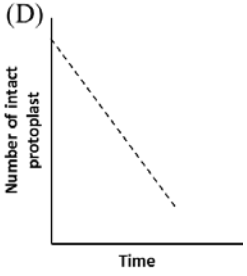
(A) Ratio of unsaturated to saturated fatty acids would increase
(B) Ratio of unsaturated to saturated fatty acids would decrease
(C) Absolute amount of both fatty acids would increase keeping the ratio same
(D) Absolute amount of both fatty acids would remain unchanged

Q.51 If protoplasts are placed in distilled water, they swell and burst as a result of endosmosis. The plot representing the kinetics of burst is

(A)


(B)


(C)


(D)


Q.52 Which of the following statements with respect to the orientation of the nitrogenous bases to the pentose sugars, and the puckering of the sugar, is correct?

(A) Anti, and 2'-endo in A form DNA
(B) Anti, and 2'-endo in B form DNA
(C) Syn, and 3'-endo in A form DNA
(D) Syn, and 3' -endo in B form DNA

Q.53 A dioecious plant has XX sexual genotype for female and XY for male. After double fertilization, what would be the genotype of the embryos and endosperms?

(A) 100% ovules will have XXX endosperm and XX embryo
(B) 100% ovules will have XXY endosperm and XY embryo
(C) 50% ovules will have XYY endosperm and XY embryo, while other 50% will have XXY endosperm and YY embryo
(D) 50% ovules will have XXX endosperm and XX embryo, while the other 50% will have XXY endosperm and XY embryo

Q.54 The amino acid substitution matrices in decreasing order of stringency for comparing protein sequences are

(A) PAM250, PAM120, PAM100
(B) PAM100, PAM120, PAM250
(C) PAM250, PAM100, PAM120
(D) PAM120, PAM250, PAM100

Q.55 The active site in the alpha/beta barrel structures is usually located

(A) inside the barrel
(B) at the amino side of the strands
(C) at the carboxy side of the strands
(D) at any arbitrary site

END OF THE QUESTION PAPER

GATE BT

GATE 2014
Answer Keys for BT - Biotechnology

Section	Q. No.	Key / Range	Marks
GA	1	D	1
GA	2	A	1
GA	3	C	1
GA	4	25 to 25	1
GA	5	A	1
GA	6	A	2
GA	7	C	2
GA	8	D	2
GA	9	C	2
GA	10	0.48 to 0.49	2
BT	1	C	1
BT	2	191 to 199	1
BT	3	B	1
BT	4	0 to 0	1
BT	5	D	1
BT	6	C	1
BT	7	B	1
BT	8	C	1
BT	9	C	1
BT	10	B	1
BT	11	A	1
BT	12	6 to 6	1
BT	13	A	1
BT	14	3 to 3	1
BT	15	B	1
BT	16	A	1
BT	17	A	1
BT	18	C	1
BT	19	A	1
BT	20	A	1
BT	21	B	1
BT	22	C	1
BT	23	D	1
BT	24	C	1

Section	Q. No.	Key / Range	Marks
BT	25	C	1
BT	26	B	2
BT	27	C	2
BT	28	D	2
BT	29	A	2
BT	30	2 to 2	2
BT	31	0.795 to 0.805	2
BT	32	B	2
BT	33	2.46 to 2.50	2
BT	34	A	2
BT	35	C	2
BT	36	A	2
BT	37	0.6 to 0.6	2
BT	38	A	2
BT	39	A	2
BT	40	B	2
BT	41	1.38 to 1.40	2
BT	42	B	2
BT	43	-52 to -52	2
BT	44	C	2
BT	45	D	2
BT	46	A	2
BT	47	C	2
BT	48	C	2
BT	49	B	2
BT	50	A	2
BT	51	A	2
BT	52	B	2
BT	53	D	2
BT	54	B	2
BT	55	C	2

GATE BT

Q. 1 – Q. 25 carry one mark each.

Q.1 Which one of the following complement proteins is the initiator of the membrane attack complex?

(A) C3a (B) C3b (C) C5a (D) C5b

Q.2 Levinthal's paradox is related to

(A) protein secretion
(B) protein degradation
(C) protein folding
(D) protein trafficking

Q.3 Which one of the following is a second generation genetically engineered crop?

(A) Bt brinjal
(B) Roundup soyabean
(C) Golden rice
(D) Bt rice

Q.4 Based on the heavy chain, which one of the following antibodies has multiple subtypes?

(A) IgM (B) IgD (C) IgE (D) IgG

Q.5 The cytokinetic organelle in plant cells is

(A) centriole (B) phragmoplast (C) proplastid (D) chromoplastid

Q.6 Anergy refers to

(A) mitochondrial dysfunction
(B) allergy to environmental antigens
(C) unresponsiveness to antigens
(D) a state of no energy

Q.7 ABO blood group antigens in humans are differentiated from each other on the basis of

(A) sialic acid (B) lipids (C) spectrin (D) glycoproteins

Q.8 Which one of the following organisms is used for the determination of phenol coefficient of a disinfectant?

(A) *Salmonella typhi*
(B) *Escherichia coli*
(C) *Candida albicans*
(D) *Bacillus psychrophilus*

Q.9 A single subunit enzyme converts $420\ \mu\text{moles}$ of substrate to product in one minute. The activity of the enzyme is _____ $\times\ 10^{-6}$ Katal.

Q.10 Which one of the following amino acids has the highest probability to be found on the surface of a typical globular protein in aqueous environment?

(A) Ala (B) Val (C) Arg (D) Ile

Q.11 Which one of the following is **NOT** a product of denitrification in *Pseudomonas*?

(A) N_2 (B) N_2O (C) NO_2^- (D) NH_4^+

Q.12 The determinant of the matrix $\begin{bmatrix} 3 & 0 & 0 \\ 2 & 5 & 0 \\ 6 & -8 & -4 \end{bmatrix}$ is _____.

Q.13 Which one of the following features is **NOT** required in a prokaryotic expression vector?

(A) *oriC* (B) Selection marker (C) CMV promoter (D) Ribosome binding site

Q.14 Production of monoclonal antibodies by hybridoma technology requires

(A) splenocytes (B) osteocytes (C) hepatocytes (D) thymocytes

Q.15 Which one of the following is **INCORRECT** about a typical apoptotic cell?

(A) Phosphatidylserine is presented on the outer cell surface
(B) Cytochrome *c* is released from mitochondria
(C) Mitochondrial membrane potential does not change
(D) Annexin-V binds to the cell surface

Q.16 Identify the file format given below:

>P1; JMFD
Protein X – *Homo sapiens*
MKALTARQQEVFDLIRDHISRTLRQQGDWL

(A) GDE (B) FASTA (C) NBRF (D) GCG

Q.17 Which one of the following relations holds true for the specific growth rate (μ) of a microorganism in the death phase?

(A) $\mu = 0$ (B) $\mu < 0$
(C) $\mu = \mu_{max}$ (D) $0 < \mu < \mu_{max}$

Q.18 How many 3-tuples are possible for the following amino acid sequence?
MADCMWDISEASE

(A) 4 (B) 5 (C) 11 (D) 12

Q.19 How many different protein sequences of 100 residues can be generated using 20 standard amino acids?

(A) 100^{20} (B) 100×20 (C) 20^{100} (D) $100! \times 20!$

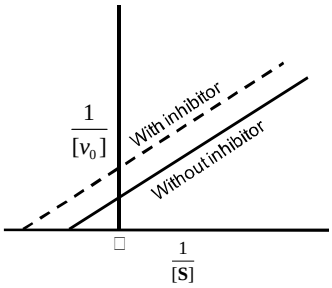
Q.20 In DNA sequencing reactions using the chain termination method, the ratio of ddNTPs to dNTPs should be

(A) 0 (B) < 1
(C) 1 (D) > 1

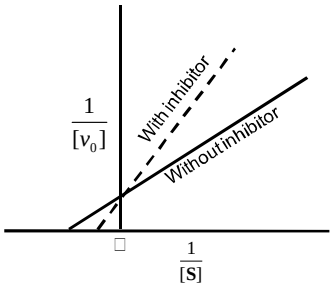
GATE BT

Q.21 Which one of the following graphs represents uncompetitive inhibition?

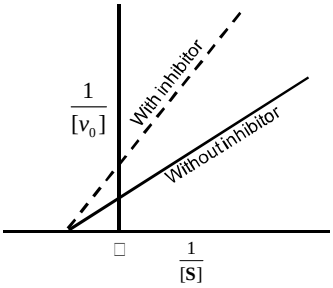
(A)



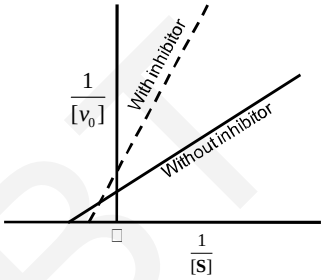
(B)



(C)



(D)



Q.22 Choose the appropriate pair of primers to amplify the following DNA fragment by the polymerase chain reaction (PCR).

```
5′-GACCTGTGG----------------------------ATACGGGAT-3′
3′-CTGGACACC----------------------------TATGCCCTA-5′
```

Primers

P. 5′-GACCTGTGG-3′
Q. 5′-CCACAGGTC-3′
R. 5′-TAGGGCATA-3′
S. 5′-ATCCCGTAT-3′

(A) **P** and **R** (B) **P** and **S** (C) **Q** and **R** (D) **Q** and **S**

Q.23 Consider the following infinite series:

$$1 + r + r^2 + r^3 + \ldots\ldots\ldots\infty$$

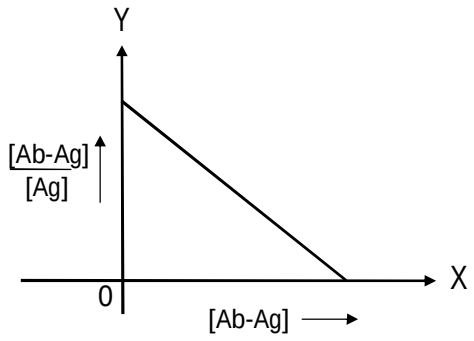
If $r = 0.3$, then the sum of this infinite series is _____.

Q.24

$$2x_1 + x_2 = 3$$
$$5x_1 + bx_2 = 7.5$$

The system of linear equations in two variables shown above will have infinite solutions, if and only if, b is equal to _____.

Q.25 The interaction between an antigen (Ag) and a single-chain antibody (Ab) was studied using Scatchard analysis. The result is shown below.



The affinity of interaction and the total concentration of antibody, respectively, can be determined from

(A) slope and Y-intercept
(B) Y-intercept and slope
(C) X-intercept and slope
(D) slope and X-intercept

Q. 26 – Q. 55 carry two marks each.

Q.26 An isolated population on an island has the following genotypic frequencies:

Genotype	*AA*	*Aa*	*aa*
Frequency	0.3	0.4	0.3

Assuming that there are only two alleles (*A* and *a*) for the gene, the genotypic frequency of *AA* in the next generation will be _____.

Q.27 How many rooted and unrooted phylogenetic trees, respectively, are possible with four different sequences?

(A) 3 and 15 (B) 15 and 3 (C) 15 and 12 (D) 12 and 3

Q.28 Match the compounds in **Group I** with the correct entries in **Group II**.

Group I	**Group II**
P) Cyanide	1) K^+ ionophore
Q) Antimycin A	2) Electron transfer from cytochrome *b* to cytochrome c_1
R) Valinomycin	3) F_1 subunit of ATP synthase
S) Aurovertin	4) Cytochrome oxidase
	5) Adenine nucleotide translocase

(A) P-5, Q-2, R-3, S-1
(B) P-5, Q-2, R-1, S-3
(C) P-4, Q-2, R-1, S-3
(D) P-4, Q-5, R-3, S-1

Q.29 What are the eigenvalues of the following matrix?

$$\begin{bmatrix} 1 & 1 \\ -2 & 4 \end{bmatrix}$$

(A) 2 and 3 (B) -2 and 3 (C) 2 and -3 (D) -2 and -3

Q.30 For a discrete random variable X, $ran(X) = \{0,1,2,3\}$ and the cumulative probability $F(X)$ is shown below:

X	0	1	2	3
$F(X)$	0.5	0.6	0.8	1.0

The mean value of X is _____.

Q.31 Match the drugs in **Group I** with their mechanism of action in **Group II**.

Group I	**Group II**
P) Paclitaxel	1) Inhibits protein translation
Q) Colchicine	2) Inhibits microtubule depolymerization
R) Etoposide	3) Inhibits DNA replication
S) Methotrexate	4) Alkylates DNA
	5) Inhibits dihydrofolate reductase
	6) Inhibits microtubule polymerization

(A) P-1, Q-6, R-3, S-4 (B) P-2, Q-6, R-3, S-5
(C) P-1, Q-3, R-6, S-5 (D) P-2, Q-3, R-6, S-4

Q.32 The limit of the function $(1 + \frac{x}{n})^n$ as n $\rightarrow \infty$ is

(A) $\ln x$ (B) $\ln \frac{1}{x}$

(C) e^{-x} (D) e^{x}

Q.33 Match the cells in **Group I** with their corresponding entries in **Group II**.

Group I	**Group II**
P) Mast cells	1) Activation of the complement pathway
Q) Natural killer cells	2) Expression of CD56
R) Neutrophils	3) Contains azurophilic granules
S) Dendritic cells	4) Defense against helminthic infection
	5) Production of antibodies specific to bacteria
	6) Contains long membranous projections

(A) P-4, Q-2, R-3, S-5 (B) P-4, Q-2, R-3, S-6
(C) P-3, Q-1, R-2, S-5 (D) P-3, Q-1, R-2, S-6

Q.34 Oxygen transfer was measured in a stirred tank bioreactor using dynamic method. The dissolved oxygen tension was found to be 80% air saturation under steady state conditions. The measured oxygen tensions at 7 s and 17 s were 55% and 68% air saturation, respectively. The volumetric mass transfer coefficient K_{La} is _____s^{-1}.

Q.35 Match the microorganisms in **Group I** with their fermentation products in **Group II**.

Group I	**Group II**
P) *Leuconostoc mesenteroides*	1) Cobalamin
Q) *Rhizopus oryzae*	2) Sorbose
R) *Gluconobacter suboxydans*	3) Dextran
S) *Streptomyces olivaceus*	4) Lactic acid
	5) Butanol

(A) P-5, Q-4, R-2, S-1 (B) P-5, Q-3, R-2, S-4
(C) P-3, Q-4, R-1, S-2 (D) P-3, Q-4, R-2, S-1

Q.36 Plasmid DNA (0.5 μg) containing an ampicillin resistance marker was added to 200 μl of competent cells. The transformed competent cells were diluted 10,000 times, out of which, 50 μl was plated on agar plates containing ampicillin. A total of 35 colonies were obtained. The transformation efficiency is _____ $\times 10^6$ cfu·μg^{-1}.

Q.37 Match the reagents in **Group I** with their preferred cleavage sites in **Group II**.

Group I	**Group II**
P) Cyanogen bromide	1) Carboxyl side of methionine
Q) *o*-Iodosobenzoate	2) Amino side of methionine
R) Hydroxylamine	3) Carboxyl side of tryptophan
S) 2-Nitro-5-thiocyanobenzoate	4) Amino side of cysteine
	5) Asparagine-glycine bonds

(A) P-1, Q-3, R-5, S-4 (B) P-2, Q-3, R-1, S-4
(C) P-1, Q-2, R-5, S-4 (D) P-4, Q-2, R-5, S-3

Q.38 *Saccharomyces cerevisiae* produces ethanol by fermentation. The theoretical yield of ethanol from 2.5 g of glucose is _____ g.

Q.39 Choose the **CORRECT** sequence of steps involved in cytoplast production.

(A) Digestion of cell wall → protoplast viability → cybrid formation → osmotic stabilizer
(B) Osmotic stabilizer → digestion of cell wall → protoplast viability → cybrid formation
(C) Protoplast viability → osmotic stabilizer → digestion of cell wall → cybrid formation
(D) Osmotic stabilizer → digestion of cell wall → cybrid formation → protoplast viability

Q.40 Match the antibiotics in **Group I** with their modes of action in **Group II**.

Group I	**Group II**
P) Chloramphenicol	1) Inhibits protein synthesis by acting on 30S ribosomal subunit
Q) Rifampicin	2) Interferes with DNA replication by inhibiting DNA gyrase
R) Tetracycline	3) Inhibits protein synthesis by acting on 50S ribosomal subunit
S) Quinolone	4) Interferes with RNA polymerase activity
	5) Inhibits β-lactamase activity

(A) P-1, Q-2, R-3, S-5 (B) P-3, Q-4, R-1, S-2
(C) P-3, Q-2, R-1, S-4 (D) P-1, Q-4, R-3, S-2

Q.41 The diameters of a large and a small vessel are 1.62 m and 16.2 cm, respectively. The vessels are geometrically similar and operated under similar volumetric agitated power input. The mixing time in the small vessel was found to be 15 s. Determine the mixing time (in seconds) in the large vessel.

(A) 15 (B) 30 (C) 61 (D) 122

Q.42 If $A = \begin{bmatrix} 4 & 2 \\ 1 & 3 \end{bmatrix}$, then $A^2 + 3A$ will be

(A) $\begin{bmatrix} 30 & 20 \\ 10 & 20 \end{bmatrix}$ (B) $\begin{bmatrix} 28 & 10 \\ 4 & 18 \end{bmatrix}$

(C) $\begin{bmatrix} 31 & 13 \\ 7 & 21 \end{bmatrix}$ (D) $\begin{bmatrix} 20 & 10 \\ 5 & 15 \end{bmatrix}$

Q.43 Consider the following multiple sequence alignment of four DNA sequences.

$$\begin{matrix} A & C & T & A \\ A & C & T & G \\ A & G & T & C \\ A & G & C & T \end{matrix}$$

Shannon's entropy of the above alignment is _____.

Q.44 The K_i of a novel competitive inhibitor designed against an enzyme is 2.5 μM. The enzyme was assayed in the absence or presence of the inhibitor (5 μM) under identical conditions. The K_m in the presence of the inhibitor was found to be 30 μM. The K_m in the absence of the inhibitor is _____ μM.

Q.45 A heterozygous tall plant (Tt) was crossed with a homozygous dwarf plant (tt). The resultant seeds were collected. If five seeds are chosen at random, then the probability (in %) that exactly two of these seeds will yield dwarf plants is _____.

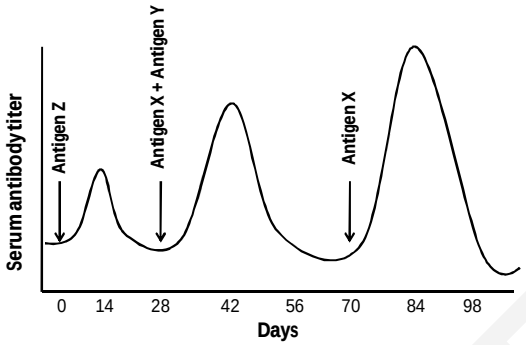
Q.46 Assuming random distribution of nucleotides, the average number of fragments generated upon digestion of a circular DNA of size 4.3×10^5 bp with *Alu*I (5′-AG↓CT-3′) is _____ $\times 10^3$.

Q.47 A synchronous culture containing 1.8×10^5 monkey kidney cells was seeded into three identical flasks. The doubling time of these cells is 24 h. After 24 h, the cells from all the three flasks were pooled and dispensed equally into each well of three 6-well plates. The number of cells in each well will be _____ $\times 10^4$.

Q.48 An *in vitro* translation system can synthesize peptides in all three reading frames of the RNA template. When 5′- UCUCUCUC----$(UC)_n$---UCUCUCUC - 3′ was used as the template in this *in vitro* translation system, the synthesized peptides contained 50% each of serine and leucine. When 5′- CCUCCUCCU---$(CCU)_n$--CCUCCU - 3′ was used as the template, the synthesized peptides contained 33.3% each of serine, leucine, and proline. Deduce the codon for proline.

(A) UCU (B) CUC (C) CCU (D) UCC

Q.49 Three distinct antigens **X**, **Y** and **Z** were used to raise antibodies. Antigen **Z** was injected in a mouse on day zero followed by the administration of antigens **X** and **Y** on day 28. A second injection of antigen **X** was administered on day 70. The antibody titers were monitored in the serum every day and the results are shown below:



Which one of the following statements regarding the antibody titers in the serum is **INCORRECT**?

(A) **Z**-specific IgG will be high on day 14
(B) **X**-specific antibody titer will be high on day 84
(C) **X**-specific IgG will be high on day 42
(D) **Y**-specific IgG will be high on day 84

Q.50 The standard free energy change $(\Delta G'^{\circ})$ for ATP hydrolysis is $-30\ \text{kJ}\cdot\text{mole}^{-1}$. The *in vivo* concentrations of ATP, ADP and P_i in *E. coli* are 7.90, 1.04 and 7.90 mM, respectively. When *E. coli* cells are cultured at 37 ºC, the free energy change (ΔG) for ATP hydrolysis *in vivo* is _____ $\text{kJ}\cdot\text{mole}^{-1}$.

Q.51 In a fed-batch culture, $200\ \text{g}\cdot\text{L}^{-1}$ glucose solution is added at a flow rate of $50\ \text{L}\cdot\text{h}^{-1}$. The initial culture volume (at quasi steady state) and the initial cell concentration are $600\ \text{L}$ and $20\ \text{g}\cdot\text{L}^{-1}$, respectively. The yield coefficient ($Y_{x/s}$) is 0.5 g cell mass·g substrate^{-1}. The cell concentration ($\text{g}\cdot\text{L}^{-1}$) at quasi steady state at $t = 8\ \text{h}$ is

(A) 40 (B) 52 (C) 60 (D) 68

Q.52 Cytoplasmic extract from the wild type strain of a bacterium has the ability to convert a colorless substrate (S) to a colored product (P) via three colorless intermediates X, Y and Z, in that order. Each step of the pathway involves a specific enzyme coded by a distinct gene. Four mutant strains (a^-, b^-, c^-, d^-) were isolated, whose extracts are incapable of producing the colored product in the presence of S. In a series of experiments, extracts from the individual mutants were incubated with X, Y, or Z and scored for color development. The data are summarized in the table below. (Yes: color developed, No: no color developed)

Mutants \ Compounds	**X**	**Y**	**Z**
$\mathbf{a}^-$	No	No	No
$\mathbf{b}^-$	No	Yes	Yes
$\mathbf{c}^-$	Yes	Yes	Yes
$\mathbf{d}^-$	No	No	Yes

Based on the data, which one of the following is the correct order of enzymes involved in the pathway?

(A) $S \xrightarrow{d} X \xrightarrow{c} Y \xrightarrow{b} Z \xrightarrow{a} P$

(B) $S \xrightarrow{a} X \xrightarrow{d} Y \xrightarrow{b} Z \xrightarrow{c} P$

(C) $S \xrightarrow{b} X \xrightarrow{a} Y \xrightarrow{c} Z \xrightarrow{d} P$

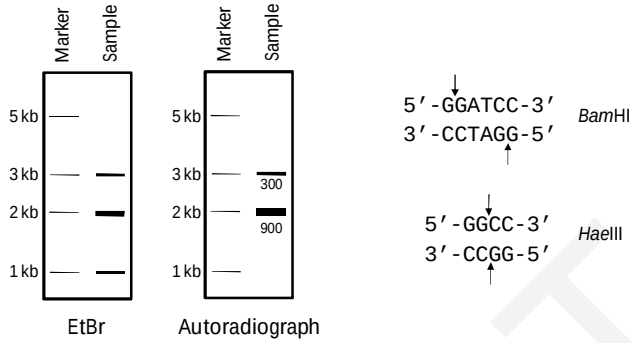
(D) $S \xrightarrow{c} X \xrightarrow{b} Y \xrightarrow{d} Z \xrightarrow{a} P$

Q.53 Samples of bacterial culture taken at 5 PM and then the next day at 5 AM were found to have 10^4 and 10^7 cells·mL^{-1}, respectively. Assuming that both the samples were taken during the log phase of cell growth, the generation time of this bacterium will be _____h.

Q.54 Biomass is being produced in a continuous stirred tank bioreactor of 750 L capacity. The sterile feed containing 8 $g \cdot L^{-1}$ glucose as substrate was fed at a flow rate of 150 $L \cdot h^{-1}$. The microbial system follows Monod's model with μ_m = 0.4 h^{-1}, K_s = 1.5 $g \cdot L^{-1}$ and $Y_{x/s}$ = 0.5 g cell mass·g substrate^{-1}. Determine the cell productivity ($g \cdot L^{-1} \cdot h^{-1}$) at steady state.

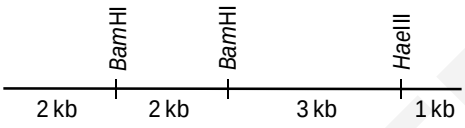
(A) 0.85 (B) 0.65 (C) 0.45 (D) 0.25

Q.55 A linear double stranded DNA of length 8 kbp has three restriction sites. Each of these can either be a *Bam*HI or a *Hae*III site. The DNA was digested completely with both enzymes. The products were purified and subjected to an end-filling reaction using the Klenow fragment and [α-^{32}P]-dCTP. The products of the end-filling reaction were purified, resolved by electrophoresis, stained with ethidium bromide (EtBr) and then subjected to autoradiography. The corresponding images are shown below.



The numbers below each band in the sample lane in the autoradiograph represent their mean signal intensity in arbitrary units. Which one of the following options is the correct restriction map of the DNA?

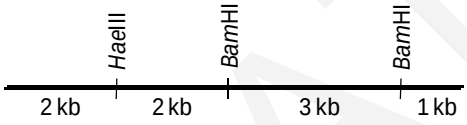
(A)



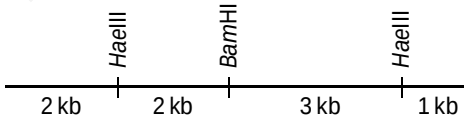
(B)



(C)



(D)



END OF THE QUESTION PAPER

GATE BT

Question Number : 1 Question Type : MCQ

Choose the most appropriate word from the options given below to complete the following sentence.

The principal presented the chief guest with a _____, as token of appreciation.

(A) momento (B) memento (C) momentum (D) moment

Question Number : 2 Question Type : MCQ

Choose the appropriate word/phrase, out of the four options given below, to complete the following sentence:

Frogs _____.

(A) croak (B) roar (C) hiss (D) patter

Question Number : 3 Question Type : MCQ

GATE BT

Choose the word most similar in meaning to the given word:

Educe

(A) Exert (B) Educate (C) Extract (D) Extend

Question Number : 4 Question Type : MCQ

Operators □, ◊ and → are defined by: $a \square b = \frac{a-b}{a+b}$; $a \lozenge b = \frac{a+b}{a-b}$; $a \rightarrow b = ab$.

Find the value of $(66 \square 6) \rightarrow (66 \lozenge 6)$.

(A) −2 (B) −1 (C) 1 (D) 2

Question Number : 5 Question Type : MCQ

If $\log_x (5/7) = -1/3$, then the value of x is

(A) 343/125
(B) 125/343
(C) −25/49
(D) −49/25

Question Number : 6 Question Type : MCQ

The following question presents a sentence, part of which is underlined. Beneath the sentence you find four ways of phrasing the underlined part. Following the requirements of the standard written English, select the answer that produces the most effective sentence.

Tuberculosis, together with its effects, ranks one of the leading causes of death in India.

(A) ranks as one of the leading causes of death
(B) rank as one of the leading causes of death
(C) has the rank of one of the leading causes of death
(D) are one of the leading causes of death

GATE BT

Question Number : 7 Question Type : MCQ

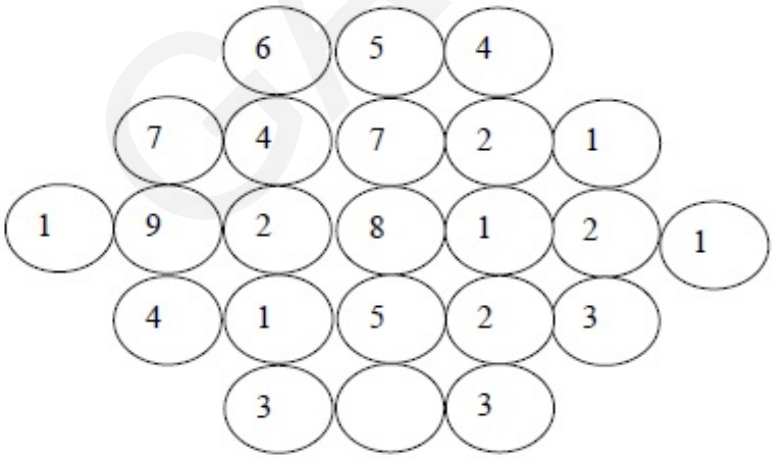
Read the following paragraph and choose the correct statement.

Climate change has reduced human security and threatened human well being. An ignored reality of human progress is that human security largely depends upon environmental security. But on the contrary, human progress seems contradictory to environmental security. To keep up both at the required level is a challenge to be addressed by one and all. One of the ways to curb the climate change may be suitable scientific innovations, while the other may be the Gandhian perspective on small scale progress with focus on sustainability.

(A) Human progress and security are positively associated with environmental security.
(B) Human progress is contradictory to environmental security.
(C) Human security is contradictory to environmental security.
(D) Human progress depends upon environmental security.

Question Number : 8 Question Type : NAT

Fill in the missing value



Correct Answer:

Question Number : 9 Question Type : MCQ

A cube of side 3 units is formed using a set of smaller cubes of side 1 unit. Find the proportion of the number of faces of the smaller cubes visible to those which are NOT visible.

(A) 1 : 4 (B) 1 : 3 (C) 1 : 2 (D) 2 : 3

Question Number : 10 Question Type : MCQ

Humpty Dumpty sits on a wall every day while having lunch. The wall sometimes breaks. A person sitting on the wall falls if the wall breaks.

Which one of the statements below is logically valid and can be inferred from the above sentences?

(A) Humpty Dumpty always falls while having lunch
(B) Humpty Dumpty does not fall sometimes while having lunch
(C) Humpty Dumpty never falls during dinner
(D) When Humpty Dumpty does not sit on the wall, the wall does not break

GATE BT

GATE BT 2015 Answer Key

1	D	14	A	27	B	40	B	53	1.20 to 1.22	
2	C	15	C	28	C	41	C	54	B	
3	C	16	C	29	A	42	A	55	A	
4	D	17	B	30	1.1	43	3.80 to 3.82	1	B	GA
5	B	18	C	31	B	44	10	2	A	GA
6	C	19	C	32	D	45	31.0 to 31.3	3	C	GA
7	D	20	B	33	B	46	1.6 to 1.7	4	C	GA
8	A	21	A	34	.068 to .078	47	2	5	A	GA
9	7	22	B	35	D	48	C	6	A	GA
10	C	23	1.40 to 1.43	36	2.8	49	D	7	B	GA
11	D	24	2.5	37	A	50	-48 to -46	8	3	GA
12	-60	25	D	38	1.20 to 1.30	51	B	9	C	GA
13	C	26	0.25	39	B	52	D	10	B	GA

GATE BT

Q. 1 – Q. 5 carry one mark each.

Q.1 The volume of a sphere of diameter 1 unit is _____ than the volume of a cube of side 1 unit.

(A) least (B) less (C) lesser (D) low

Q.2 The unruly crowd demanded that the accused be _____ without trial.

(A) hanged (B) hanging (C) hankering (D) hung

Q.3 Choose the statement(s) where the underlined word is used correctly:

(i) A prone is a dried plum.
(ii) He was lying prone on the floor.
(iii) People who eat a lot of fat are prone to heart disease.

(A) (i) and (iii) only (B) (iii) only (C) (i) and (ii) only (D) (ii) and (iii) only

Q.4 **Fact:** If it rains, then the field is wet.

Read the following statements:
(i) It rains
(ii) The field is not wet
(iii) The field is wet
(iv) It did not rain

Which one of the options given below is **NOT** logically possible, based on the given fact?

(A) If (iii), then (iv). (B) If (i), then (iii).

(C) If (i), then (ii). (D) If (ii), then (iv).

Q.5 A window is made up of a square portion and an equilateral triangle portion above it. The base of the triangular portion coincides with the upper side of the square. If the perimeter of the window is 6 m, the area of the window in m^2 is _____.

(A) 1.43 (B) 2.06 (C) 2.68 (D) 2.88

GATE BT

Q. 6 – Q. 10 carry two marks each.

Q.6 Students taking an exam are divided into two groups, **P** and **Q** such that each group has the same number of students. The performance of each of the students in a test was evaluated out of 200 marks. It was observed that the mean of group **P** was 105, while that of group **Q** was 85. The standard deviation of group **P** was 25, while that of group **Q** was 5. Assuming that the marks were distributed on a normal distribution, which of the following statements will have the highest probability of being **TRUE**?

(A) No student in group **Q** scored less marks than any student in group **P**.

(B) No student in group **P** scored less marks than any student in group **Q**.

(C) Most students of group **Q** scored marks in a narrower range than students in group **P**.

(D) The median of the marks of group **P** is 100.

Q.7 A smart city integrates all modes of transport, uses clean energy and promotes sustainable use of resources. It also uses technology to ensure safety and security of the city, something which critics argue, will lead to a surveillance state.

Which of the following can be logically inferred from the above paragraph?

(i) All smart cities encourage the formation of surveillance states.
(ii) Surveillance is an integral part of a smart city.
(iii) Sustainability and surveillance go hand in hand in a smart city.
(iv) There is a perception that smart cities promote surveillance.

(A) (i) and (iv) only (B) (ii) and (iii) only

(C) (iv) only (D) (i) only

Q.8 Find the missing sequence in the letter series.

B, FH, LNP, _ _ _ _.

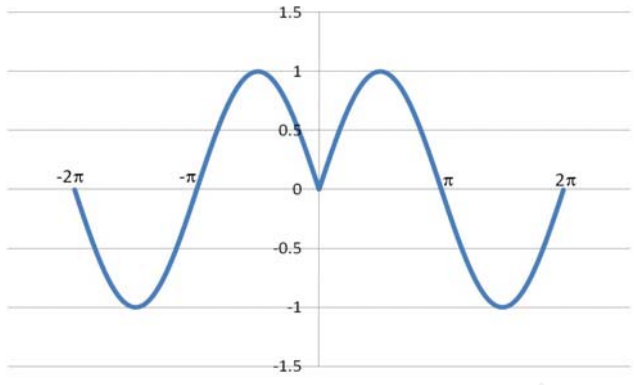
(A) SUWY (B) TUVW (C) TVXZ (D) TWXZ

Q.9 The binary operation □ is defined as $a \square b = ab+(a+b)$, where a and b are any two real numbers. The value of the identity element of this operation, defined as the number x such that $a \square x = a$, for any a, is _____.

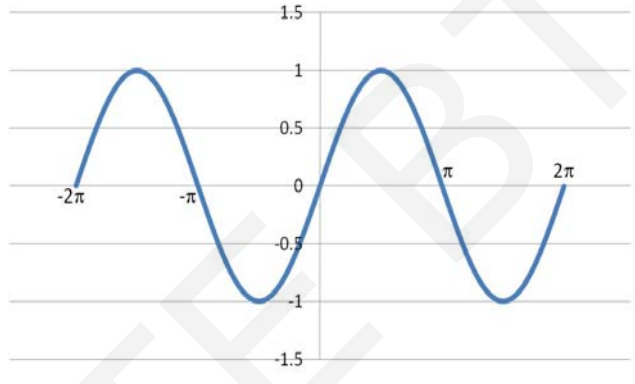
(A) 0 (B) 1 (C) 2 (D) 10

Q.10 Which of the following curves represents the function $y = \ln(|e^{[|\sin(|x|)|]}|)$ for $|x| < 2\pi$? Here, x represents the abscissa and y represents the ordinate.

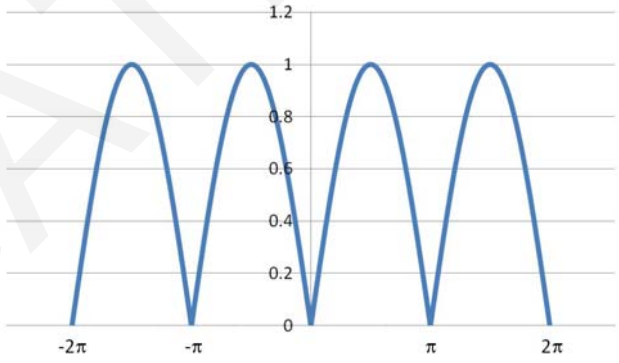
(A)



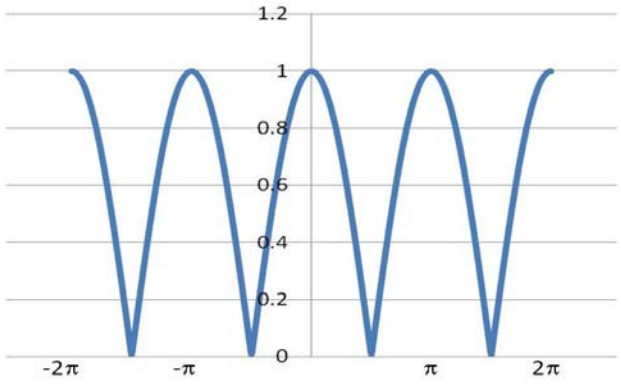
(B)



(C)



(D)



END OF THE QUESTION PAPER

GATE BT

Q. 1 – Q. 25 carry one mark each.

Q.1 Bacteria with two or more flagella at one or both ends are called

(A) amphitrichous (B) peritrichous (C) lophotrichous (D) atrichous

Q.2 Which family of viruses has single stranded DNA?

(A) Herpesviridae (B) Poxviridae (C) Retroviridae (D) Parvoviridae

Q.3 What will be the binding status of regulatory proteins in *lac* operon when concentrations of both lactose and glucose are very low in the culture medium?

(A) Only the repressor remains bound to the operator
(B) Only the cyclic AMP-Catabolic Activator Protein (cAMP-CAP) complex remains bound to the CAP binding site
(C) Neither the repressor nor cAMP-CAP complex remain bound to their respective binding sites
(D) Both the repressor and cAMP-CAP complex remain bound to their respective binding sites

Q.4 Which of the following are TRUE for *Treponema pallidum*?

P. It is the causative agent of syphilis
Q. It is a spirochete
R. It is a non-motile bacterium
S. It is generally susceptible to penicillin

Choose the correct combination.

(A) P, Q and R only (B) P, Q and S only (C) P, R and S only (D) Q, R and S only

Q.5 In a typical mitotic cell division cycle in eukaryotes, **M** phase occurs immediately after the

(A) $\mathbf{G_0}$ phase (B) **S** phase (C) $\mathbf{G_1}$ phase (D) $\mathbf{G_2}$ phase

Q.6 Which one of the following is **NOT** a therapeutic agent based on nucleic acid for the treatment of genetic disorders?

(A) Antisense oligonucleotide (B) Ribozyme
(C) Aptamer (D) Avidin

Q.7 ATP biosynthesis takes place utilizing the H^+ gradient in mitochondria and chloroplasts. Identify the correct sites of H^+ gradient formation.

(A) Across the outer membrane of mitochondria and across the inner membrane of chloroplast
(B) Across the inner membrane of mitochondria and across the thylakoid membrane of chloroplast
(C) Within the matrix of mitochondria and across the inner membrane of chloroplast
(D) Within the matrix of mitochondria and within the stroma of chloroplast

Q.8 Which one of the following is **NOT** an algorithm for building phylogenetic trees?

(A) Maximum parsimony (B) Neighbor joining
(C) Maximum likelihood (D) Bootstrap

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Q.9 Cesium chloride density gradient centrifugation is commonly used for the separation of DNA molecules. The buoyant density, ρ, of a double stranded Cs^+DNA is given by the equation
$\rho = 1.66 + 0.098X_{G+C}$
where X_{G+C} denotes

(A) total number of G and C
(B) mole fraction of G+C
(C) number of GC repeats
(D) ratio of G+C to A+T content

Q.10 Disaccharide molecules that contain β(1 → 4) glycosidic linkage are

(A) sucrose and maltose
(B) sucrose and isomaltose
(C) maltose and isomaltose
(D) lactose and cellobiose

Q.11 Junctional diversity of antibody molecules results from

(A) the addition of switch region nucleotides
(B) the addition of N and P nucleotides
(C) the joining of V, D and J segments
(D) mutations in complementarity-determining regions

Q.12 Which one of the following is **NOT** used for the measurement of cell viability in animal cell culture?

(A) Trypan blue dye exclusion
(B) Tetrazolium (MTT) assay
(C) LDH activity in the culture medium
(D) Coulter counter

Q.13 Which one of the following techniques relies on the spin angular momentum of a photon?

(A) CD spectroscopy
(B) Fluorescence spectroscopy
(C) IR spectroscopy
(D) Raman spectroscopy

Q.14 Which one of the following statements is **NOT** true?

(A) In competitive inhibition, substrate and inhibitor compete for the same active site of an enzyme
(B) Addition of a large amount of substrate to an enzyme cannot overcome uncompetitive inhibition
(C) A transition state analogue in enzyme catalyzed reaction increases the rate of product formation
(D) In non-competitive inhibition, K_m of an enzyme for its substrate remains constant as the concentration of the inhibitor increases

Q.15 Based on their function, find the **ODD** one out.

(A) miRNA (B) siRNA (C) shRNA (D) snRNA

Q.16 Prandtl number is the ratio of

(A) thermal diffusivity to momentum diffusivity
(B) mass diffusivity to momentum diffusivity
(C) momentum diffusivity to thermal diffusivity
(D) thermal diffusivity to mass diffusivity

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Q.17 Fed batch cultivation is suitable for which of the following?

P. Processes with substrate inhibition
Q. Processes with product inhibition
R. High cell density cultivation

(A) P and Q only (B) P and R only (C) Q and R only (D) P, Q and R

Q.18 A biological process is involved in the _____ treatment of industrial effluent.

(A) primary (B) secondary (C) tertiary (D) quaternary

Q.19 In dead-end filtration, rate of filtration is

(A) directly proportional to the square root of pressure drop across the filter medium
(B) inversely proportional to the pressure drop across the filter medium
(C) inversely proportional to the viscosity of the solution
(D) inversely proportional to the square of viscosity of the solution

Q.20 The power required for agitation of non-aerated medium in fermentation is _____ kW.

Operating conditions are as follows:
Fermentor diameter = 3 m
Number of impellers = 1
Mixing speed = 300 rpm
Diameter of the Rushton turbine = 1 m
Viscosity of the broth = 0.001 Pa.s
Density of the broth = 1000 $kg.m^{-3}$
Power number = 5

Q.21 Which one of the following is the most suitable type of impeller for mixing high viscosity (viscosity > 10^5 cP) fluids?

(A) Propeller (B) Helical ribbon (C) Paddle (D) Flat blade turbine

Q.22 Runs scored by a batsman in five one-day matches are 55, 75, 67, 88 and 15. The standard deviation is _____.

Q.23 The **positive** Eigen value of the following matrix is _____.

$$\begin{bmatrix} 2 & 1 \\ 5 & -2 \end{bmatrix}$$

Q.24 The Laplace transform $F(s)$ of the function $f(t) = \cos(at)$, where a is constant, is _____.

(A) $\frac{s^2}{s^2+a^2}$ (B) $\frac{a}{s^2+a^2}$ (C) $\frac{s}{s^2+a^2}$ (D) $\frac{s}{s^2-a^2}$

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Q.25 The value of the integral $\int_0^{0.9} \frac{dx}{(1-x)(2-x)}$ is _____.

Q. 26 – Q. 55 carry two marks each.

Q.26 Which combination of the following statements is **CORRECT** for cyanobacteria?

P. They can perform oxygenic photosynthesis
Q. Usually filamentous forms are involved in nitrogen fixation
R. Nitrogen fixation occurs in heterocysts
S. They cannot grow in a mineral medium exposed to light and air

(A) P, Q and R (B) P, S and R (C) Q, R and S (D) P, Q and S

Q.27 Which set of the following events occurs during the elongation step of translation?

P. Attachment of mRNA with the smaller subunit of ribosome
Q. Loading of correct aminoacyl-tRNA into the A site
R. Formation of a peptide bond between the amino acyl-tRNA in the A site and the peptide chain that is attached to the peptidyl-tRNA in the P site
S. Dissociation of the ribosomal subunits
T. Translocation of peptidyl-tRNA from the A site to the P site of the ribosome

(A) P, Q and R (B) P, Q and T (C) Q, R and T (D) R, S and T

Q.28 A DNA sequence, 5'-ATGGACGTGCTTCCCAAAGCATCGGGC-3', is mutated to obtain

P. 5′-ATGGACGTGCTTC**a**CAAAGCATCGGGC-3′
Q. 5′-ATGGACGTGCTTCCC**g**AAAGCATCGGGC-3′
R. 5′-ATGGACGTGCTTCC**-**AAAGCATCGGGC-3′
S. 5′-ATGGACGTGCTTCCCAA**t**GCATCGGGC-3′
T. 5′-ATGGACG**a**GCTTCCCAAAGCATCGGGC-3′

[Point mutations are shown in the **lower case** or '**-**' within the sequences]

Which of the above mutant sequences **DO NOT** have frame-shift?

(A) P, Q and S (B) P, S and T (C) Q, R and S (D) Q, S and T

Q.29 Which of the following events occur during the stationary phase of bacterial growth?

P. Rise in cell number stops
Q. Spore formation in some Gram-positive bacteria such as *Bacillus subtilis*
R. Cell size increases in some Gram-negative bacteria such as *Escherichia coli*
S. Growth rate of bacterial cells nearly equals their death rate
T. Decrease in peptidoglycan crosslinking

(A) P, Q and S only (B) P, S and T only (C) Q, R and S only (D) P, R and T only

Q.30 Select the **CORRECT** combination of genetic components that are essential for the transfer of T-DNA segment from *Agrobacterium tumefaciens* to plant cells.

(A) Border repeat sequences and oncogenes
(B) Border repeat sequences and *vir* genes
(C) Opine biosynthetic genes and *vir* genes
(D) Opine biosynthetic genes and oncogenes

Q.31 Match the secondary metabolites (**Column-I)** with the corresponding plant species (**Column-II**).

Column-I	**Column-II**
P. Morphine	1. *Datura stramonium*
Q. Pyrethrins	2. *Catharanthus roseus*
R. Scopolamine	3. *Papaver somniferum*
S. Vincristine	4. *Tagetes erecta*

(A) P-4, Q-3, R-1, S-2
(B) P-3, Q-4, R-1, S-2
(C) P-2, Q-3, R-4, S-1
(D) P-4, Q-1, R-2, S-3

Q.32 A variety of genetic elements are used in the transgenic plant research. Match the genetic elements (**Column-I)** with their corresponding source (**Column-II)**.

Column-I	**Column-II**
P. *Ubiquitin1* promoter	1. *Agrobacterium tumefaciens*
Q. *Nos* transcriptional terminator	2. *Streptomyces hygroscopicus*
R. *bar* selection marker gene	3. *Escherichia coli*
S. *gus* reporter gene	4. *Zea mays*

(A) P-2, Q-1, R-3, S-4
(B) P-2, Q-3, R-4, S-1
(C) P-3, Q-4, R-1, S-2
(D) P-4, Q-1, R-2, S-3

Q.33 Match the type of chromosomal inheritance (**Column-I)** with the corresponding genetic disease or trait (**Column-II)**.

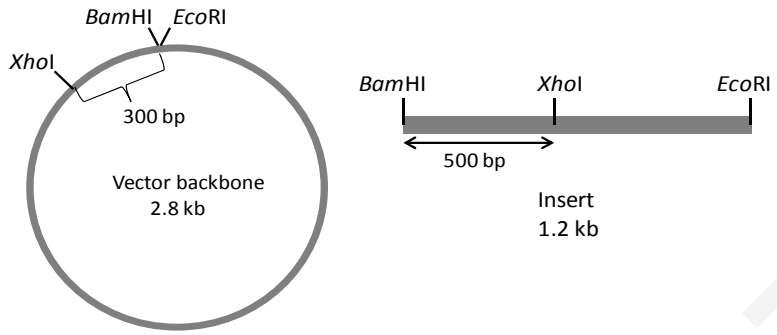
Column-I	**Column-II**
P. Autosomal recessive inheritance	1. Huntington disease
Q. Autosomal dominant inheritance	2. Hairy ears
R. X-linked inheritance	3. Cystic fibrosis
S. Y-linked inheritance	4. Hemophilia

(A) P-1, Q-4, R-3, S-2
(B) P-4, Q-3, R-2, S-1
(C) P-3, Q-1, R-4, S-2
(D) P-4, Q-2, R-3, S-1

Q.34 A crossing was performed between the genotypes ***DdEeFfgg*** and ***ddEeFfGg***. Assuming that the allelic pairs of all genes assort independently, the proportion of progeny having the genotype ***ddeeffgg*** is expected to be _____%.

Q.35 The equilibrium potential of a biological membrane for Na^+ is 55 mV at 37 °C. Concentration of Na^+ inside the cell is 20 mM. Assuming the membrane is permeable to Na^+ only, the Na^+ concentration outside the membrane will be _____ mM.
(Faraday constant: 23062 $cal.V^{-1}.mol^{-1}$, Gas constant: 1.98 $cal.mol^{-1}.K^{-1}$)

Q.36 A 1.2 kb DNA fragment was cloned into *Bam*HI and *Eco*RI sites located on a 2.8 kb cloning vector. The *Bam*HI and *Eco*RI sites are adjacent to each other on the vector backbone. The vector contains an *Xho*I site located 300 bp upstream of the *Bam*HI site. An internal *Xho*I site is present in the gene sequence as shown in the figure. The resultant recombinant plasmid is digested with *Eco*RI and *Xho*I and analyzed through 1% agarose gel electrophoresis. Assuming complete digestion with *Eco*RI and *Xho*I, the DNA fragments (in base pairs) visible on the agarose gel will correspond to:



(A) 2800, 700 and 500
(B) 2800, 700 and 800
(C) 2500, 700 and 800
(D) 2500, 1200 and 300

Q.37 Find the **INCORRECT** combination.

(A) Surface immunoglobulins − B cell antigen receptor
(B) Affinity maturation − isotype switching
(C) Fc region of antibodies − binding to complement proteins
(D) Spleen, the secondary lymphoid organ − no connection with the lymphatic system

Q.38 Which of the following statement(s) is/are **CORRECT** for antigen activated effector T cells?

P. $CD4^+$ cells make contact with macrophages and stimulate their microbicidal activity
Q. $CD4^+$ cells make contact with B cells and stimulate them to differentiate into plasma cells
R. $CD8^+$ cells make contact with B cells and stimulate them to differentiate into plasma cells
S. $CD8^+$ cells make contact with virus infected cells and kill them

(A) Q only (B) Q and S only (C) P, Q and S only (D) P, Q, R and S

Q.39 Which one of the following statements regarding G proteins is **INCORRECT**?

(A) GDP is bound to G protein in the resting stage
(B) GTP bound α subunit cannot reassemble with $\beta\gamma$ dimer
(C) All G proteins are trimeric
(D) Activation of G protein may result in activation or inhibition of the target enzymes

Q.40 In animal cell culture, a CO_2 enriched atmosphere in the incubator chamber is used to maintain the culture pH between 6.9 and 7.4. Which one of the following statements is **CORRECT**?

(A) Higher the bicarbonate concentration in the medium, higher should be the requirement of gaseous CO_2
(B) Lower the bicarbonate concentration in the medium, higher should be the requirement of gaseous CO_2
(C) Higher the bicarbonate concentration in the medium, lower should be the requirement of gaseous CO_2
(D) CO_2 requirement is independent of bicarbonate concentration in the medium

Q.41 Choose the **CORRECT** combination of True (T) and False (F) statements about microcarriers used in animal cell culture.

P. Higher cell densities can be achieved using microcarriers
Q. Microcarriers increase the surface area for cell growth
R. Microcarriers are used for both anchorage- and nonanchorage-dependent cells
S. Absence of surface charge on microcarriers enhances attachment of cells

(A) P-T, Q-F, R-T and S-F
(B) P-T, Q-T, R-F and S-F
(C) P-F, Q-F, R-T and S-T
(D) P-F, Q-T, R-F and S-T

Q.42 In an assay of the type II dehydroquinase of molecular mass 18 kDa, it is found that the V_{max} of the enzyme is 0.0134 $\mu mol.min^{-1}$ when 1.8 µg enzyme is added to the assay mixture. If the K_m for the substrate is 25 µM, the k_{cat}/K_m ratio will be _____ $\times 10^4$ $M^{-1}.s^{-1}$.

Q.43 The molar extinction coefficients of Trp and Tyr at 280 nm are 5690 and 1280 $M^{-1}.cm^{-1}$, respectively. The polypeptide chain of yeast alcohol dehydrogenase (37 kDa) contains 5 Trp and 14 Tyr residues. The absorbance at 280 nm of a 0.32 $mg.mL^{-1}$ solution of yeast alcohol dehydrogenase measured in a cuvette of 1 cm pathlength will be _____.

(Assume that the molar extinction coefficient values for Trp and Tyr apply to these amino acids in the yeast alcohol dehydrogenase).

Q.44 The activity of lactate dehydrogenase can be measured by monitoring the following reaction:

$$\text{Pyruvate} + \text{NADH} \longrightarrow \text{Lactate} + \text{NAD}^+$$

The molar extinction coefficient of NADH at 340 nm is 6220 $M^{-1}.cm^{-1}$. NAD^+ does not absorb at this wavelength. In an assay, 25 µL of a sample of enzyme (containing 5 µg protein per mL) was added to a mixture of pyruvate and NADH to give a total volume of 3 mL in a cuvette of 1 cm pathlength. The rate of decrease in absorbance at 340 nm was 0.14 min^{-1}. The specific activity of the enzyme will be _____ $\mu mol.min^{-1}.mg^{-1}$.

Q.45 Analysis of a hexapeptide using enzymatic cleavage reveals the following result:

- Amino acid composition of the peptide is: 2R, A,V, S, Y
- Trypsin digestion yields two fragments and the compositions are: (R, A, V) and (R, S, Y)
- Chymotrypsin digestion yields two fragments and the compositions are: (A, R, V, Y) and (R, S)
- Digestion with carboxypeptidase A yields no cleavage product.

Given: Trypsin cleaves at carboxyl side of R.
Chymotrypsin cleaves at carboxyl side of Y.
Carboxypeptidase A cleaves at amino side of the C-terminal amino acid (except R and K) of the peptide.

The correct amino acid sequence of the peptide is:

(A) RSYRVA (B) AVRYSR (C) SRYVAR (D) SVRRYA

Q.46 The empirical formula for biomass of an unknown organism is $CH_{1.8}O_{0.5}N_{0.2}$. To grow this organism, ethanol (C_2H_5OH) and ammonia are used as carbon and nitrogen sources, respectively. Assume no product formation other than biomass. To produce 1 mole of biomass from 1 mole of ethanol, the number of moles of oxygen required will be _____.

Q.47 *Saccharomyces cerevisiae* is cultured in a chemostat (continuous fermentation) at a dilution rate of 0.5 h^{-1}. The feed substrate concentration is 10 $g.L^{-1}$. The biomass concentration in the chemostat at steady state will be _____ $g.L^{-1}$.

Assumptions: Feed is sterile, maintenance is negligible and maximum biomass yield with respect to substrate is 0.4 (g biomass per g ethanol).

Microbial growth kinetics is given by $\mu = \frac{\mu_m s}{K_s + s}$

where μ is specific growth rate (h^{-1}), μ_m = 0.7 h^{-1}, K_s = 0.3 $g.L^{-1}$ and s is substrate concentration ($g.L^{-1}$).

Q.48 Decimal reduction time of bacterial spores is 23 min at 121 °C and the death kinetics follow first order. One liter medium containing 10^5 spores per mL was sterilized for 10 min at 121 °C in a batch sterilizer. The number of spores in the medium after sterilization (assuming destruction of spores in heating and cooling period is negligible) will be _____$\times 10^7$.

Q.49 A bioreactor is scaled up based on equal impeller tip speed. Consider the following parameters for small and large bioreactors:

Parameters	**Small bioreactor**	**Large bioreactor**
Impeller speed	N_1	N_2
Diameter of impeller	D_1	D_2
Power consumption	P_1	P_2

Assuming geometrical similarity and the bioreactors are operated in turbulent regime, what will be P_2/P_1?

(A) $(D_1/D_2)^2$ (B) $(D_2/D_1)^2$ (C) $(D_1/D_2)^5$ (D) $(D_2/D_1)^5$

Q.50 An enzyme converts substrate A to product B. At a given liquid feed stream of flow rate 25 L.min^{-1} and feed substrate concentration of 2 mol.L^{-1}, the volume of continuous stirred tank reactor needed for 95% conversion will be _____ L.

Given the rate equation: $-r_A = \dfrac{0.1C_A}{1+0.5C_A}$

where $-r_A$ is the rate of reaction in mol.L^{-1}.min^{-1} and C_A is the substrate concentration in mol.L^{-1}

Assumptions: Enzyme concentration is contant and does not undergo any deactivation during the reaction.

Q.51 A protein is to be purified using ion-exchange column chromatography. The relationship between HETP (Height Equivalent to Theoretical Plate) and the linear liquid velocity of mobile phase is given by:

$$H = \frac{A}{u} + Bu + C$$

where H is HETP (m) and u is linear liquid velocity of mobile phase (m.s^{-1}). The values of A, B and C are 3×10^{-8} m^2.s^{-1}, 3 s and 6×10^{-5} m, respectively. The number of theoretical plates based on **minimum** HETP for a column of 66 cm length will be _____.

Q.52 An enzyme is immobilized on the surface of a **non-porous** spherical particle of 2 mm diameter. The immobilized enzyme is suspended in a solution having bulk substrate concentration of 10 mM. The enzyme follows first order kinetics with rate constant 10 s^{-1} and the external mass transfer coefficient is 1 cm.s^{-1}. Assume steady state condition wherein rate of enzyme reaction (mmol.L^{-1}.s^{-1}) at the surface is equal to mass transfer rate (mmol.L^{-1}.s^{-1}). The substrate concentration at the surface of the immobilized particle will be _____ mM.

Q.53 $\dfrac{d^2y}{dx^2} - y = 0$. The initial conditions for this second order homogeneous differential equation are $y(0) = 1$ and $\dfrac{dy}{dx} = 3$ at $x = 0$

The value of y when $x = 2$ is _____.

Q.54 The value of determinant A given below is _____ .

$$A = \begin{pmatrix} 5 & 16 & 81 \\ 0 & 2 & 2 \\ 0 & 0 & 16 \end{pmatrix}$$

Q.55 Consider the equation

$$V = \frac{aS}{b + S + \frac{S^2}{c}}$$

Given $a = 4$, $b = 1$ and $c = 9$, the **positive** value of S at which V is maximum, will be _____.

END OF THE QUESTION PAPER

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GATE BT 2016 Answer Key

Q. No	Type	Section	Key	Marks
1	MCQ	GA	B	1
2	MCQ	GA	A	1
3	MCQ	GA	D	1
4	MCQ	GA	C	1
5	MCQ	GA	B	1
6	MCQ	GA	C	2
7	MCQ	GA	C	2
8	MCQ	GA	C	2
9	MCQ	GA	A	2
10	MCQ	GA	C	2
1	MCQ	BT	C	1
2	MCQ	BT	D	1
3	MCQ	BT	D	1
4	MCQ	BT	B	1
5	MCQ	BT	D	1
6	MCQ	BT	D	1
7	MCQ	BT	B	1
8	MCQ	BT	D	1
9	MCQ	BT	B	1
10	MCQ	BT	D	1
11	MCQ	BT	B	1
12	MCQ	BT	D	1
13	MCQ	BT	A	1
14	MCQ	BT	C	1
15	MCQ	BT	D	1
16	MCQ	BT	C	1
17	MCQ	BT	B	1
18	MCQ	BT	B	1
19	MCQ	BT	C	1
20	NAT	BT	625.0 : 625.0	1
21	MCQ	BT	B	1
22	NAT	BT	24.5 : 28.5	1
23	NAT	BT	3.0 : 3.0	1
24	MCQ	BT	C	1
25	NAT	BT	1.65 : 1.75	1
26	MCQ	BT	A	2
27	MCQ	BT	C	2
28	MCQ	BT	B	2
29	MCQ	BT	A	2
30	MCQ	BT	B	2
31	MCQ	BT	B	2
32	MCQ	BT	D	2
33	MCQ	BT	C	2
34	NAT	BT	1.3 : 1.8	2
35	NAT	BT	147.0 : 170.0	2
36	MCQ	BT	C	2
37	MCQ	BT	B	2
38	MCQ	BT	C	2
39	MCQ	BT	C	2

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40	MCQ	BT	A	2
41	MCQ	BT	B	2
42	NAT	BT	8.6 : 9.4	2
43	NAT	BT	0.37 : 0.43	2
44	NAT	BT	525.0 : 555.0	2
45	MCQ	BT	B	2
46	NAT	BT	1.9 : 2.0	2
47	NAT	BT	3.65 : 3.75	2
48	NAT	BT	3.6 : 3.8	2
49	MCQ	BT	B	2
50	NAT	BT	4986.0 : 4989.0	2
51	NAT	BT	1000.0 : 1000.0	2
52	NAT	BT	7.5 : 7.5	2
53	NAT	BT	14.55 : 14.75	2
54	NAT	BT	160.0 : 160.0	2
55	NAT	BT	3.0 : 3.0	2

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Graduate Aptitude Test in Engineering 2017

Question Paper Name: Biotechnology 5th Feb 2017
Subject Name: Biotechnology
Duration: 180
Total Marks: 100



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Organizing Institute:

Indian Institute of Technology Roorkee



Question Number : 1 **Correct : 1 Wrong : -0.33**

An enzyme catalyzes a reaction by

(A) decreasing the energy of the substrate.
(B) decreasing the activation energy of the reaction.
(C) decreasing product stability.
(D) increasing the activation barrier of the reaction.

Question Number : 2 **Correct : 1 Wrong : -0.33**

Natural proteins are composed primarily of 20 α-amino acids. Which one of the following statements is true for any of these amino acids in a solution of pH 1.5?

(A) Only the amino group is ionized.
(B) Only the carboxylic acid group is ionized.
(C) Both amino and carboxylic acid groups are ionized.
(D) Both amino and carboxylic acid groups are neutral.

Question Number : 3 **Correct : 1 Wrong : -0.33**

Which one of the following organisms is an indicator of fecal contamination?

(A) *Escherichia coli* (B) *Streptococcus lactis*
(C) *Bacillus subtilis* (D) *Lactobacillus acidophilus*

Question Number : 4 **Correct : 1 Wrong : 0**

The surface area (in m^2) of the largest sphere that can fit into a hollow cube with edges of length 1 meter is _____.

Given data: $\pi = 3.14$

Question Number : 5 **Correct : 1 Wrong : 0**

In a thin layer chromatography experiment using a silica gel plate, a compound showed migration of 12.5 cm and the solvent front showed migration of 18 cm. The R_f value for the compound is _____.

Question Number : 6 **Correct : 1 Wrong : 0**

$\lim_{x \to 0} \frac{\sin(x)}{x}$ is _____ .

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Question Number : 7 **Correct : 1 Wrong : -0.33**

Which one of the following mechanisms is used by human pathogens to evade host immune responses?

(A) Somatic hypermutation
(B) Antibody production
(C) Antigenic variation
(D) Complement activation

Question Number : 8 **Correct : 1 Wrong : -0.33**

If the nucleotide composition (%) of a viral genome is A = 10, U = 20, C = 40, and G = 30, which one of the following is this genome?

(A) Double stranded RNA
(B) Single stranded RNA
(C) Single stranded DNA
(D) Double stranded DNA

Question Number : 9 **Correct : 1 Wrong : -0.33**

Which one of the following techniques can be used to determine the structure of a 15 kDa globular protein at atomic resolution?

(A) Raman spectroscopy
(B) IR spectroscopy
(C) UV spectroscopy
(D) NMR spectroscopy

Question Number : 10 **Correct : 1 Wrong : -0.33**

Assertion [a]: Gram negative bacteria show staining with saffranin.
Reason [r]: Gram negative bacteria have an outer membrane with lipopolysaccharides.

(A) Both [a] and [r] are true and [r] is the correct reason for [a].
(B) Both [a] and [r] are true but [r] is not the correct reason for [a].
(C) Both [a] and [r] are false.
(D) [a] is true but [r] is false.

Question Number : 11 **Correct : 1 Wrong : -0.33**

The plant hormone indole-3-acetic acid is derived from

(A) histidine (B) tyrosine (C) tryptophan (D) proline

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Question Number : 12 **Correct : 1 Wrong : -0.33**

Enzyme-linked immunosorbent assay (ELISA) is used for

(A) quantifying antibody levels in blood
(B) determining the molecular weight of an antigen
(C) purifying proteins from biological fluids
(D) determining the molecular weight of an antibody

Question Number : 13 **Correct : 1 Wrong : -0.33**

Macrophages eliminate pathogenic bacteria upon activation by

(A) NK cells (B) basophils (C) CD4+ T cells (D) plasma cells

Question Number : 14 **Correct : 1 Wrong : 0**

If a protein contains four cysteine residues, the number of different ways they can simultaneously form two intra-molecular disulphide bonds is _____.

Question Number : 15 **Correct : 1 Wrong : -0.33**

Secretory proteins synthesized by ER-associated ribosomes traverse through

(A) mitochondria
(B) peroxisomes
(C) the Golgi apparatus
(D) the nucleus

Question Number : 16 **Correct : 1 Wrong : 0**

For y = f(x), if $\frac{d^2y}{dx^2} = 0, \frac{dy}{dx} = 0$ at x = 0, and y = 1 at x =1, the value of y at x = 2 is _____.

Question Number : 17 **Correct : 1 Wrong : -0.33**

During protein synthesis, tRNAs are NOT involved in

(A) charging (B) initiation (C) elongation (D) termination

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Question Number : 18 **Correct : 1 Wrong : -0.33**

In eukaryotes, cytokinesis is inhibited by

(A) cytochalasin D (B) vinblastine (C) nocodazole (D) colchicine

Question Number : 19 **Correct : 1 Wrong : -0.33**

A proto-oncogene is suspected to have undergone duplication in a certain type of cancer. Of the following techniques, which one would verify the gene duplication?

(A) Northern blotting
(B) Southern blotting
(C) South western blotting
(D) Western blotting

Question Number : 20 **Correct : 1 Wrong : -0.33**

A polymerase chain reaction (PCR) was set up with the following reagents: DNA template, *Taq* polymerase, buffer, dNTPs, and Mg^{2+}. Which one of the following is missing in the reaction mixture?

(A) Helicase
(B) Single-stranded binding proteins
(C) Primers
(D) Reverse transcriptase

Question Number : 21 **Correct : 1 Wrong : -0.33**

An enzymatic reaction exhibits Michaelis-Menten kinetics. For this reaction, on doubling the concentration of enzyme while maintaining $[S] >> [E_0]$,

(A) both K_m and V_{max} will remain the same.
(B) K_m will remain the same but V_{max} will increase.
(C) K_m will increase but V_{max} will remain the same.
(D) both K_m and V_{max} will increase.

Question Number : 22 **Correct : 1 Wrong : 0**

A 5 liter chemostat is fed fresh medium at 0.2 liters/minute having a substrate concentration of 25 grams/liter. At steady state, the outgoing stream has substrate concentration of 2.5 grams/liter. The rate of consumption (grams/liter/minute) of the substrate in the reactor is _____.

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Question Number : 23 Correct : 1 Wrong : -0.33

The transcription factor X binds a 10 base pair DNA stretch. In the DNA of an organism, X was found to bind at 20 distinct sites. An analysis of these 20 binding sites showed the following distribution:

Base	Position in the binding site									
	1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**
A	11	0	0	0	16	2	4	0	4	3
T	3	0	19	0	1	3	4	20	2	4
G	4	20	0	0	2	4	6	0	12	2
C	2	0	1	20	1	11	6	0	2	11

What is the consensus sequence for the binding site of X?

(A) NGTCNNNTNN
(B) AGTCACNTGC
(C) CACCTANCTG
(D) ANNNACGNGC

Question Number : 24 Correct : 1 Wrong : 0

A bacterium has a genome of size 6 million base pairs. If the average rate of DNA synthesis is 1000 base pairs/second, the time taken (in minutes) for replication of the genome will be _____.

Question Number : 25 Correct : 1 Wrong : 0

At the transcription start site of a gene, any of the four nucleotides can occur with equal probability p. The Shannon Entropy S, given by $S = -\sum_{i=1}^{4} p_i \ln p_i$, for this start site is _____.

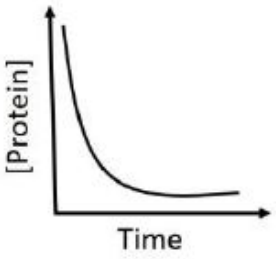
Given data: ln(2) = 0.69

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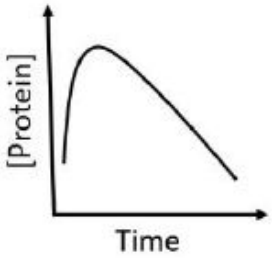
Question Number : 26 **Correct : 2 Wrong : -0.66**

Which one of the following graphs represents the kinetics of protein precipitation by addition of ammonium sulphate? On the Y-axis, [Protein] represents the concentration of free protein in solution.

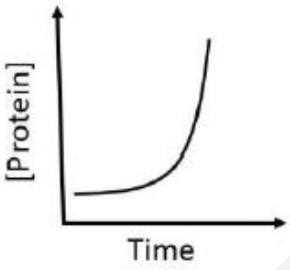
(A)



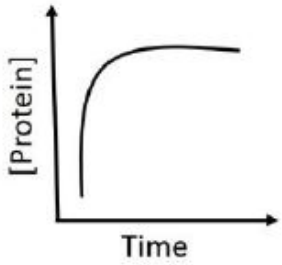
(B)



(C)



(D)



Question Number : 27 **Correct : 2 Wrong : -0.66**

The distribution of marks scored by a large class in an exam can be represented as a normal distribution with mean μ and standard deviation σ. In a follow-up exam in the same class, everyone scored 5 marks more than their respective score in the earlier exam. For this follow-up exam, the distribution of marks can be represented as a normal distribution with mean μ_2 and standard deviation σ_2. Which one of the following is correct?

(A) $\mu = \mu_2; \sigma > \sigma_2$ (B) $\mu < \mu_2; \sigma > \sigma_2$ (C) $\mu > \mu_2; \sigma < \sigma_2$ (D) $\mu < \mu_2; \sigma = \sigma_2$

Question Number : 28 **Correct : 2 Wrong : 0**

The angle (in degrees) between the vectors $\vec{x} = \hat{\imath} - \hat{\jmath} + 2\hat{k}$ and $\vec{y} = 2\hat{\imath} - \hat{\jmath} - 1.5\hat{k}$ is _____ .

Question Number : 29 **Correct : 2 Wrong : -0.66**

Match the proteins in Group I with cellular processes in Group II

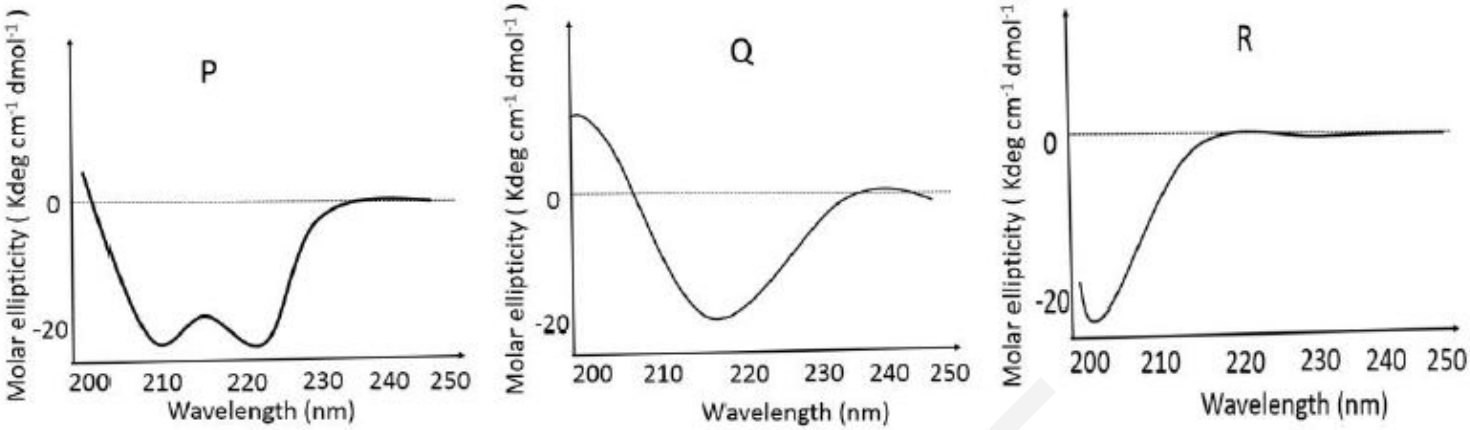
Group I	Group II
P. p53	1. DNA packaging
Q. Lysozyme	2. Apoptosis
R. Tubulins	3. Hydrolysis of polysaccharides
S. Histones	4. Chromosome segregation

(A) P-4, Q-2, R-3, S-1
(B) P-2, Q-3, R-1, S-4
(C) P-4, Q-3, R-1, S-2
(D) P-2, Q-3, R-4, S-1

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Question Number : 30 **Correct : 2 Wrong : -0.66**

The circular dichroism spectra of three proteins P, Q, and R are given below:



Choose the correct combination.

(A) P: α-helix, Q: β-sheet, and R: Random coil
(B) P: β-sheet, Q: α-helix, and R: Random coil
(C) P: α-helix, Q: Random coil, and R: β-sheet
(D) P: Random coil, Q: α-helix, and R: β-sheet

Question Number : 31 **Correct : 2 Wrong : -0.66**

Match the organisms in Column I with the characteristics in Column II

Column I	**Column II**
P. *Methanococcus*	1. Halophile
Q. *Dunaliella*	2. Acidophile
R. *Sulfolobus*	3. Mesophile
S. *Escherichia*	4. Barophile

(A) P-4, Q-3, R-2, S-1
(B) P-3, Q-2, R-4, S-1
(C) P-2, Q-1, R-4, S-3
(D) P-4, Q-1, R-2, S-3

Question Number : 32 **Correct : 2 Wrong : -0.66**

Which one of the following amino acids has three ionizable groups?

(A) Glycine (B) Leucine (C) Valine (D) Lysine

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Question Number : 33 **Correct : 2 Wrong : 0**

Consider an infinite number of cylinders. The first cylinder has a radius of 1 meter and height of 1 meter. The second one has a radius of 0.5 meter and height of 0.5 meter. Every subsequent cylinder has half the radius and half the height of the preceding cylinder. The sum of the volumes (in cubic meters) of these infinite number of cylinders is _____.

Given data: $\pi = 3.14$.

Question Number : 34 **Correct : 2 Wrong : 0**

The concentration (in micromolar) of NADH in a solution with $A_{340} = 0.50$ is _____.

Given data: Path length = 1 cm; Molar extinction coefficient of NADH $\varepsilon_{340} = 6220\ M^{-1}cm^{-1}$.

Question Number : 35 **Correct : 2 Wrong : 0**

The specific activity of an enzyme in a crude extract of *E. coli* is 9.5 units/mg of protein. The specific activity increased to 68 units/mg of protein upon ion-exchange chromatography. The fold purification is _____.

Question Number : 36 **Correct : 2 Wrong : -0.66**

Which one of the following organisms is responsible for crown gall disease in plants?

(A) *Xanthomonas campestris*
(B) *Rhizobium etli*
(C) *Agrobacterium tumefaciens*
(D) *Erwinia stewartii*

Question Number : 37 **Correct : 2 Wrong : 0**

The value of c for which the following system of linear equations has an infinite number of solutions is _____.

$$\begin{bmatrix} 1 & 2 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} c \\ 4 \end{bmatrix}$$

Question Number : 38 **Correct : 2 Wrong : -0.66**

Match the plant hormones in Column I with functions in Column II

Column I	Column II
P. Gibberellic acid	1. Seed and bud dormancy
Q. Zeatin	2. Fruit ripening
R. Ethylene	3. Delaying leaf senescence
S. Abscisic acid	4. Regulation of plant height

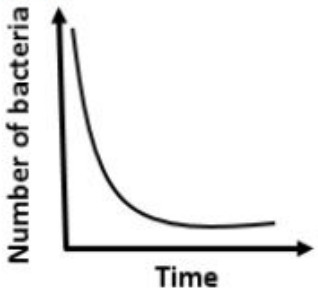
(A) P-4, Q-3, R-2, S-1
(B) P-4, Q-2, R-3, S-1
(C) P-3, Q-1, R-2, S-4
(D) P-2, Q-1, R-4, S-3

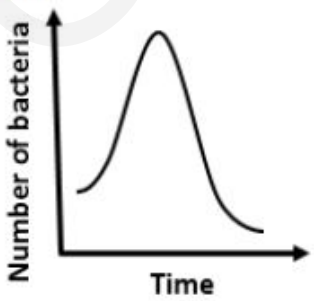
Question Number : 39 **Correct : 2 Wrong : 0**

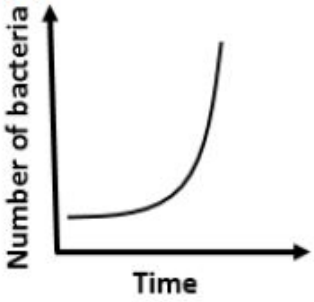
For an *E. coli* culture in the exponential phase of growth, optical density (OD) at 540 nm is 0.3 at 2 hours and 0.6 at 4 hours. Assuming that the measured OD is linearly proportional to the number of *E. coli* cells, the growth rate (per hour) for this culture is _____.

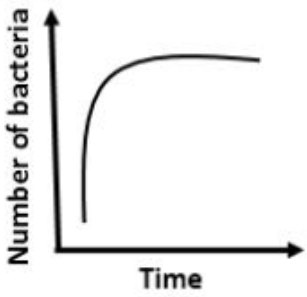
Question Number : 40 **Correct : 2 Wrong : -0.66**

An immunocompetent person becomes infected with a pathogenic strain of bacteria. Which one of the following graphs correctly depicts bacterial load in this person over time?

(A)


(B)


(C)


(D)


Question Number : 41 **Correct : 2 Wrong : -0.66**

The genome is diploid at the end of which phases of a human mitotic cell cycle?

(A) G2 & S (B) G1 & M (C) M & S (D) G1 & G2

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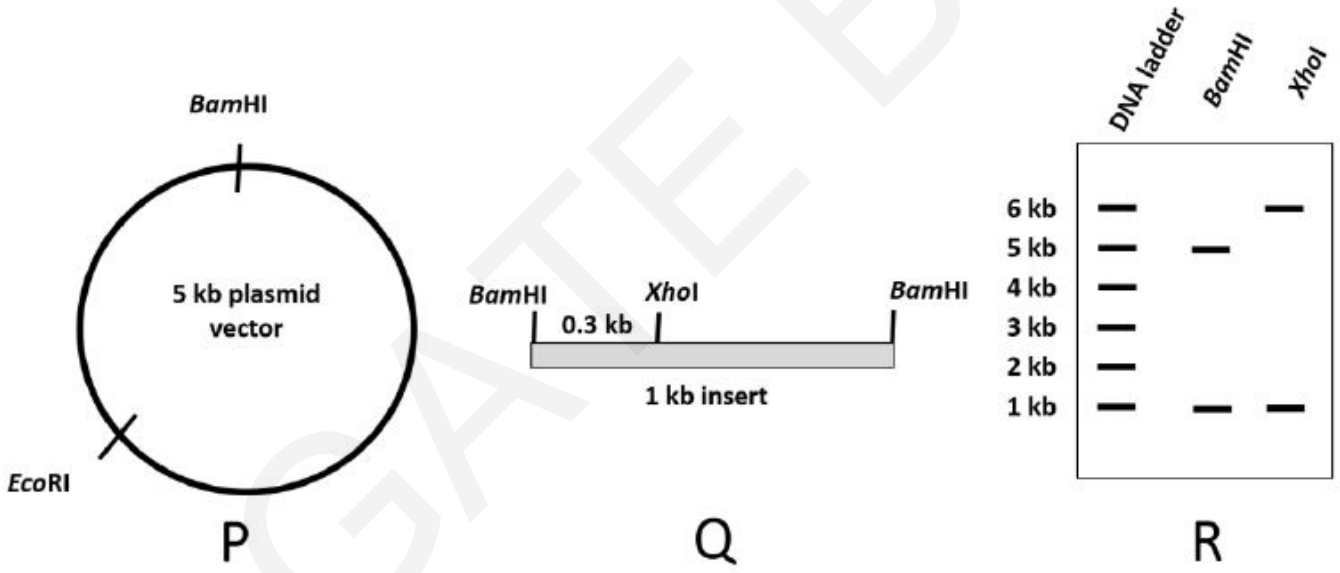
Question Number : 42 Correct : 2 Wrong : -0.66

A recombinant protein is to be expressed under the control of the *lac* promoter and operator in a strain of *E. coli* having the genotype $lacI^+$ crp^+. Even in the absence of inducer IPTG, low levels of expression of the recombinant protein are seen (leaky expression). Which one of the following should be done to minimize such leaky expression?

(A) Addition of lactose to the medium
(B) Removal of all glucose from the medium
(C) Addition of excess glucose to the medium
(D) Addition of *allo*-lactose to the medium

Question Number : 43 Correct : 2 Wrong : -0.66

Shown below is a plasmid vector (P) and an insert (Q). The insert was cloned into the *Bam*HI site of the vector. The recombinant plasmid was isolated and digested with *Bam*HI or *Xho*I. The results from the digestion experiments are shown in (R).



Which one of the following explains the digestion results shown in (R)?

(A) The insert did not ligate to the vector.
(B) One copy of the insert ligated to the vector.
(C) The insert ligated to the vector as two tandem copies.
(D) The insert ligated to the vector as two copies but not in tandem.

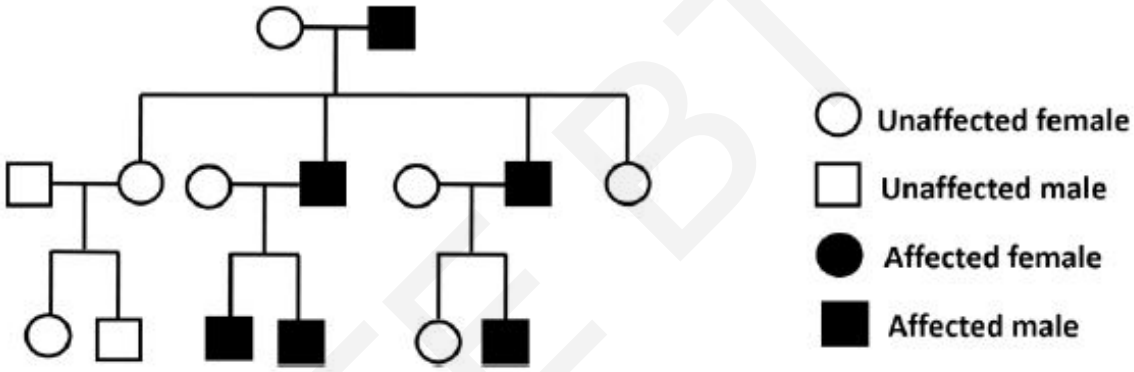
Question Number : 44 **Correct : 2 Wrong : -0.66**

Which one of the following CANNOT be a recognition sequence for a Type II restriction enzyme?

(A) GAATTC
(B) AGCT
(C) GCGGCCGC
(D) ATGCCT

Question Number : 45 **Correct : 2 Wrong : -0.66**

A pedigree of an inheritable disease is shown below.



This inheritable disease is

(A) X-linked dominant
(B) X-linked recessive or Y-linked
(C) only Y-linked
(D) only X-linked recessive

Question Number : 46 **Correct : 2 Wrong : 0**

If the chemical composition of proteins in an organism is $CH_{1.5}O_{0.3}N_{0.3}S_{0.004}$, the mass percentage of carbon in the proteins is _____.

Given data: Atomic weights (Da) of C = 12, H = 1, O = 16, N = 14, and S = 32.

Question Number : 47 **Correct : 2 Wrong : 0**

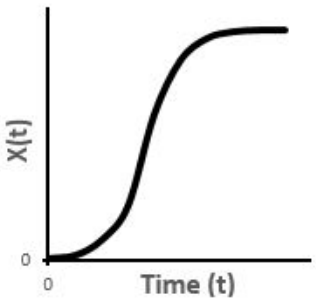
For the probability density $P(x) = 0.5e^{-0.5x}$, the integral $\int_0^\infty P(x)dx =$_____.

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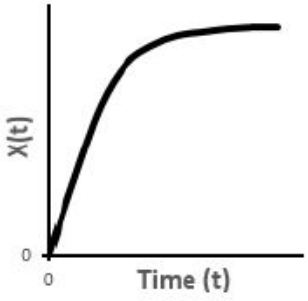
Question Number : 48 Correct : 2 Wrong : -0.66

Growth of a microbe in a test tube is modeled as $\frac{dX}{dt} = rX\left(1 - \frac{X}{K}\right)$, where, X is the biomass, r is the growth rate, and K is the carrying capacity of the environment ($r \neq 0$; $K \neq 0$). If the value of starting biomass is $\frac{K}{100}$, which one of the following graphs qualitatively represents the growth dynamics?

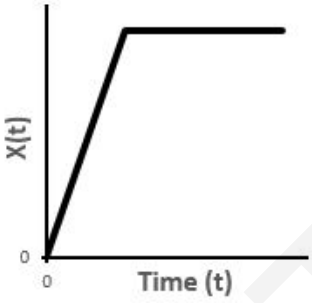
(A)



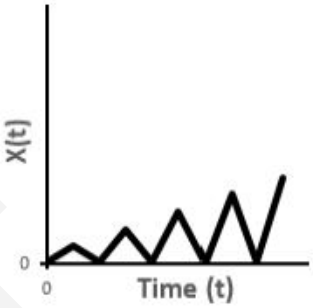
(B)



(C)



(D)



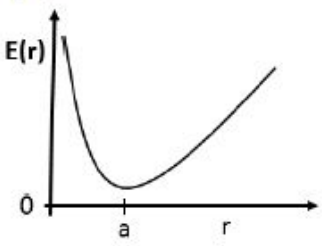
Question Number : 49 Correct : 2 Wrong : 0

A zero-order liquid phase reaction A $\xrightarrow{k}$ B is being carried out in a batch reactor with $k = 10^{-3}$ mol/min. If the starting concentration of A is 0.1 moles/liter, the time (in minutes) taken by the system before A is exhausted in a 100 liter reactor is _____.

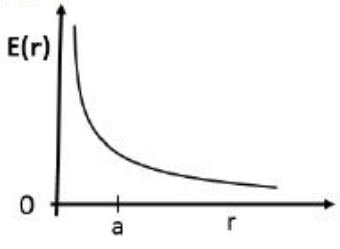
Question Number : 50 Correct : 2 Wrong : -0.66

The interaction energy E between two spherical particles is plotted as a function of the distance (r) between them. When r<a, where a is a constant, the net force between the spherical particles is repulsive. When r≥a, they attract via van der Waals attraction. Which one of the following plots correctly represents the interaction energy between the above two particles?

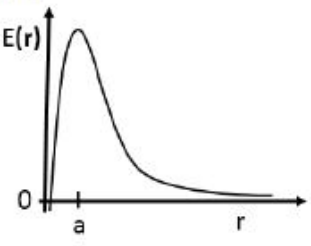
(A)



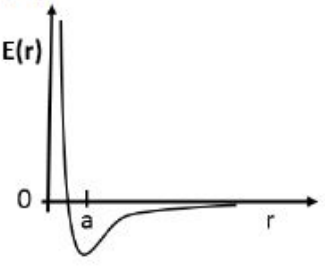
(B)



(C)



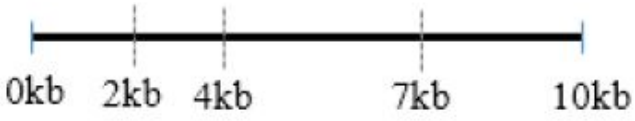
(D)



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Question Number : 51 **Correct : 2 Wrong : -0.66**

*Eco*RI restriction sites on a 10kb DNA fragment are shown below.



Upon partial digestion, what are the lengths (in kb) of all the possible DNA fragments obtained?

(A) 2, 3, 4, 5, 6, 7, 8, and 10
(B) 2, 3, 4, 5, 6, and 7
(C) 2, 3, 4, and 7
(D) 2 and 3

Question Number : 52 **Correct : 2 Wrong : 0**

A DNA strand of length 25 nm wraps diametrically around the circumference of a spherical histone-octamer once. The radius (nm) of the histone-octamer is _____.

Given data: $\pi = 3.14$

Question Number : 53 **Correct : 2 Wrong : 0**

One hundred *E. coli* cells are each infected by a single λ phage particle. The ratio of the number of phage particles committing to lysogeny to those committing to lysis is 4:1. Assuming that the average burst size is 80, the number of free phage particles released after one round of infection is _____.

Question Number : 54 **Correct : 2 Wrong : 0**

During anaerobic growth, an organism converts glucose (P) into biomass (Q), ethanol (R), acetaldehyde (S), and glycerol (T). Every mole of carbon present in glucose gets distributed among the products as follows:

1 (C-mole P) → 0.14 (C-mole Q) + 0.25 (C-mole R) + 0.3 (C-mole S) + 0.31 (C-mole T)

From 1800 grams of glucose fed to the organism, the ethanol produced (in grams) is _____.

Given data: Atomic weights (Da) of C = 12, H = 1, O = 16, and N = 14

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Question Number : 55 **Correct : 2 Wrong : -0.66**

Which of the following conditions promote the development of human autoimmune disorders?

P. Inability to eliminate self-reactive lymphocytes
Q. Generation of auto-antibodies
R. Ability to eliminate self-reactive T-cells
S. Induction of regulatory T-cells in the thymus

(A) P, R (B) Q, S (C) P, Q (D) R, S

Question Number : 56 **Correct : 1 Wrong : -0.33**

She has a sharp tongue and it can occasionally turn_____

(A) hurtful (B) left (C) methodical (D) vital

Question Number : 57 **Correct : 1 Wrong : -0.33**

I _____ made arrangements had I _____ informed earlier.

(A) could have, been
(B) would have, being
(C) had, have
(D) had been, been

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Question Number : 58 **Correct : 1 Wrong : -0.33**

In the summer, water consumption is known to decrease overall by 25%. A Water Board official states that in the summer household consumption decreases by 20%, while other consumption increases by 70%.

Which of the following statements is correct?

(A) The ratio of household to other consumption is 8/17
(B) The ratio of household to other consumption is 1/17
(C) The ratio of household to other consumption is 17/8
(D) There are errors in the official's statement.

Question Number : 59 **Correct : 1 Wrong : -0.33**

40% of deaths on city roads may be attributed to drunken driving. The number of degrees needed to represent this as a slice of a pie chart is

(A) 120 (B) 144 (C) 160 (D) 212

Question Number : 60 **Correct : 1 Wrong : -0.33**

Some tables are shelves. Some shelves are chairs. All chairs are benches. Which of the following conclusions can be deduced from the preceding sentences?

i. At least one bench is a table
ii. At least one shelf is a bench
iii. At least one chair is a table
iv. All benches are chairs

(A) Only i (B) Only ii (C) Only ii and iii (D) Only iv

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Question Number : 61 **Correct : 2 Wrong : -0.66**

"If you are looking for a history of India, or for an account of the rise and fall of the British Raj, or for the reason of the cleaving of the subcontinent into two mutually antagonistic parts and the effects this mutilation will have in the respective sections, and ultimately on Asia, you will not find it in these pages; for though I have spent a lifetime in the country, I lived too near the seat of events, and was too intimately associated with the actors, to get the perspective needed for the impartial recording of these matters".

Here, the word 'antagonistic' is closest in meaning to

(A) impartial (B) argumentative (C) separated (D) hostile

Question Number : 62 **Correct : 2 Wrong : -0.66**

S, T, U, V, W, X, Y, and Z are seated around a circular table. T's neighbours are Y and V. Z is seated third to the left of T and second to the right of S. U's neighbours are S and Y; and T and W are not seated opposite each other. Who is third to the left of V?

(A) X (B) W (C) U (D) T

Question Number : 63 **Correct : 2 Wrong : -0.66**

Trucks (10 m long) and cars (5 m long) go on a single lane bridge. There must be a gap of at least 20 m after each truck and a gap of at least 15 m after each car. Trucks and cars travel at a speed of 36 km/h. If cars and trucks go alternately, what is the maximum number of vehicles that can use the bridge in one hour?

(A) 1440 (B) 1200 (C) 720 (D) 600

Question Number : 64 **Correct : 2 Wrong : -0.66**

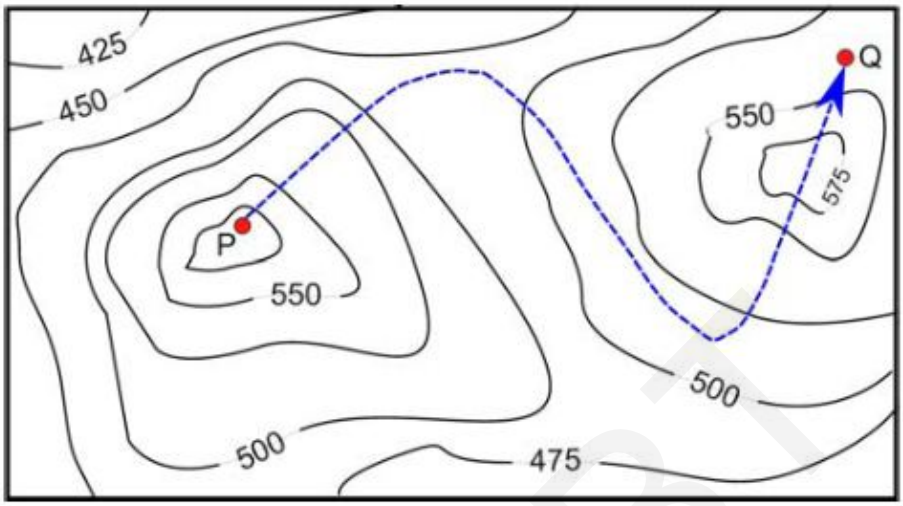
There are 3 Indians and 3 Chinese in a group of 6 people. How many subgroups of this group can we choose so that every subgroup has at least one Indian?

(A) 56 (B) 52 (C) 48 (D) 44

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Question Number : 65 **Correct : 2 Wrong : -0.66**

A contour line joins locations having the same height above the mean sea level. The following is a contour plot of a geographical region. Contour lines are shown at 25 m intervals in this plot.



The path from P to Q is best described by

(A) Up-Down-Up-Down
(B) Down-Up-Down-Up
(C) Down-Up-Down
(D) Up-Down-Up

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GATE BT 2017 Answer Key

Q. No.	Type	Section	Key	Marks
1	MCQ	BT	B	1
2	MCQ	BT	A	1
3	MCQ	BT	A	1
4	NAT	BT	3.10 to 3.20	1
5	NAT	BT	0.68 to 0.70	1
6	NAT	BT	1 to 1	1
7	MCQ	BT	C	1
8	MCQ	BT	B	1
9	MCQ	BT	D	1
10	MCQ	BT	A	1
11	MCQ	BT	C	1
12	MCQ	BT	A	1
13	MCQ	BT	C	1
14	NAT	BT	3 to 3	1
15	MCQ	BT	C	1
16	NAT	BT	1 to 1	1
17	MCQ	BT	D	1
18	MCQ	BT	A or C	1
19	MCQ	BT	B	1
20	MCQ	BT	C	1
21	MCQ	BT	B	1
22	NAT	BT	0.85 to 0.95	1
23	MCQ	BT	B	1
24	NAT	BT	100 to 100	1
25	NAT	BT	1.35 to 1.42	1
26	MCQ	BT	A	2
27	MCQ	BT	D	2
28	NAT	BT	90 to 90	2
29	MCQ	BT	D	2
30	MCQ	BT	A	2
31	MCQ	BT	D	2
32	MCQ	BT	D	2
33	NAT	BT	3.50 to 3.70	2
34	NAT	BT	78 to 82	2
35	NAT	BT	7.00 to 7.40	2
36	MCQ	BT	C	2

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37	NAT	BT	4 to 4	2
38	MCQ	BT	A	2
39	NAT	BT	0.34 to 0.35	2
40	MCQ	BT	B	2
41	MCQ	BT	B	2
42	MCQ	BT	C	2
43	MCQ	BT	C	2
44	MCQ	BT	D	2
45	MCQ	BT	C	2
46	NAT	BT	52.00 to 54.00	2
47	NAT	BT	1 to 1	2
48	MCQ	BT	A	2
49	NAT	BT	100 to 100 or 10000 to 10000	2
50	MCQ	BT	D	2
51	MCQ	BT	A	2
52	NAT	BT	3.90 to 4.10	2
53	NAT	BT	1600 to1600	2
54	NAT	BT	345 to 345	2
55	MCQ	BT	C	2
56	MCQ	GA	A	1
57	MCQ	GA	A	1
58	MCQ	GA	D	1
59	MCQ	GA	B	1
60	MCQ	GA	B	1
61	MCQ	GA	D	2
62	MCQ	GA	A	2
63	MCQ	GA	A	2
64	MCQ	GA	A	2
65	MCQ	GA	C	2

Q. 1 – Q. 5 carry one mark each.

Q.1 "When she fell down the _____, she received many _____ but little help."

The words that best fill the blanks in the above sentence are

(A) stairs, stares
(B) stairs, stairs
(C) stares, stairs
(D) stares, stares

Q.2 "In spite of being warned repeatedly, he failed to correct his _____ behaviour."

The word that best fills the blank in the above sentence is

(A) rational (B) reasonable (C) errant (D) good

Q.3 For $0 \leq x \leq 2\pi$, $\sin x$ and $\cos x$ are both decreasing functions in the interval _____.

(A) $\left(0, \frac{\pi}{2}\right)$ (B) $\left(\frac{\pi}{2}, \pi\right)$ (C) $\left(\pi, \frac{3\pi}{2}\right)$ (D) $\left(\frac{3\pi}{2}, 2\pi\right)$

Q.4 The area of an equilateral triangle is $\sqrt{3}$. What is the perimeter of the triangle?

(A) 2 (B) 4 (C) 6 (D) 8

Q.5 Arrange the following three-dimensional objects in the descending order of their volumes:

(i) A cuboid with dimensions 10 cm, 8 cm and 6 cm
(ii) A cube of side 8 cm
(iii) A cylinder with base radius 7 cm and height 7 cm
(iv) A sphere of radius 7 cm

(A) (i), (ii), (iii), (iv)
(B) (ii), (i), (iv), (iii)
(C) (iii), (ii), (i), (iv)
(D) (iv), (iii), (ii), (i)

Q. 6 – Q. 10 carry two marks each.

Q.6 An automobile travels from city A to city B and returns to city A by the same route. The speed of the vehicle during the onward and return journeys were constant at 60 km/h and 90 km/h, respectively. What is the average speed in km/h for the entire journey?

(A) 72 (B) 73 (C) 74 (D) 75

Q.7 A set of 4 parallel lines intersect with another set of 5 parallel lines. How many parallelograms are formed?

(A) 20 (B) 48 (C) 60 (D) 72

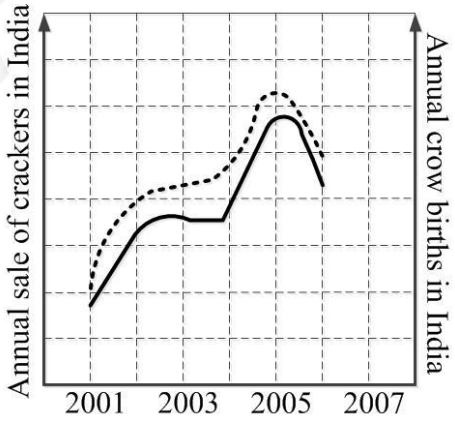
Q.8 To pass a test, a candidate needs to answer at least 2 out of 3 questions correctly. A total of 6,30,000 candidates appeared for the test. Question A was correctly answered by 3,30,000 candidates. Question B was answered correctly by 2,50,000 candidates. Question C was answered correctly by 2,60,000 candidates. Both questions A and B were answered correctly by 1,00,000 candidates. Both questions B and C were answered correctly by 90,000 candidates. Both questions A and C were answered correctly by 80,000 candidates. If the number of students answering all questions correctly is the same as the number answering none, how many candidates failed to clear the test?

(A) 30,000 (B) 2,70,000 (C) 3,90,000 (D) 4,20,000

Q.9 If $x^2 + x - 1 = 0$ what is the value of $x^4 + \frac{1}{x^4}$?

(A) 1 (B) 5 (C) 7 (D) 9

Q.10 In a detailed study of annual crow births in India, it was found that there was relatively no growth during the period 2002 to 2004 and a sudden spike from 2004 to 2005. In another unrelated study, it was found that the revenue from cracker sales in India which remained fairly flat from 2002 to 2004, saw a sudden spike in 2005 before declining again in 2006. The solid line in the graph below refers to annual sale of crackers and the dashed line refers to the annual crow births in India. Choose the most appropriate inference from the above data.



(A) There is a strong correlation between crow birth and cracker sales.
(B) Cracker usage increases crow birth rate.
(C) If cracker sale declines, crow birth will decline.
(D) Increased birth rate of crows will cause an increase in the sale of crackers.

END OF THE QUESTION PAPER

Q. 1 – Q. 25 carry one mark each.

Q.1 Consider an unfair coin. The probability of getting heads is 0.6. If you toss this coin twice, what is the probability that the first or the second toss is heads?

(A) 0.56 (B) 0.64 (C) 0.84 (D) 0.96

Q.2 If serum is removed from the growth medium of human embryonic kidney cell line (HEK), then the cells will

(A) proliferate faster
(B) proliferate normally
(C) undergo cell cycle arrest
(D) undergo immediate apoptosis

Q.3 The repeat sequence of telomere in humans is

(A) 5′-TATAAT-3′ (B) 5′-TTAGGG-3′ (C) 5′-GGGCCC-3′ (D) 5′-AAAAAA-3′

Q.4 If a segment of a sense strand of DNA is 5′-ATGGACCAGA-3′, then the resulting RNA sequence after transcription is

(A) 5′-AGACCAGGTA-3′
(B) 5′-UCUGGUCCAU-3′
(C) 5′-UACCUGGUCU-3′
(D) 5′-AUGGACCAGA-3′

Q.5 Which one of the following is an example of a neurotoxin?

(A) Cholera toxin
(B) Streptolysin-O
(C) Botulinum toxin
(D) Diphtheria toxin

Q.6 Which of the following components constitute a molecular mechanics force field?

P. Bond stretching
Q. Bond angle bending
R. Torsional bond rotation
S. Non-bonded interactions

(A) P and Q only
(B) P, Q and R only
(C) P, Q and S only
(D) P, Q, R and S

Q.7 Which one of the following BLAST search programs is used to identify homologs of a genomic DNA query in a protein sequence database?

(A) blastp (B) blastn (C) blastx (D) tblastn

Q.8 A mixture contains three similarly sized peptides P, Q and R. The peptide P is positively charged, Q is weakly negative and R is strongly negative. If this mixture is passed through an ion-exchange chromatography column containing an anionic resin, their order of elution will be

(A) P, Q, R
(B) R, Q, P
(C) Q, R, P
(D) P, Q and R elute together

Q.9 Which one of the following is **INCORRECT** about protein structures?

(A) A protein fold is stabilized by favorable non-covalent interactions
(B) All parts of a fold can be classified as helices, strands or turns
(C) Two non-covalent atoms cannot be closer than the sum of their van der Waals radii
(D) The peptide bond is nearly planar

Q.10 Which one of the following metabolic processes in mammalian cells does **NOT** occur in the mitochondria?

(A) Citric acid cycle
(B) Oxidative phosphorylation
(C) Fatty acid β-oxidation
(D) Glycolysis

Q.11 Which one of the following is **NOT** a principal component of innate immunity?

(A) Mucosal epithelia
(B) Dendritic cells
(C) Complement system
(D) Memory B-cells

Q.12 Which of the following technique(s) can be used to study conformational changes in myoglobin?

P. Mass spectrometry
Q. Fluorescence spectroscopy
R. Circular dichroism spectroscopy
S. Light microscopy

(A) P only (B) P and S only (C) Q and R only (D) S only

Q.13 Which one of the following bioreactor configurations is the basis for a trickling biological filter?

(A) Stirred tank
(B) Packed bed
(C) Air lift
(D) Fluidized bed

GATE BT

Q.14 Cell type A secretes molecule X into the culture medium. Cell type B in the same culture responds to the molecule X by expressing protein Y. Which one of the following modes of signaling represents the interaction between A and B?

(A) Autocrine
(B) Juxtacrine
(C) Paracrine
(D) Intracrine

Q.15 Which one of the following statements is true for actin?

(A) Actin filament is structurally polarized and the two ends are not identical
(B) *De novo* actin polymerization is a single-step process
(C) The pointed end of the actin filaments is the fast growing end
(D) Actin forms spindle fibers during mitosis

Q.16 Standard error is

(A) the probability of a type I error in a statistical test
(B) the error in estimating a sample standard deviation
(C) the standard deviation of a variable that follows standard normal distribution
(D) the standard deviation of distribution of sample means

Q.17 Which one of the following techniques is used to monitor RNA transcripts, both temporally and spatially?

(A) Northern blotting
(B) *In situ* hybridization
(C) Southern blotting
(D) Western blotting

Q.18 Identify the character based method(s) used for the construction of a phylogenetic tree.

P. Maximum parsimony
Q. Neighbor joining
R. Maximum likelihood
S. Bootstrapping

(A) Q only
(B) P and R only
(C) Q and S only
(D) S only

Q.19 Which one of the following is the solution for $cos^2x + 2cosx + 1 = 0$, for values of x in the range of $0^\circ < x < 360^\circ$?

(A) 45°
(B) 90°
(C) 180°
(D) 270°

Q.20 Which one of the following plant secondary metabolites is a natural insecticide?

(A) Digitoxin
(B) Pyrethrin
(C) Salicylic acid
(D) Avenacin A-1

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Q.21 The determinant of the matrix $\begin{pmatrix} 4 & -6 \\ -3 & 2 \end{pmatrix}$ is _____

Q.22 The variable z has a standard normal distribution. If $P(0 \leq z \leq 1) = 0.34$, then $P(z^2 > 1)$ is equal to (up to two decimal places) _____

Q.23 The absorbance of a solution of tryptophan measured at 280 nm in a cuvette of 2.0 cm path length is 0.56 at pH 7. The molar extinction coefficient (ε) for tryptophan at 280 nm is 5600 $M^{-1}cm^{-1}$ at pH 7. The concentration of tryptophan (in μM) in the solution is _____

Q.24 A single stem cell undergoes 10 asymmetric cell divisions. The number of stem cells at the end is _____

Q.25 Genomic DNA isolated from a bacterium was digested with a restriction enzyme that recognizes a 6-base pair (bp) sequence. Assuming random distribution of bases, the average length (in bp) of the fragments generated is _____

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Q. 26 – Q. 55 carry two marks each.

Q.26 In leguminous plants, both the rhizobium genes and the plant genes influence nodulation and nitrogen fixation. Which one of the following functions is **NOT** encoded by the host plant genes?

(A) Production of inducers that modify rhizobial cell wall
(B) Production of flavonoid inducers
(C) Establishment of contact between bacteria and legume
(D) Root hair curling

Q.27 Which of the following cytokines are endogenous pyrogens?

P. Tumor necrosis factor-α
Q. Interleukin-1
R. Transforming growth factor-β
S. Interleukin-10

(A) P and Q only
(B) P and R only
(C) R and S only
(D) Q and S only

Q.28 Match the classes of RNA molecules in Group I with their functions in Group II.

Group I	**Group II**
P. snoRNA	1. Protects germline from transposable elements
Q. piRNA	2. Blocks translation of selected mRNA
R. miRNA	3. Template for telomere elongation
S. snRNA	4. Modification and processing of rRNA
	5. Splicing of RNA transcripts

(A) P-3, Q-5, R-2, S-4
(B) P-1, Q-3, R-2, S-5
(C) P-1, Q-4, R-5, S-2
(D) P-4, Q-1, R-2, S-5

Q.29 Determine the correctness or otherwise of the following Assertion [a] and the Reason [r]

Assertion: *Ab initio* gene finding algorithms that predict protein coding genes in eukaryotic genomes are not completely accurate.

Reason: Eukaryotic splice sites are difficult to predict.

(A) Both [a] and [r] are false
(B) [a] is true but [r] is false
(C) Both [a] and [r] are true and [r] is the correct reason for [a]
(D) Both [a] and [r] are true but [r] is not the correct reason for [a]

Q.30 Which one of the following amino acids is catalyzed by activated macrophages to produce reactive nitrogen species?

(A) Arginine (B) Asparagine (C) Cysteine (D) Histidine

Q.31 Determine the correctness or otherwise of the following Assertion [a] and the Reason [r]

Assertion: The association constant in water for the G-C base pair is three times lower than that for the A-T base pair.

Reason: There are three hydrogen bonds in the G-C base pair and two in the A-T base pair.

(A) Both [a] and [r] are true and [r] is the correct reason for [a]
(B) [a] is false but [r] is true
(C) Both [a] and [r] are false
(D) Both [a] and [r] are true and [r] is not the correct reason for [a]

Q.32 Which one of the combinations of the following statements is true about antibody structure?

P. Limited proteolysis of rabbit IgG with the enzyme pepsin generates two antigen-binding regions (Fab) and an Fc fragment
Q. Limited proteolysis of rabbit IgG with the enzyme papain generates a single bivalent antigen-binding region $F(ab')_2$ and peptide fragments
R. The Fc fragment of IgG can self-associate and crystallize into a lattice
S. The $F(ab')_2$ fragment of IgG is composed of both light and heavy chains

(A) P and Q only (B) P and R only
(C) R and S only (D) Q and S only

Q.33 Which one of the following statements is true with regard to processing and presentation of protein antigens?

(A) In the class II MHC pathway, protein antigens in the cytosol are processed by proteasomes
(B) In the class I MHC pathway, extracellular protein antigens are endocytosed into vesicles and processed
(C) In the class I MHC pathway, transporter associated antigen processing (TAP) protein is required for translocating processed peptides generated in the cytosol
(D) Invariant chain in endoplasmic reticulum is involved in transport of peptides and loading of class I MHC

Q.34 Which of the following are true about bacterial superoxide dismutase?

P. Present in obligate aerobes
Q. Present in facultative anaerobes
R. Present in aerotolerant anaerobes
S. Absent in obligate aerobes

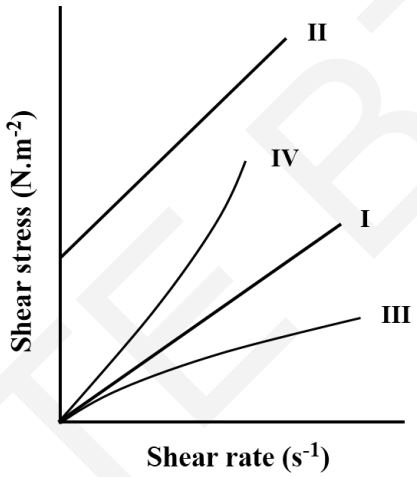
(A) P and Q only (B) P, Q and R only
(C) P and R only (D) Q and S only

Q.35 Which of the following are true with regard to anaerobic respiration in bacteria?

P. The final electron acceptor is an inorganic substance other than molecular oxygen
Q. The number of ATP molecules produced per glucose molecule is more than that produced in aerobic respiration
R. The number of ATP molecules produced per glucose molecule is less than that produced in aerobic respiration
S. Only substrate level phosphorylation is used to generate ATP

(A) P and S only
(B) Q and S only
(C) P and R only
(D) P, Q and S only

Q.36 Shear stress versus shear rate behavior of four different types of fluids (I, II, III and IV) are shown in the figure below.

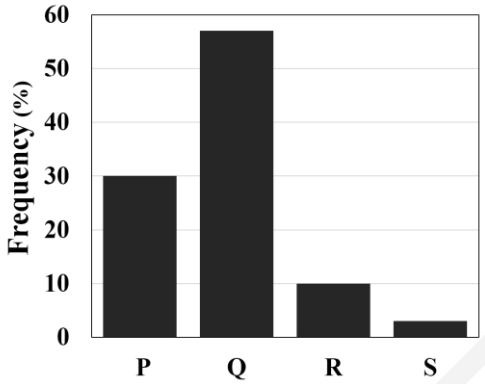


Which one of the following options is correct?

(A) I-Newtonian, II-Bingham plastic, III-Dilatant, IV-Pseudoplastic
(B) I-Pseudoplastic, II-Dilatant, III-Newtonian, IV-Bingham plastic
(C) I-Newtonian, II-Pseudoplastic, III-Bingham plastic, IV-Dilatant
(D) I-Newtonian, II-Bingham plastic, III-Pseudoplastic, IV-Dilatant

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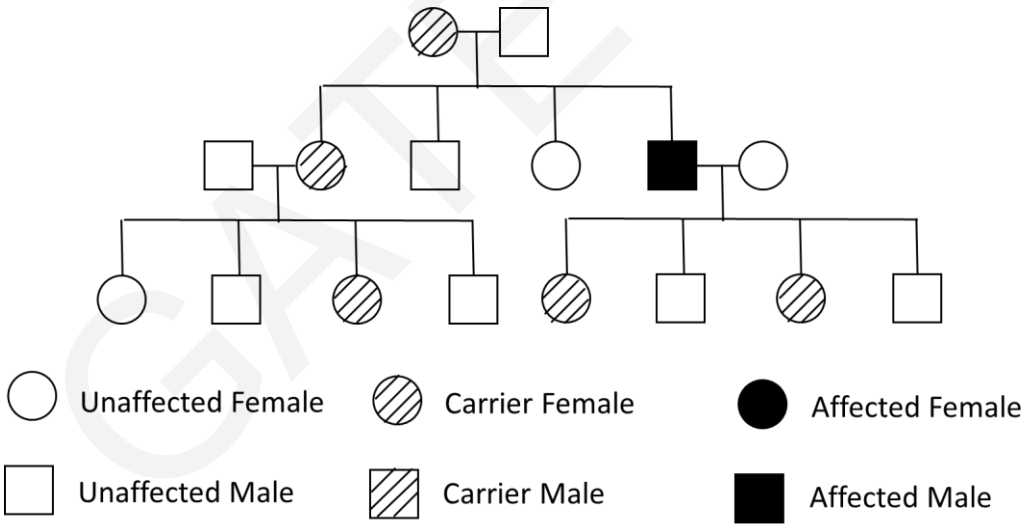
Q.37 An analysis of DNA-protein interactions was carried out using all DNA-protein complexes in the protein data bank (PDB). The frequency distribution of four amino acid residues, represented as P, Q, R and S, occurring in non-covalent interactions between the protein and DNA backbone is shown below.



Which one of the following is correct?

(A) P-Lys, Q-Arg, R-Gln, S-Glu
(B) P-Gln, Q-Glu, R-Lys, S-Arg
(C) P-Asn, Q-Asp, R-Arg, S-Lys
(D) P-His, Q-Glu, R-Gln, S-Lys

Q.38 A pedigree of an inheritable disease is shown below.



What type of inheritance does the disease follow?

(A) Autosomal dominant
(B) X-linked dominant
(C) X-linked recessive
(D) Autosomal recessive

Q.39 Match the industrial products mentioned in Group I with their producer organisms in Group II

Group I	**Group II**
P. Citric acid	1. *Trichoderma viride*
Q. Cellulase	2. *Clostridium acetobutylicum*
R. Vitamin B_{12}	3. *Aspergillus niger*
S. Butanol	4. *Propionibacterium freudenreichii*

(A) P-4, Q-3, R-1, S-2
(B) P-3, Q-1, R-2, S-4
(C) P-2, Q-1, R-4, S-3
(D) P-3, Q-1, R-4, S-2

Q.40 5′ capping of mRNA transcripts in eukaryotes involves the following events:

P. Addition of GMP on the 5′ end
Q. Removal of γ-phosphate of the triphosphate on first base at the 5′ end
R. 5′-5′ linkage between GMP and the first base at 5′ end
S. Addition of methyl group to N7 position of guanine

Which one of the following is the correct sequence of events?

(A) P, Q, R, S
(B) P, R, Q, S
(C) Q, P, R, S
(D) Q, P, S, R

Q.41 Calculate the following integral (up to two decimal places)

$$\int_0^1 (x+3)(x+1)dx = _____$$

Q.42 The probability distribution for a discrete random variable X is given below.

X	1	2	3	4
$P(X)$	0.3	0.4	0.2	0.1

The expectation value of X is (up to one decimal place) _____

Q.43 If $1 + r + r^2 + r^3 + \cdots \infty = 1.5$,
then, $1 + 2r + 3r^2 + 4r^3 + \cdots \infty =$ (up to two decimal places) _____

Q.44 Moist heat sterilization of spores at 121 ^{o}C follows first order kinetics as per the expression:

$$\frac{dN}{dt} = -k_d N$$

where, N is the number of viable spores, t is the time, k_d is the rate constant and $\frac{dN}{dt}$ is the rate of change of viable spores.

If k_d value is 1.0 min^{-1}, the time (in minutes) required to reduce the number of viable spores from an initial value of 10^{10} to a final value of 1 is (up to two decimal places) _____

Q.45 An aqueous solution containing 6.8 mg/L of an antibiotic is extracted with amyl acetate. If the partition coefficient of the antibiotic is 170 and the ratio of water to solvent is 85, then the extraction factor is _____

Q.46 A microbial strain is cultured in a 100 L stirred fermenter for secondary metabolite production. If the specific rate of oxygen uptake is 0.4 h^{-1} and the oxygen solubility in the broth is 8 mg/L, then the volumetric mass transfer coefficient (K_La) (in s^{-1}) of oxygen required to achieve a maximum cell concentration of 12 g/L is (up to two decimal places) _____

Q.47 In a chemostat, the feed flow rate and culture volume are 100 ml/h and 1.0 L, respectively. With glucose as substrate, the values of μ_{max} and K_s are 0.2 h^{-1} and 1 g/L, respectively. For a glucose concentration of 10 g/L in the feed, the effluent substrate concentration (in g/L) is _____

Q.48 Mammalian cells in active growth phase were seeded at a density of 1×10^5 cells/ml. After 72 hours, 1×10^6 cells/ml were obtained. The population doubling time of the cells in hours is (up to two decimal places) _____

Q.49 Yeast converts glucose to ethanol and carbon dioxide by glycolysis as per the following reaction:

$$C_6H_{12}O_6 \rightarrow 2C_2H_5OH + 2CO_2$$

Assuming complete conversion, the amount of ethanol produced (in g) from 200 g of glucose is (up to two decimal places) _____

Q.50 At the end of a batch culture, glucose solution is added at a flow rate of 200 ml/h. If the culture volume after 2 h of glucose addition is 1000 ml, the initial culture volume (in ml) is _____

Q.51 Consider the following alignment of two DNA sequences:

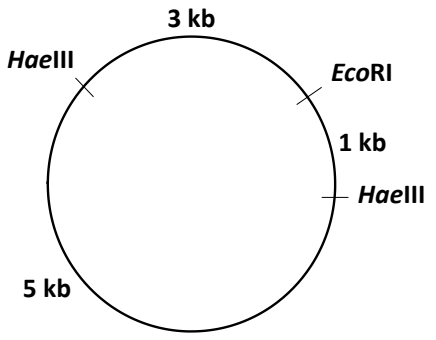
AGTAAC
AA--AC

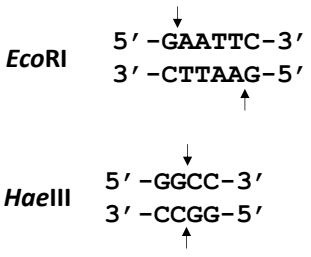
Assuming an affine gap scoring scheme of an identity matrix for substitution, a gap initiation penalty of 1 and a gap extension penalty of 0.1, the score of the alignment is (up to one decimal place) _____

Q.52 First order deactivation rate constants for soluble and immobilized amyloglucosidase enzyme are 0.03 min^{-1} and 0.005 min^{-1}, respectively. The ratio of half-life of the immobilized enzyme to that of the soluble enzyme is (rounded off to the nearest integer) _____

Q.53 Consider a simple uni-substrate enzyme that follows Michaelis-Menten kinetics. When the enzyme catalyzed reaction was carried out in the presence of 10 nM concentration of an inhibitor, there was no change in the maximal velocity. However, the slope of the Lineweaver-Burk plot increased 3-fold. The dissociation constant for the enzyme-inhibitor complex (in nM) is _____

Q.54 The product of complete digestion of the plasmid shown below with *Eco*RI and *Hae*III was purified and used as a template in a reaction containing Klenow fragment of DNA polymerase, dNTPs and [α-^{32}P]-dATP in a suitable reaction buffer. The product thus obtained was purified and subjected to gel electrophoresis followed by autoradiography.





The number of bands that will appear on the X-ray film is _____

Q.55 A rod shaped bacterium has a length of 2 μm, diameter of 1 μm and density the same as that of water. If proteins constitute 15% of the cell mass and the average protein has a mass of 50 kDa, the number of proteins in the cell is _____
(1 Da = 1.6×10^{-24}g)

END OF THE QUESTION PAPER

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GATE BT 2018 Answer Key

Q.No.	Type	Section	Key/Range	Marks
1	MCQ	GA	A	1
2	MCQ	GA	C	1
3	MCQ	GA	B	1
4	MCQ	GA	C	1
5	MCQ	GA	D	1
6	MCQ	GA	A	2
7	MCQ	GA	C	2
8	MCQ	GA	D	2
9	MCQ	GA	C	2
10	MCQ	GA	A	2
1	MCQ	BT	C	1
2	MCQ	BT	C	1
3	MCQ	BT	B	1
4	MCQ	BT	D	1
5	MCQ	BT	C	1
6	MCQ	BT	D	1
7	MCQ	BT	C	1
8	MCQ	BT	B	1
9	MCQ	BT	B	1
10	MCQ	BT	D	1
11	MCQ	BT	D	1
12	MCQ	BT	C	1

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Q.No.	Type	Section	Key/Range	Marks
13	MCQ	BT	B	1
14	MCQ	BT	C	1
15	MCQ	BT	A	1
16	MCQ	BT	D	1
17	MCQ	BT	B	1
18	MCQ	BT	B	1
19	MCQ	BT	C	1
20	MCQ	BT	B	1
21	NAT	BT	-10 to -10	1
22	NAT	BT	0.32 to 0.32	1
23	NAT	BT	50 to 50	1
24	NAT	BT	1 to 1	1
25	NAT	BT	4096 to 4096	1
26	MCQ	BT	D	2
27	MCQ	BT	A	2
28	MCQ	BT	D	2
29	MCQ	BT	C	2
30	MCQ	BT	A	2
31	MCQ	BT	B	2
32	MCQ	BT	C	2
33	MCQ	BT	C	2
34	MCQ	BT	B	2

GATE BT

Q.No.	Type	Section	Key/Range	Marks
35	MCQ	BT	C	2
36	MCQ	BT	D	2
37	MCQ	BT	A	2
38	MCQ	BT	C	2
39	MCQ	BT	D	2
40	MCQ	BT	C	2
41	NAT	BT	5.30 to 5.34	2
42	NAT	BT	2.1 to 2.1	2
43	NAT	BT	2.25 to 2.25	2
44	NAT	BT	23.02 to 23.04	2
45	NAT	BT	2 to 2	2
46	NAT	BT	0.16 to 0.17	2
47	NAT	BT	1 to 1	2
48	NAT	BT	21.00 to 22.00	2
49	NAT	BT	102.00 to 103.00	2
50	NAT	BT	600 to 600	2
51	NAT	BT	1.8 to 1.9	2
52	NAT	BT	6 to 6	2
53	NAT	BT	5 to 5	2
54	NAT	BT	2 to 2	2
55	NAT	BT	2900000 to 3200000	2

GATE BT

GATE 2019 General Aptitude (GA) Set-8

Q. 1 – Q. 5 carry one mark each.

Q.1 The fishermen, _____ the flood victims owed their lives, were rewarded by the government.

(A) whom (B) to which (C) to whom (D) that

Q.2 Some students were not involved in the strike.

If the above statement is true, which of the following conclusions is/are logically necessary?

1. Some who were involved in the strike were students.
2. No student was involved in the strike.
3. At least one student was involved in the strike.
4. Some who were not involved in the strike were students.

(A) 1 and 2 (B) 3 (C) 4 (D) 2 and 3

Q.3 The radius as well as the height of a circular cone increases by 10%. The percentage increase in its volume is _____.

(A) 17.1 (B) 21.0 (C) 33.1 (D) 72.8

Q.4 Five numbers 10, 7, 5, 4 and 2 are to be arranged in a sequence from left to right following the directions given below:
1. No two odd or even numbers are next to each other.
2. The second number from the left is exactly half of the left-most number.
3. The middle number is exactly twice the right-most number.

Which is the second number from the right?

(A) 2 (B) 4 (C) 7 (D) 10

Q.5 Until Iran came along, India had never been _____ in kabaddi.

(A) defeated (B) defeating (C) defeat (D) defeatist

Q. 6 – Q. 10 carry two marks each.

Q.6 Since the last one year, after a 125 basis point reduction in repo rate by the Reserve Bank of India, banking institutions have been making a demand to reduce interest rates on small saving schemes. Finally, the government announced yesterday a reduction in interest rates on small saving schemes to bring them on par with fixed deposit interest rates.

Which one of the following statements can be inferred from the given passage?

(A) Whenever the Reserve Bank of India reduces the repo rate, the interest rates on small saving schemes are also reduced

(B) Interest rates on small saving schemes are always maintained on par with fixed deposit interest rates

(C) The government sometimes takes into consideration the demands of banking institutions before reducing the interest rates on small saving schemes

(D) A reduction in interest rates on small saving schemes follow only after a reduction in repo rate by the Reserve Bank of India

Q.7 In a country of 1400 million population, 70% own mobile phones. Among the mobile phone owners, only 294 million access the Internet. Among these Internet users, only half buy goods from e-commerce portals. What is the percentage of these buyers in the country?

(A) 10.50 (B) 14.70 (C) 15.00 (D) 50.00

Q.8 The nomenclature of Hindustani music has changed over the centuries. Since the medieval period *dhrupad* styles were identified as *baanis*. Terms like *gayaki* and *baaj* were used to refer to vocal and instrumental styles, respectively. With the institutionalization of music education the term *gharana* became acceptable. *Gharana* originally referred to hereditary musicians from a particular lineage, including disciples and grand disciples.

Which one of the following pairings is NOT correct?

(A) *dhrupad*, *baani*
(B) *gayaki*, vocal
(C) *baaj*, institution
(D) *gharana*, lineage

Q.9 Two trains started at 7AM from the same point. The first train travelled north at a speed of 80km/h and the second train travelled south at a speed of 100 km/h. The time at which they were 540 km apart is _____ AM.

(A) 9 (B) 10 (C) 11 (D) 11.30

Q.10 "I read somewhere that in ancient times the prestige of a kingdom depended upon the number of taxes that it was able to levy on its people. It was very much like the prestige of a head-hunter in his own community."

Based on the paragraph above, the prestige of a head-hunter depended upon _____

(A) the prestige of the kingdom
(B) the prestige of the heads
(C) the number of taxes he could levy
(D) the number of heads he could gather

END OF THE QUESTION PAPER

GATE BT

Q. 1 – Q. 25 carry one mark each.

Q.1 The Bt toxin gene from *Bacillus thuringiensis* used to generate genetically modified crops is

(A) *cry* (B) *cro* (C) *cdc* (D) *cre*

Q.2 Which one of the following is used as a pH indicator in animal cell culture medium?

(A) Acridine orange
(B) Phenol red
(C) Bromophenol blue
(D) Coomassie blue

Q.3 Tetracycline inhibits the

(A) interaction between tRNA and mRNA
(B) translocation of mRNA through ribosome
(C) peptidyl transferase activity
(D) binding of amino-acyl tRNA to ribosome

Q.4 Which one of the following is a database of protein sequence motifs?

(A) PROSITE (B) TrEMBL (C) SWISSPROT (D) PDB

Q.5 Which one of the following enzymes is encoded by human immunodeficiency virus (HIV) genome?

(A) Reverse transcriptase
(B) Phospholipase
(C) Phosphatase
(D) ATP synthase

Q.6 DNA synthesis in eukaryotes occurs during which phase of the mitotic cell cycle?

(A) M (B) G_1 (C) S (D) G_0

Q.7 Match the human diseases in Group I with the causative agents in Group II.

Group I	**Group II**
P. Amoebiasis	1. *Leishmania donovani*
Q. African sleeping sickness	2. *Trypanosoma cruzi*
R. Kala azar	3. *Entamoeba histolytica*
S. Chagas' disease	4. *Trypanosoma gambiense*

(A) P-3, Q-4, R-2, S-1
(B) P-3, Q-2, R-1, S-4
(C) P-3, Q-4, R-1, S-2
(D) P-4, Q-3, R-1, S-2

GATE BT

Q.8 Which one of the following techniques can be used to compare the expression of a large number of genes in two biological samples in a single experiment?

(A) Polymerase chain reaction
(B) DNA microarray
(C) Northern hybridization
(D) Southern hybridization

Q.9 Which of the following processes can increase genetic diversity of bacteria in nature?

P. Conjugation
Q. Transformation
R. Transduction
S. Transfection

(A) P only (B) P and Q only (C) P, Q and R only (D) P, Q, R and S

Q.10 Which one of the following is NOT a part of the human nonspecific defense system?

(A) Interferon (B) Mucous (C) Saliva (D) Antibody

Q.11 A mutation in a gene that codes for a polypeptide results in a variant polypeptide that lacks the last three amino acids. What type of mutation is this?

(A) Synonymous mutation
(B) Nonsense mutation
(C) Missense mutation
(D) Silent mutation

Q.12 Which one of the following equations represents a one-dimensional wave equation?

(A) $\frac{\partial u}{\partial t} = c^2 \frac{\partial^2 u}{\partial x^2}$ (B) $\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}$ (C) $\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial u}{\partial x}$ (D) $\frac{\partial^2 u}{\partial t^2} + \frac{\partial^2 u}{\partial x^2} = 0$

Q.13 Which of the following are geometric series?

P. 1, 6, 11, 16, 21, 26, …
Q. 9, 6, 3, 0, -3, -6, …
R. 1, 3, 9, 27, 81, …
S. 4, -8, 16, -32, 64, …

(A) P and Q only (B) R and S only (C) Q and S only (D) P, Q and R only

Q.14 Which one of the following statements is CORRECT for enzyme catalyzed reactions? (ΔG is Gibbs free energy change, K_{eq} is equilibrium constant)

(A) Enzymes affect ΔG, but not K_{eq}
(B) Enzymes affect K_{eq}, but not ΔG
(C) Enzymes affect both ΔG and K_{eq}
(D) Enzymes do not affect ΔG or K_{eq}

GATE BT

Q.15 Which one of the following can NOT be a limiting substrate if Monod's growth kinetics is applicable?

(A) Extracellular carbon source
(B) Extracellular nitrogen source
(C) Dissolved oxygen
(D) Intracellular carbon source

Q.16 Which one of the following is the unit of heat transfer coefficient?

(A) $W\ m^2\ K^{-1}$ (B) $W\ m^{-2}\ K$ (C) $W\ m^{-2}\ K^{-1}$ (D) $W\ m^2\ K$

Q.17 Which one of the following is catabolized during endogenous metabolism in a batch bacterial cultivation?

(A) internal reserves
(B) extracellular substrates
(C) extracellular products
(D) toxic substrates

Q.18 Which one of the following need NOT be conserved in a biochemical reaction?

(A) Total mass
(B) Total moles
(C) Number of atoms of each element
(D) Total energy

Q.19 The number of possible rooted trees in a phylogeny of three species is _____.

Q.20 Matrix A = $\begin{bmatrix} 0 & 6 \\ p & 0 \end{bmatrix}$ will be skew-symmetric when p = _____.

Q.21 The solution of $\lim_{x \to 8} \left(\frac{x^2 - 64}{x - 8} \right)$ is _____.

Q.22 The median value for the dataset (12, 10, 16, 8, 90, 50, 30, 24) is _____.

Q.23 The degree of reduction for acetic acid ($C_2H_4O_2$) is _____.

Q.24 The mass of 1 kmol of oxygen molecules is _____ g (rounded off to the nearest integer).

Q.25 Protein concentration of a crude enzyme preparation was 10 mg mL^{-1}. 10 µL of this sample gave an activity of 5 µmol min^{-1} under standard assay conditions. The specific activity of this crude enzyme preparation is _____ units mg^{-1}.

GATE BT

Q. 26 – Q. 55 carry two marks each.

Q.26 In general, which one of the following statements is NOT CORRECT?

(A) Hydrogen bonds result from electrostatic interactions
(B) Hydrogen bonds contribute to the folding energy of proteins
(C) Hydrogen bonds are weaker than van der Waals interactions
(D) Hydrogen bonds are directional

Q.27 For site-directed mutagenesis, which one of the following restriction enzymes is used to digest methylated DNA?

(A) KpnI (B) DpnI (C) XhoI (D) MluI

Q.28 Match the organelles in Group I with their functions in Group II.

Group I	Group II
P. Lysosome	1. Digestion of foreign substances
Q. Smooth ER	2. Protein targeting
R. Golgi apparatus	3. Lipid synthesis
S. Nucleolus	4. Protein synthesis
	5. rRNA synthesis

(A) P-1, Q-3, R-2, S-5 (B) P-1, Q-4, R-5, S-3
(C) P-2, Q-5, R-3, S-4 (D) P-1, Q-3, R-4, S-5

Q.29 Which of the following statements are CORRECT when a protein sequence database is searched using the BLAST algorithm?

P. A larger E-value indicates higher sequence similarity
Q. E-value $< 10^{-10}$ indicates sequence homology
R. A higher BLAST score indicates higher sequence similarity
S. E-value $> 10^{10}$ indicates sequence homology

(A) P, Q and R only (B) Q and R only
(C) P, R and S only (D) P and S only

Q.30 Which one of the following is coded by the ABO blood group locus in the human genome?

(A) Acyl transferase (B) Galactosyltransferase
(C) Transposase (D) β-Galactosidase

Q.31 Which of the following factors affect the fidelity of DNA polymerase in polymerase chain reaction?

P. Mg^{2+} concentration
Q. pH
R. Annealing temperature

(A) P and Q only
(B) P and R only
(C) Q and R only
(D) P, Q and R

Q.32 Group I lists spectroscopic methods and Group II lists biomolecular applications of these methods. Match the methods in Group I with the applications in Group II.

Group I	**Group II**
P. Infrared	1. Identification of functional groups
Q. Circular Dichroism	2. Determination of secondary structure
R. Nuclear Magnetic Resonance	3. Estimation of molecular weight
	4. Determination of 3-D structure

(A) P-4, Q-3, R-1
(B) P-2, Q-1, R-3
(C) P-1, Q-2, R-4
(D) P-3, Q-2, R-4

Q.33 The hexapeptide P has an isoelectric point (pI) of 6.9. Hexapeptide Q is a variant of P that contains valine instead of glutamate at position 3. The two peptides are analyzed by polyacrylamide gel electrophoresis at pH 8.0. Which one of the following statements is CORRECT?

(A) P will migrate faster than Q towards the anode
(B) P will migrate faster than Q towards the cathode
(C) Both P and Q will migrate together
(D) Q will migrate faster than P towards the anode

Q.34 Antibody-producing hybridoma cells are generated by the fusion of a

(A) T cell with a myeloma cell
(B) B cell with a myeloma cell
(C) macrophage with a myeloma cell
(D) T cell and a B cell

Q.35 Which of the following statements are CORRECT about the function of fetal bovine serum in animal cell culture?

P. It stimulates cell growth
Q. It enhances cell attachment
R. It provides hormones and minerals
S. It maintains pH at 7.4

(A) P and Q only (B) P and S only (C) P, Q and R only (D) P, Q, R and S

Q.36 Which one of the following covalent linkages exists between 7-Methyl guanosine (m^7G) and mRNAs?

(A) 2'–3' triphosphate
(B) 3'–5' triphosphate
(C) 5'–5' triphosphate
(D) 2'–5' triphosphate

Q.37 Which one of the following amino acid residues will destabilize an α-helix when inserted in the middle of the helix?

(A) Pro (B) Val (C) Ile (D) Leu

Q.38 What is the solution of the differential equation $\frac{dy}{dx} = \frac{x}{y}$, with the initial condition, at x = 0, y = 1?

(A) $x^2 = y^2 + 1$ (B) $y^2 = x^2 + 1$ (C) $y^2 = 2x^2 + 1$ (D) $x^2 - y^2 = 0$

Q.39 The Laplace transform of the function $f(t) = t^2 + 2t + 1$ is

(A) $\frac{1}{s^3} + \frac{3}{s^2} + \frac{1}{s}$ (B) $\frac{4}{s^3} + \frac{4}{s^2} + \frac{1}{s}$ (C) $\frac{1}{s^3} + \frac{2}{s^2} + \frac{1}{s}$ (D) $\frac{2}{s^3} + \frac{2}{s^2} + \frac{1}{s}$

Q.40 Which of the following factors can influence the lag phase of a microbial culture in a batch fermentor?

P. Inoculum size
Q. Inoculum age
R. Medium composition

(A) P and Q only (B) Q and R only (C) P and R only (D) P, Q and R

Q.41 Which one of the following statements is CORRECT about proportional controllers?

(A) The initial change in control output signal is relatively slow
(B) The initial corrective action is greater for larger error
(C) They have no offset
(D) There is no corrective action if the error is a constant

Q.42 Match the instruments in Group I with their corresponding measurements in Group II.

Group I	**Group II**
P. Manometer	1. Agitator speed
Q. Rotameter	2. Pressure difference
R. Tachometer	3. Cell number
S. Haemocytometer	4. Air flow rate

(A) P-4, Q-1, R-2, S-3
(B) P-3, Q-4, R-1, S-2
(C) P-2, Q-4, R-1, S-3
(D) P-2, Q-1, R-4, S-3

Q.43 Which of the following statements is ALWAYS CORRECT about an ideal chemostat?

P. Substrate concentration inside the chemostat is equal to that in the exit stream
Q. Optimal dilution rate is lower than critical dilution rate
R. Biomass concentration increases with increase in dilution rate
S. Cell recirculation facilitates operation beyond critical dilution rate

(A) P and Q only (B) P, R and S only (C) P and S only (D) P, Q and S only

Q.44 Determine the correctness or otherwise of the following Assertion [a] and the Reason [r]

Assertion [a]: It is possible to regenerate a whole plant from a single plant cell.
Reason [r]: It is easier to introduce transgenes in to plants than animals.

(A) Both [a] and [r] are true and [r] is the correct reason for [a]
(B) Both [a] and [r] are true but [r] is not the correct reason for [a]
(C) Both [a] and [r] are false
(D) [a] is true but [r] is false

Q.45 A UV-visible spectrophotometer has a minimum detectable absorbance of 0.02. The minimum concentration of a protein sample that can be measured reliably in this instrument with a cuvette of 1 cm path length is _____ µM. The molar extinction coefficient of the protein is 10,000 L mol^{-1} cm^{-1}.

Q.46 The difference in concentrations of an uncharged solute between two compartments is 1.6-fold. The energy required for active transport of the solute across the membrane separating the two compartments is ____ cal mol^{-1} (rounded off to the nearest integer). (R = 1.987 cal mol^{-1} K^{-1}, T = 37 ºC)

Q.47 In pea plants, purple color of flowers is determined by the dominant allele while white color is determined by the recessive allele. A genetic cross between two purple flower-bearing plants results in an offspring with white flowers. The probability that the third offspring from these parents will have purple flowers is _____ (rounded off to 2 decimal places).

Q.48 The molecular mass of a protein is 22 kDa. The size of the cDNA (excluding the untranslated regions) that codes for this protein is _____ kb (rounded off to 1 decimal place).

Q.49 A new game is being introduced in a casino. A player can lose Rs. 100, break even, win Rs. 100, or win Rs. 500. The probabilities ($P(X)$) of each of these outcomes (X) are given in the following table:

X (in Rs.)	-100	0	100	500
$P\ (X)$	0.25	0.5	0.2	0.05

The standard deviation (σ) for the casino payout is Rs. _____ (rounded off to the nearest integer).

Q.50 $\int_{-1}^{1} f(x)dx$ calculated using trapezoidal rule for the values given in the table is _____ (rounded off to 2 decimal places).

x	-1	$-2/3$	$-1/3$	0	$1/3$	$2/3$	1
$f(x)$	0.37	0.51	0.71	1.0	1.40	1.95	2.71

Q.51 Yeast biomass ($C_6H_{10}O_3N$) grown on glucose is described by the stoichiometric equation given below:

$$C_6H_{12}O_6 + 0.48\ NH_3 + 3\ O_2 \rightarrow 0.48\ C_6H_{10}O_3N + 3.12\ CO_2 + 4.32\ H_2O$$

The amount of glucose needed for the production of 50 g L^{-1} of yeast biomass in a batch reactor with a working volume of 1,00,000 L is _____ kg (rounded off to the nearest integer).

Q.52 Phenolic wastewater discharged from an industry was treated with *Pseudomonas* sp. in an aerobic bioreactor. The influent and effluent concentrations of phenol were 10,000 and 10 ppm, respectively. The inlet feed rate of wastewater was 80 L h^{-1}. The kinetic properties of the organism are as follows:

Maximum specific growth rate (μ_m) = 1 h^{-1}
Saturation constant (K_S) = 100 mg L^{-1}
Cell death rate (k_d) = 0.01 h^{-1}

Assuming that the bioreactor operates under 'chemostat' mode, the working volume required for this process is _____ L (rounded off to the nearest integer).

Q.53 In a cross-flow filtration process, the pressure drop (ΔP) driving the fluid flow is 2 atm, inlet feed pressure (P_i) is 3 atm and filtrate pressure (P_f) is equal to atmospheric pressure. The average transmembrane pressure drop (ΔP_m) is _____ atm.

Q.54 An industrial fermentor containing 10,000 L of medium needs to be sterilized. The initial spore concentration in the medium is 10^6 spores mL^{-1}. The desired probability of contamination after sterilization is 10^{-3}. The death rate of spores at 121 ºC is 4 min^{-1}. Assume that there is no cell death during heating and cooling phases. The holding time of the sterilization process is _____ min (rounded off to the nearest integer).

Q.55 The dimensions and operating condition of a lab-scale fermentor are as follows:

Volume = 1 L

Diameter = 20 cm

Agitator speed = 600 rpm

Ratio of impeller diameter to fermentor diameter = 0.3

This fermentor needs to be scaled up to 8,000 L for a large scale industrial application. If the scale-up is based on constant impeller tip speed, the speed of the agitator in the larger reactor is _____ rpm. Assume that the scale-up factor is the cube root of the ratio of fermentor volumes.

END OF THE QUESTION PAPER

GATE BT

Q.No.	Type	Section	Key/Range	Marks
1	MCQ	GA	C	1
2	MCQ	GA	C	1
3	MCQ	GA	C	1
4	MCQ	GA	C	1
5	MCQ	GA	A	1
6	MCQ	GA	C	2
7	MCQ	GA	A	2
8	MCQ	GA	C	2
9	MCQ	GA	B	2
10	MCQ	GA	D	2
1	MCQ	BT	A	1
2	MCQ	BT	B	1
3	MCQ	BT	D	1
4	MCQ	BT	A	1
5	MCQ	BT	A	1
6	MCQ	BT	C	1
7	MCQ	BT	C	1
8	MCQ	BT	B	1
9	MCQ	BT	B	1
10	MCQ	BT	D	1
11	MCQ	BT	B	1
12	MCQ	BT	B	1

GATE BT

GATE BT 2019 Answer Key

Q.No.	Type	Section	Key/Range	Marks
13	MCQ	BT	B	1
14	MCQ	BT	D	1
15	MCQ	BT	D	1
16	MCQ	BT	C	1
17	MCQ	BT	A	1
18	MCQ	BT	B	1
19	NAT	BT	3 to 3	1
20	NAT	BT	-6 to -6	1
21	NAT	BT	16 to 16	1
22	NAT	BT	20 to 20	1
23	NAT	BT	4 to 4	1
24	NAT	BT	31800 to 32000	1
25	NAT	BT	50 to 50	1
26	MCQ	BT	C	2
27	MCQ	BT	B	2
28	MCQ	BT	A	2
29	MCQ	BT	B	2
30	MCQ	BT	B	2
31	MCQ	BT	D	2
32	MCQ	BT	C	2
33	MCQ	BT	A	2
34	MCQ	BT	B	2

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Q.No.	Type	Section	Key/Range	Marks
35	MCQ	BT	C	2
36	MCQ	BT	C	2
37	MCQ	BT	A	2
38	MCQ	BT	B	2
39	MCQ	BT	D	2
40	MCQ	BT	D	2
41	MCQ	BT	B	2
42	MCQ	BT	C	2
43	MCQ	BT	D	2
44	MCQ	BT	B	2
45	NAT	BT	2 to 2	2
46	NAT	BT	280 to 295	2
47	NAT	BT	0.75 to 0.75	2
48	NAT	BT	0.6 to 0.6	2
49	NAT	BT	128 to 130	2
50	NAT	BT	2.20 to 2.50	2
51	NAT	BT	12500 to 14500	2
52	NAT	BT	970 to 1010	2
53	NAT	BT	1 to 1	2
54	NAT	BT	9 to 10	2
55	NAT	BT	30 to 30	2

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